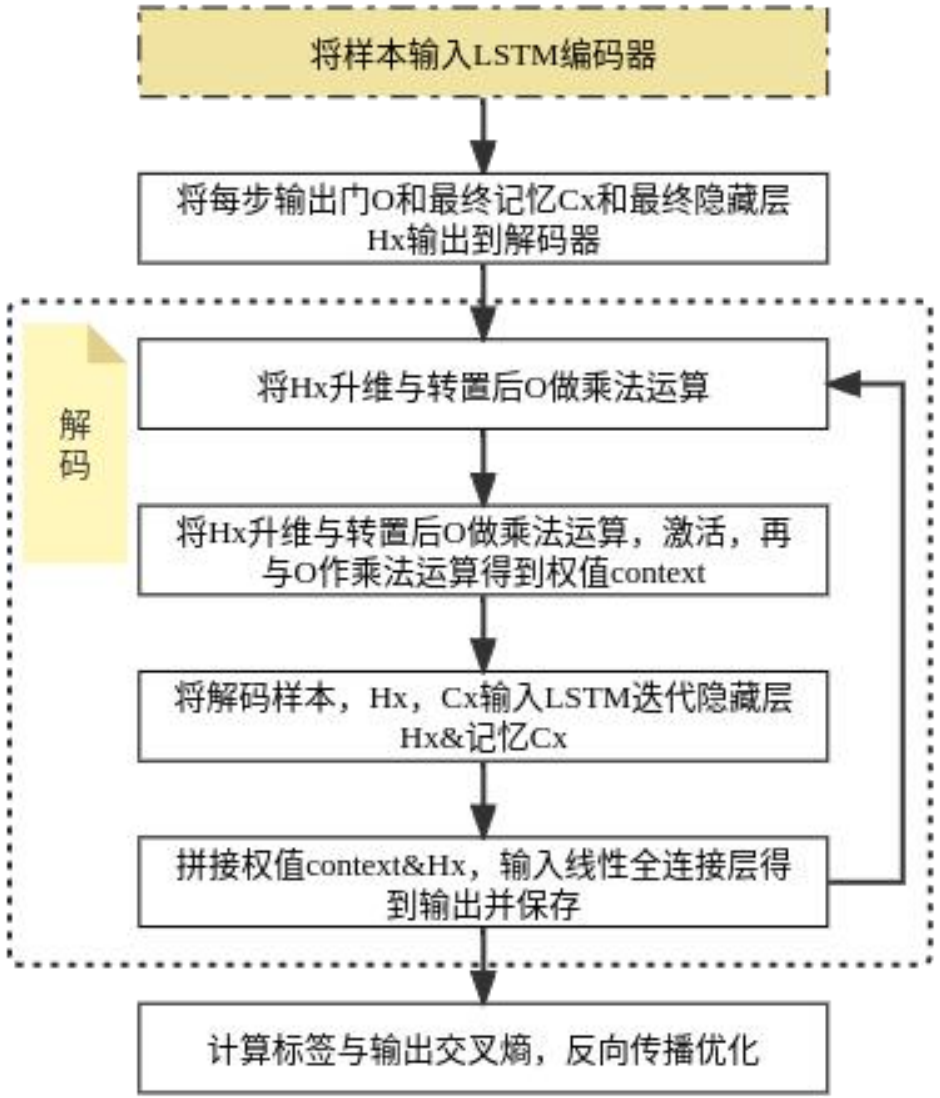
NLP_attention之后

2023.03.20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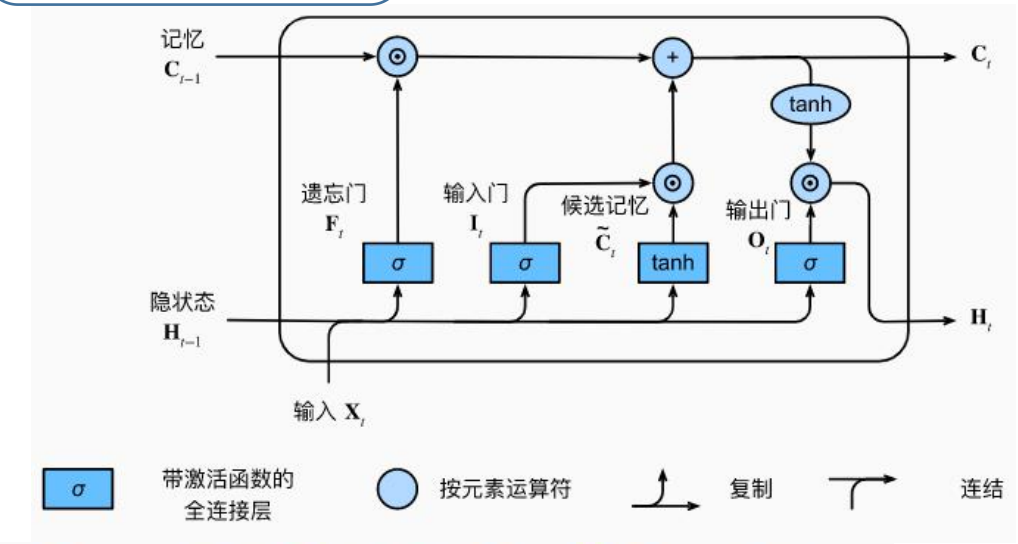
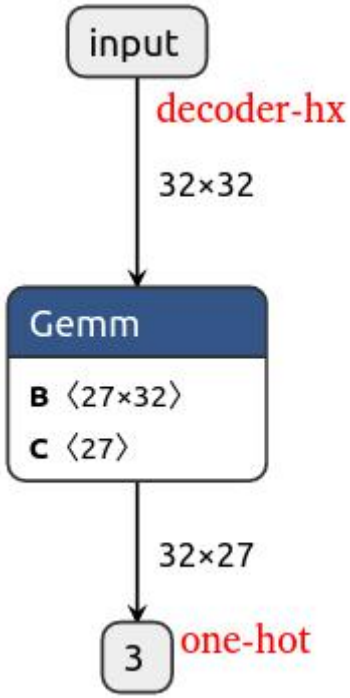
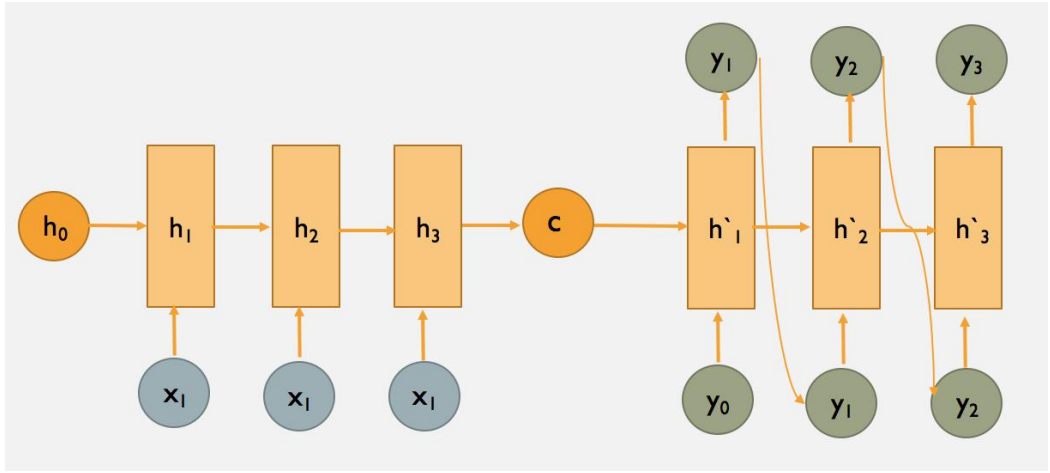
计算标签与输出交叉熵，反向传播优化

attention原理

LSTM





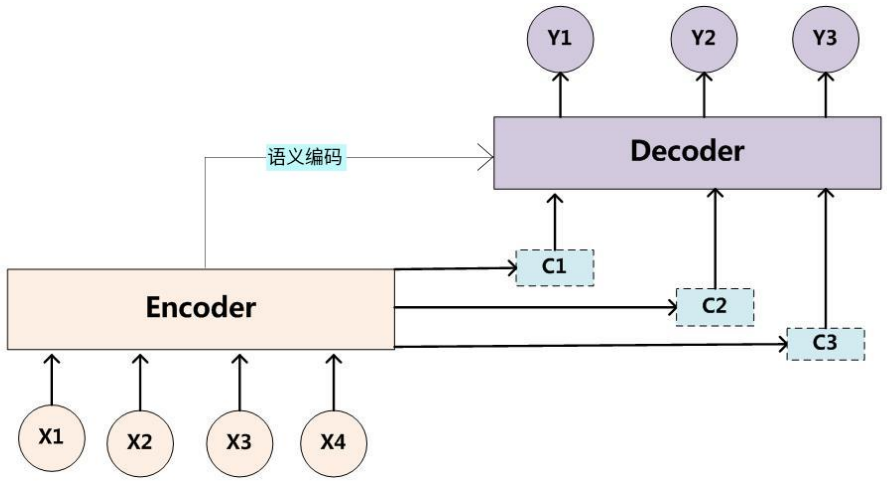


$$\mathbf{I}_t = \sigma(\mathbf{X}_t \mathbf{W}_{xi} + \mathbf{H}_{t-1} \mathbf{W}_{hi} + \mathbf{b}_i),$$
$$\mathbf{F}_t = \sigma(\mathbf{X}_t \mathbf{W}_{xf} + \mathbf{H}_{t-1} \mathbf{W}_{hf} + \mathbf{b}_f),$$
$$\mathbf{O}_t = \sigma(\mathbf{X}_t \mathbf{W}_{xo} + \mathbf{H}_{t-1} \mathbf{W}_{ho} + \mathbf{b}_o),$$

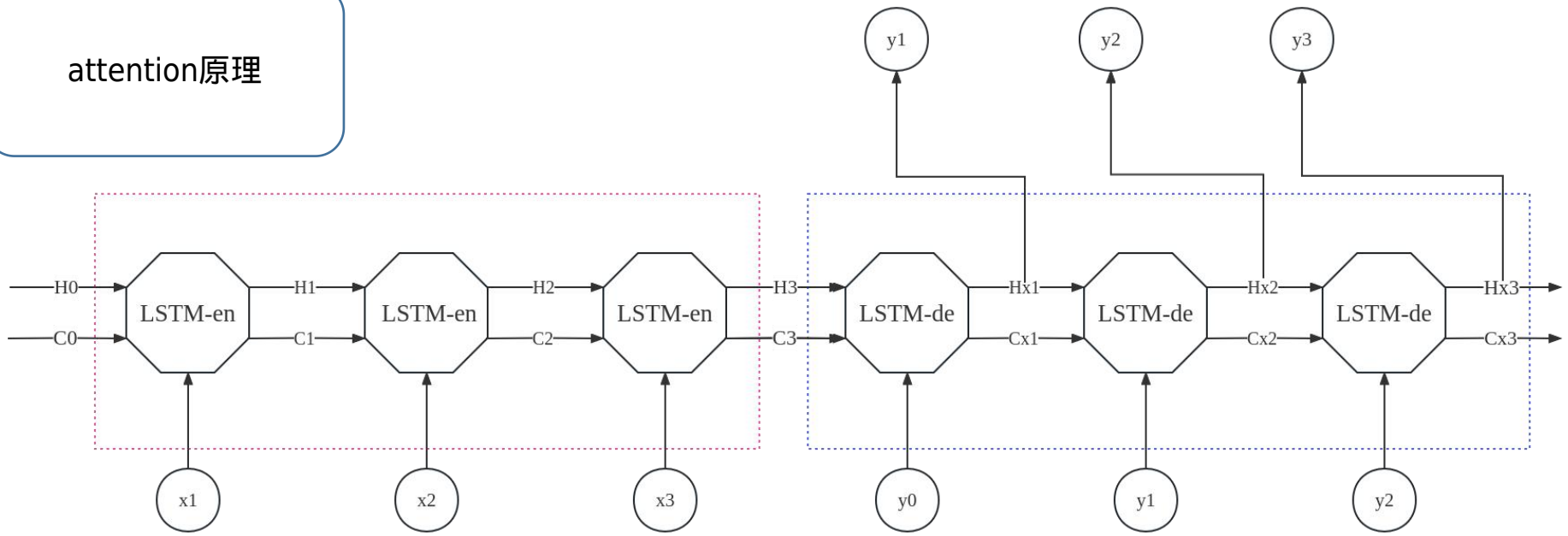
$$\mathbf{C}_t = \mathbf{F}_t \odot \mathbf{C}_{t-1} + \mathbf{I}_t \odot \tilde{\mathbf{C}}_t.$$

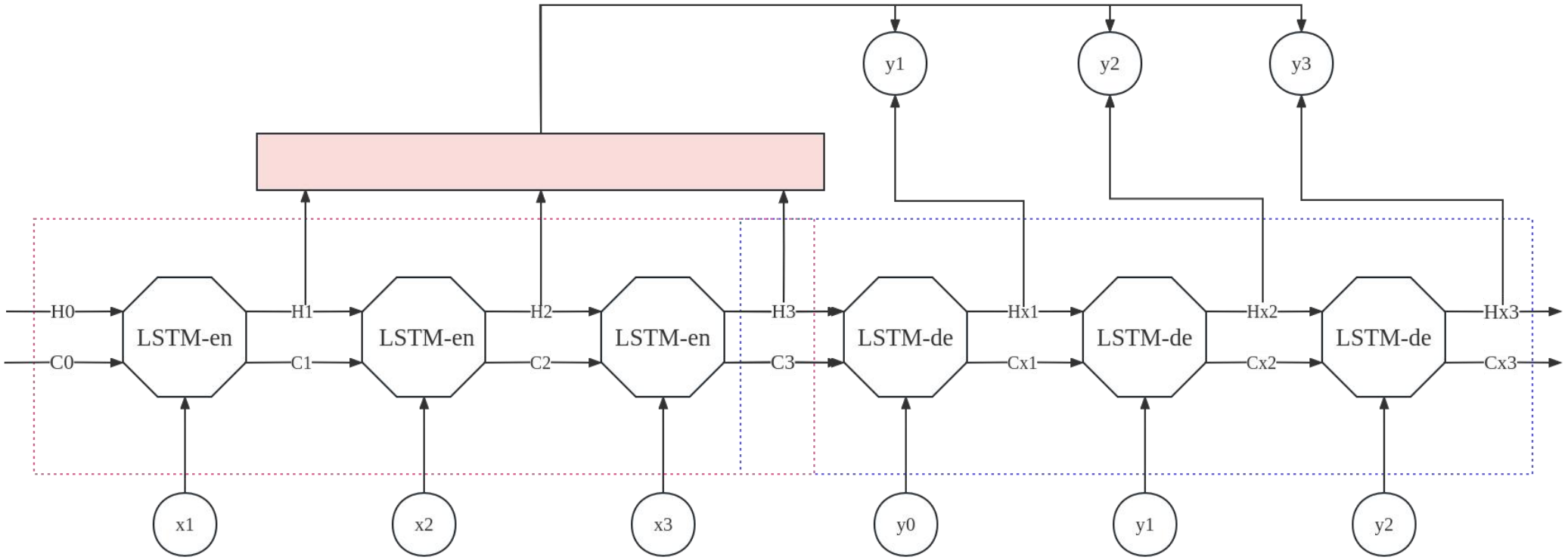
$$\tilde{\mathbf{C}}_t = \tanh(\mathbf{X}_t \mathbf{W}_{xc} + \mathbf{H}_{t-1} \mathbf{W}_{hc} + \mathbf{b}_c)$$

$$\mathbf{H}_t = \mathbf{O}_t \odot \tanh(\mathbf{C}_t)$$

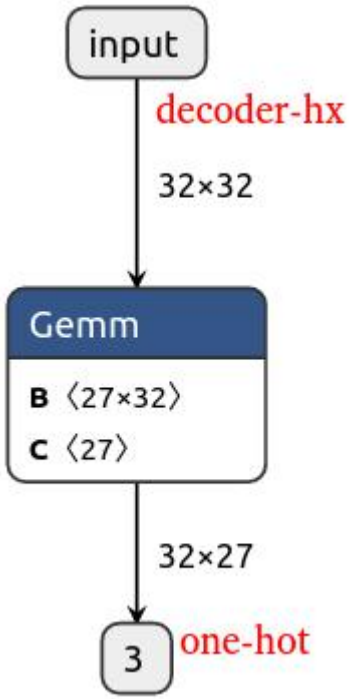


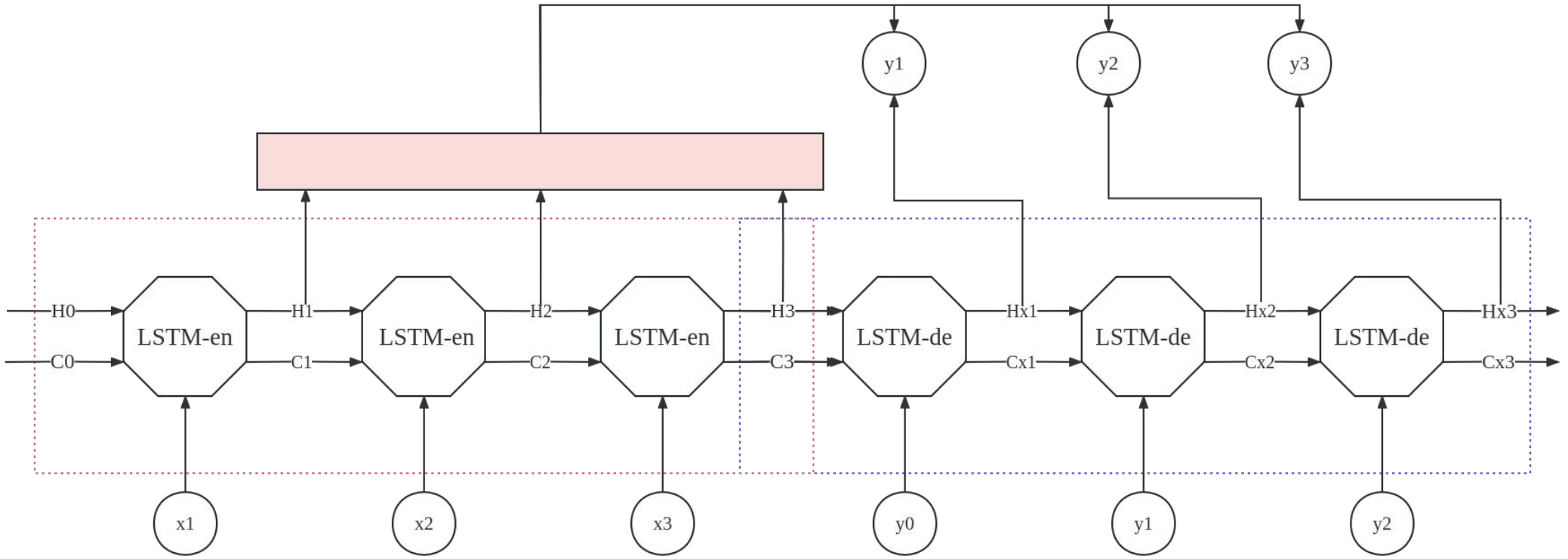
attention原理

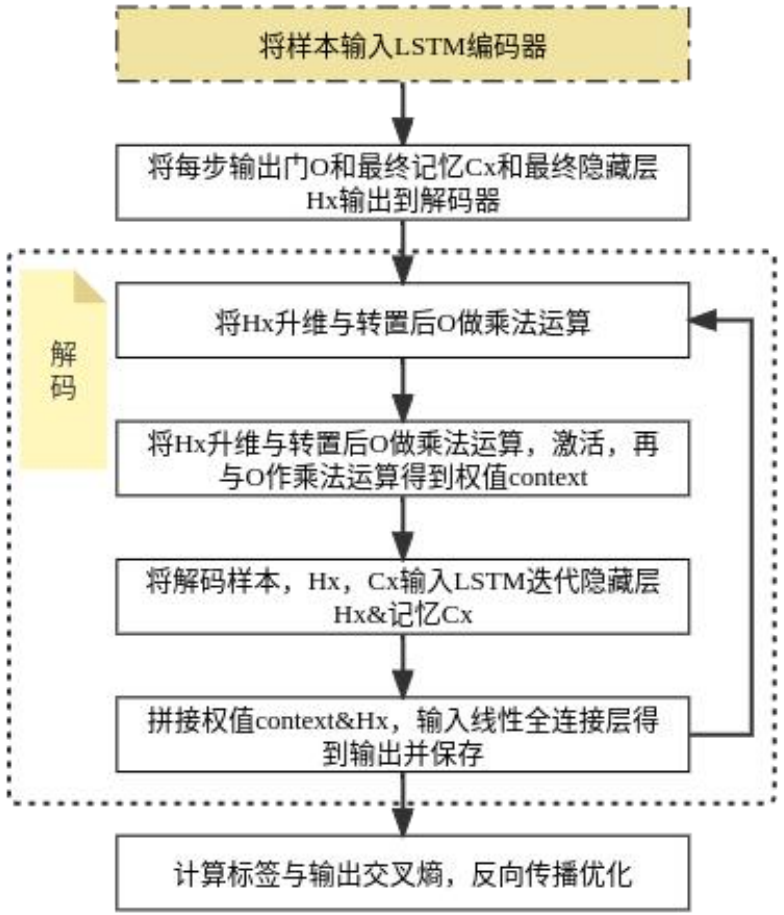




LSTM


input
decoder-hx
32×32
Gemm
B 〈27×32〉
C 〈27〉
32×27
3
one-hot


y1
y2
y3
H0
C0
LSTM-en
H1
C1
LSTM-en
H2
C2
LSTM-en
H3
C3
LSTM-de
Hx1
Cx1
LSTM-de
Hx2
Cx2
LSTM-de
Hx3
Cx3
x1
x2
x3
y0
y1
y2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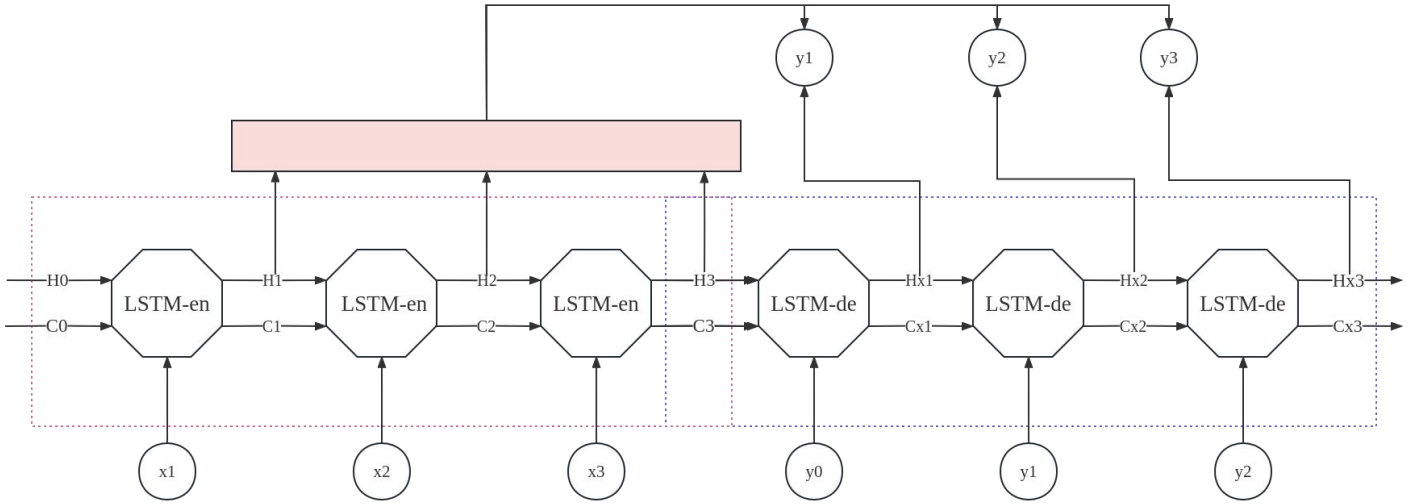
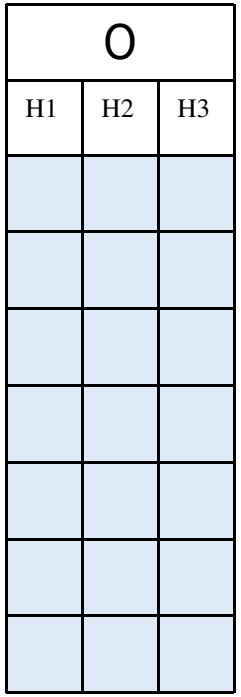
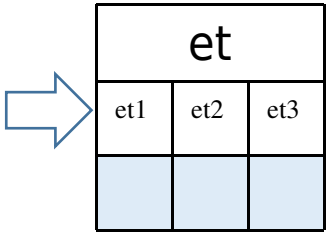

y1
y2
y3
H0
C0
LSTM-en
H1
C1
LSTM-en
H2
C2
LSTM-en
H3
C3
LSTM-de
Hx1
Cx1
LSTM-de
Hx2
Cx2
LSTM-de
Hx3
Cx3
x1
x2
x3
y0
y1
y2

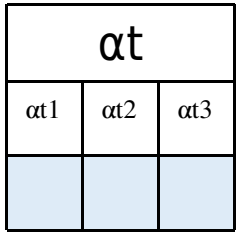
α

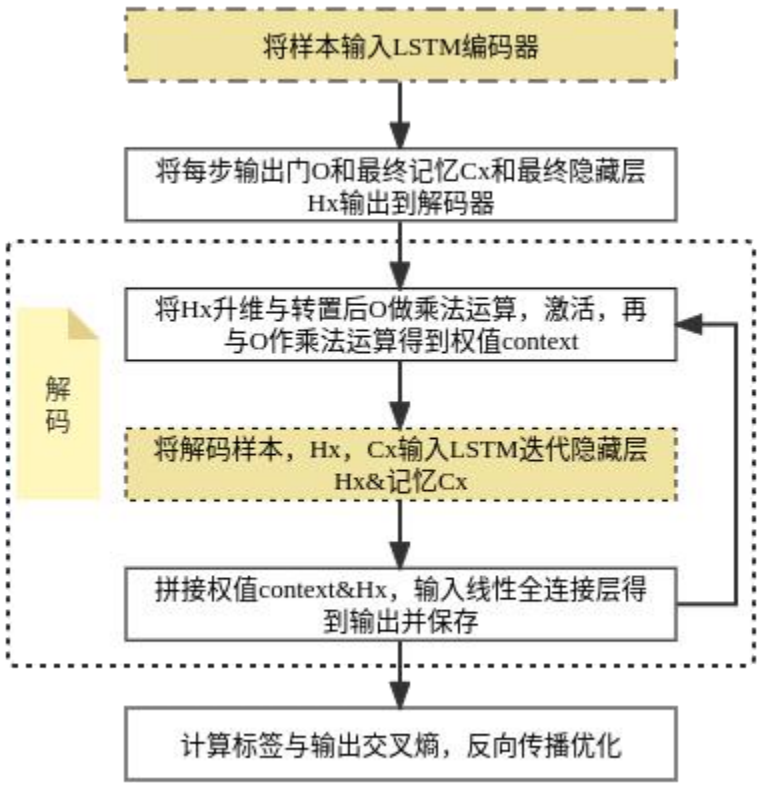

O
H1
H2
H3


Hxi


et
et1
et2
et3


softmax


αt
αt1
αt2
αt3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拼接权值context&Hx，输入线性全连接层得到输出并保存
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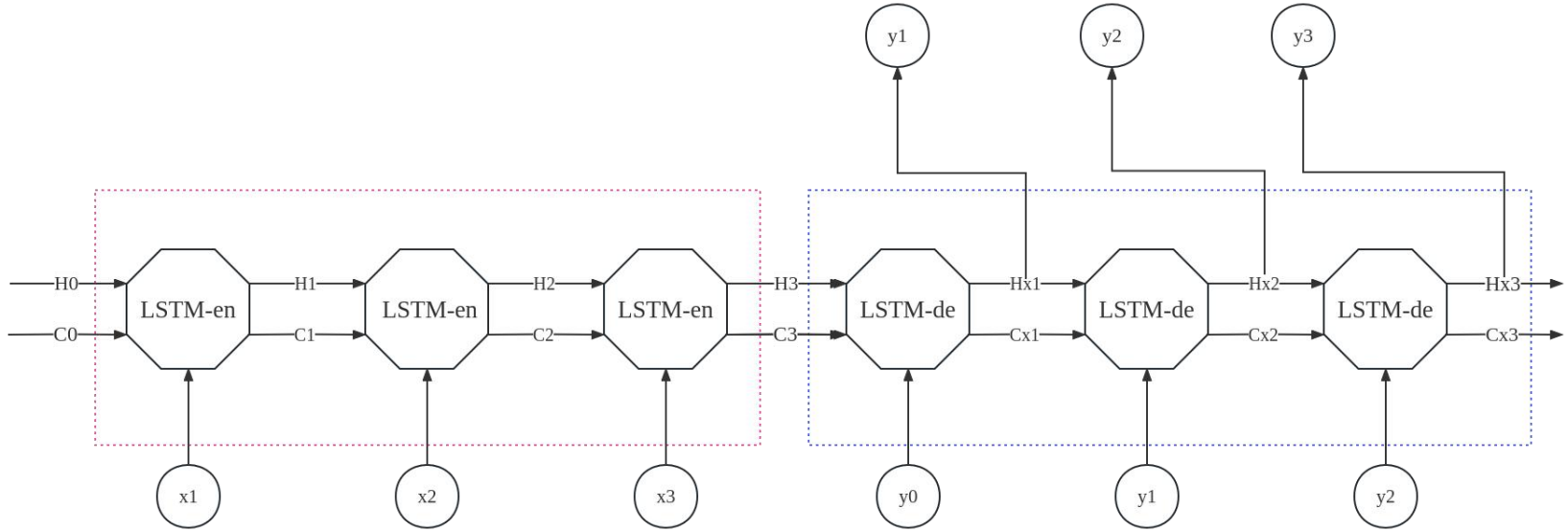
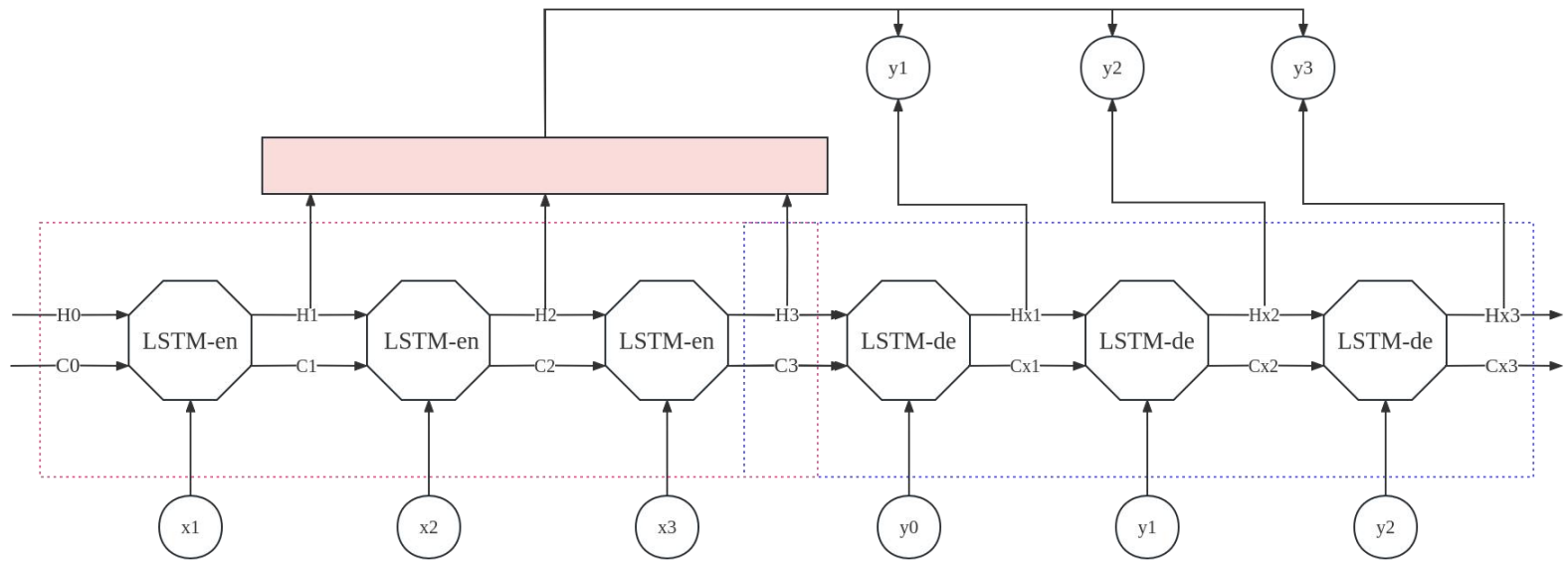
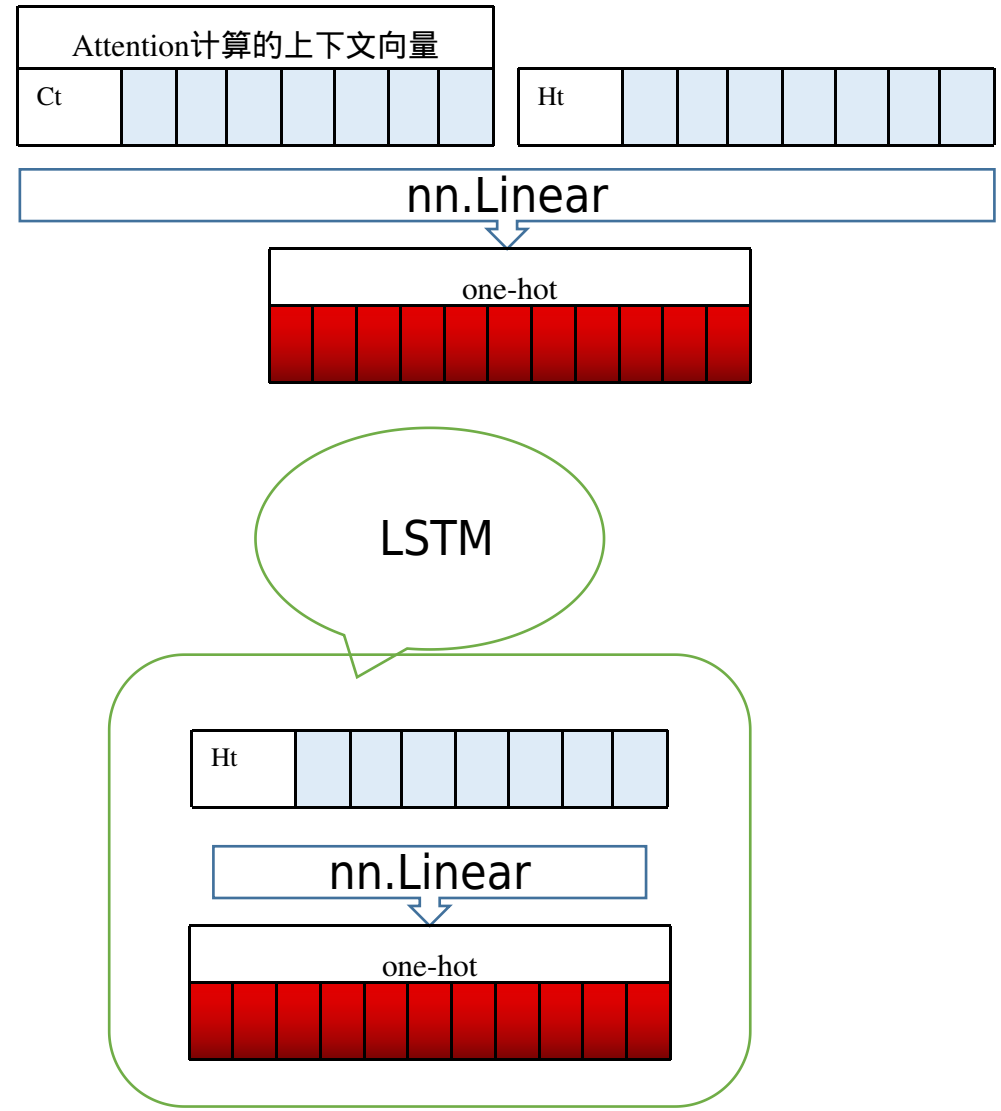

y1
y2
y3
H0
C0
LSTM-en
H1
C1
LSTM-en
H2
C2
LSTM-en
H3
C3
LSTM-de
Hx1
Cx1
LSTM-de
Hx2
Cx2
LSTM-de
Hx3
Cx3
x1
x2
x3
y0
y1
y2


y1
y2
y3
H0
C0
LSTM-en
H1
C1
LSTM-en
H2
C2
LSTM-en
H3
C3
LSTM-de
Hx1
Cx1
LSTM-de
Hx2
Cx2
LSTM-de
Hx3
Cx3
x1
x2
x3
y0
y1
y2



这里的attention本质上讲是对解码器LSTM隐藏层进行的一个修正

$$e_t = [a(h'_{t-1}, h_1), a(h'_{t-1}, h_2), \ldots, a(h'_{t-1}, h_N)]$$

函数a用于计算$Decoder\ h'_{t-1}$与$Encoder$第i个神经元的相关性

$$a(h'_{t-1}, h_i) = h_i^T h'_{t-1}$$

$$a(h'_{t-1}, h_i) = h_i^T W h'_{t-1}$$

$$a(h'_{t-1}, h_i) = \tanh(W_1 h_i + W_2 h'_{t-1})$$

Attention 相关性

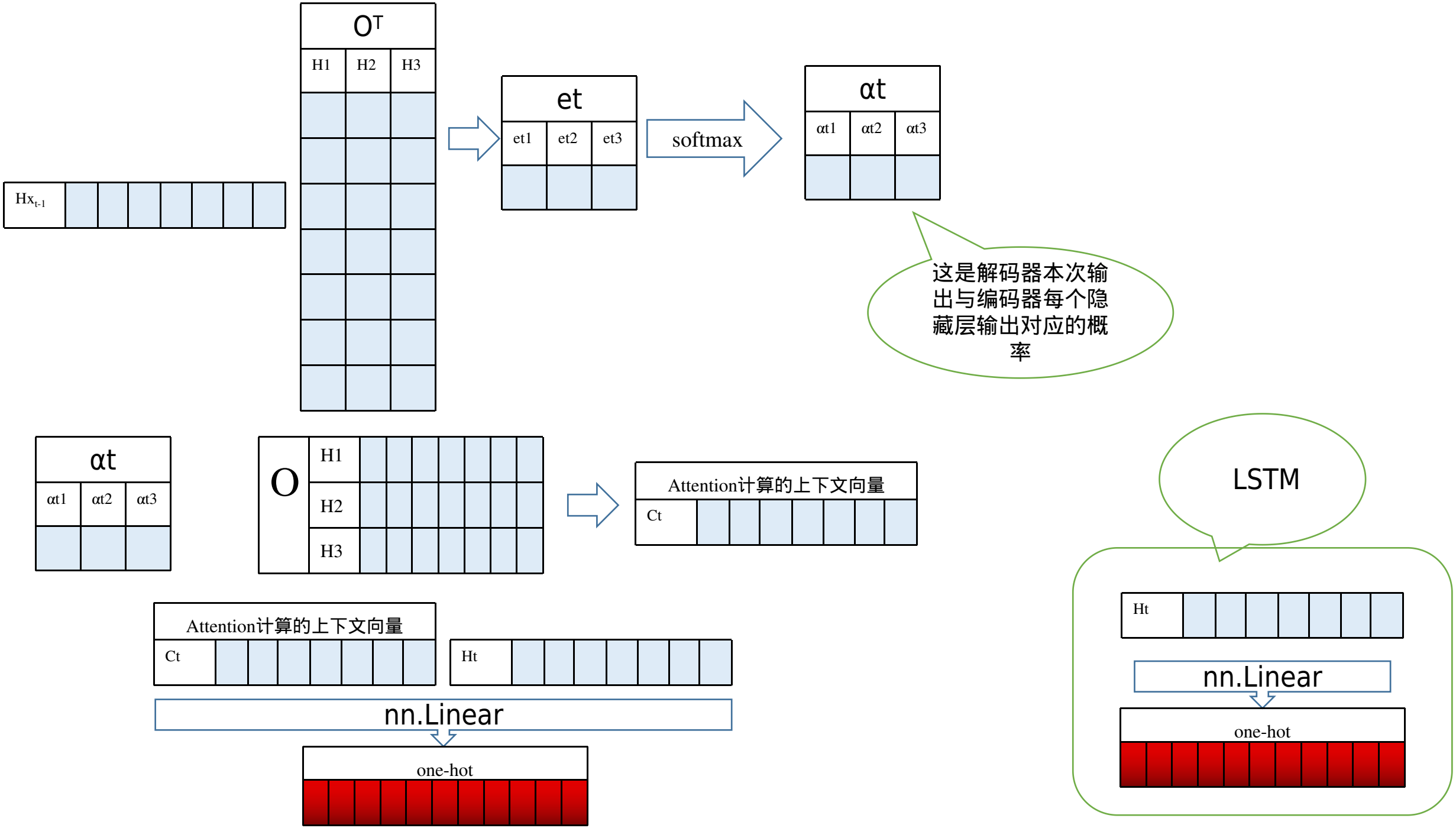
上面得到相关性向量 et 后，需要进行归一化，使用 softmax 归一化。然后用归一化后的系数融合 Encoder 的多个隐藏层向量得到 Decoder 当前神经元的上下文向量 ct：

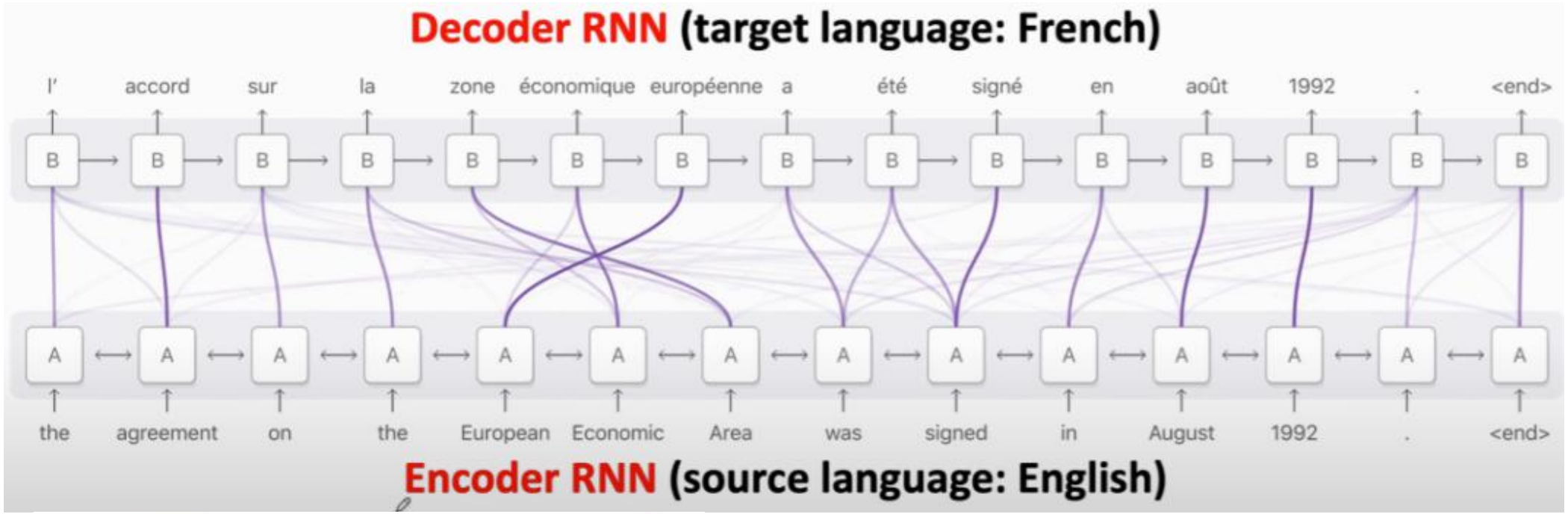
$$\alpha_t = softmax(e_t)$$

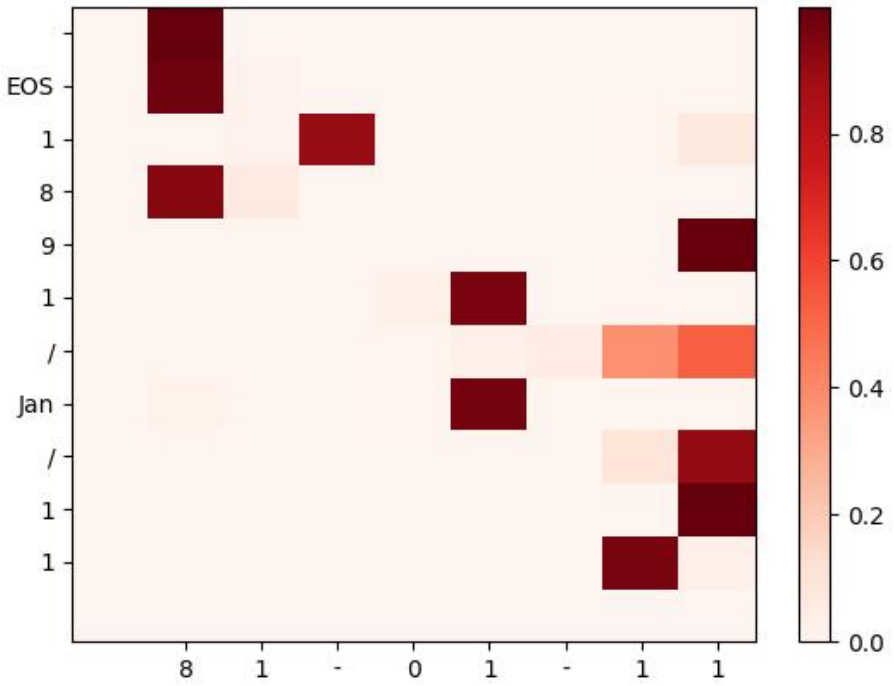
$$\alpha_{ti} = \frac{\exp(e_{ti})}{\sum_{j=1}^{N} \exp(e_{tj})}$$

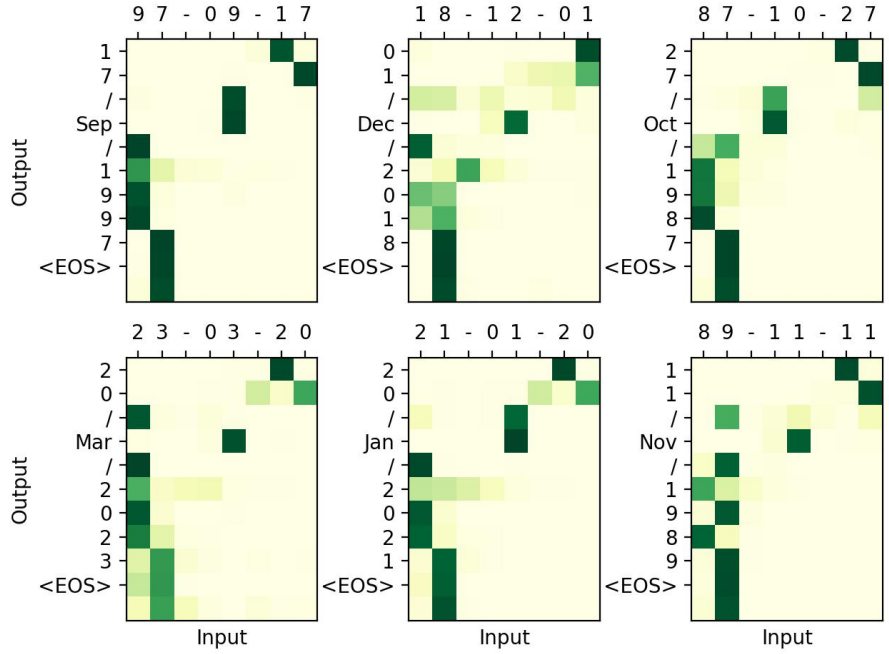
$$c_t = \sum_{i=1}^{N} \alpha_{ti} h_i$$

使用 Attention 计算上下文向量 c


O^T
H1
H2
H3
Hx_{t-1}
et
et1
et2
et3
softmax
αt
αt1
αt2
αt3
这是解码器本次输出与编码器每个隐藏层输出对应的概率
αt
αt1
αt2
αt3
O
H1
H2
H3
Attention计算的上下文向量
Ct
Attention计算的上下文向量
Ct
Ht
nn.Linear
one-hot
LSTM
Ht
nn.Linear
one-hot

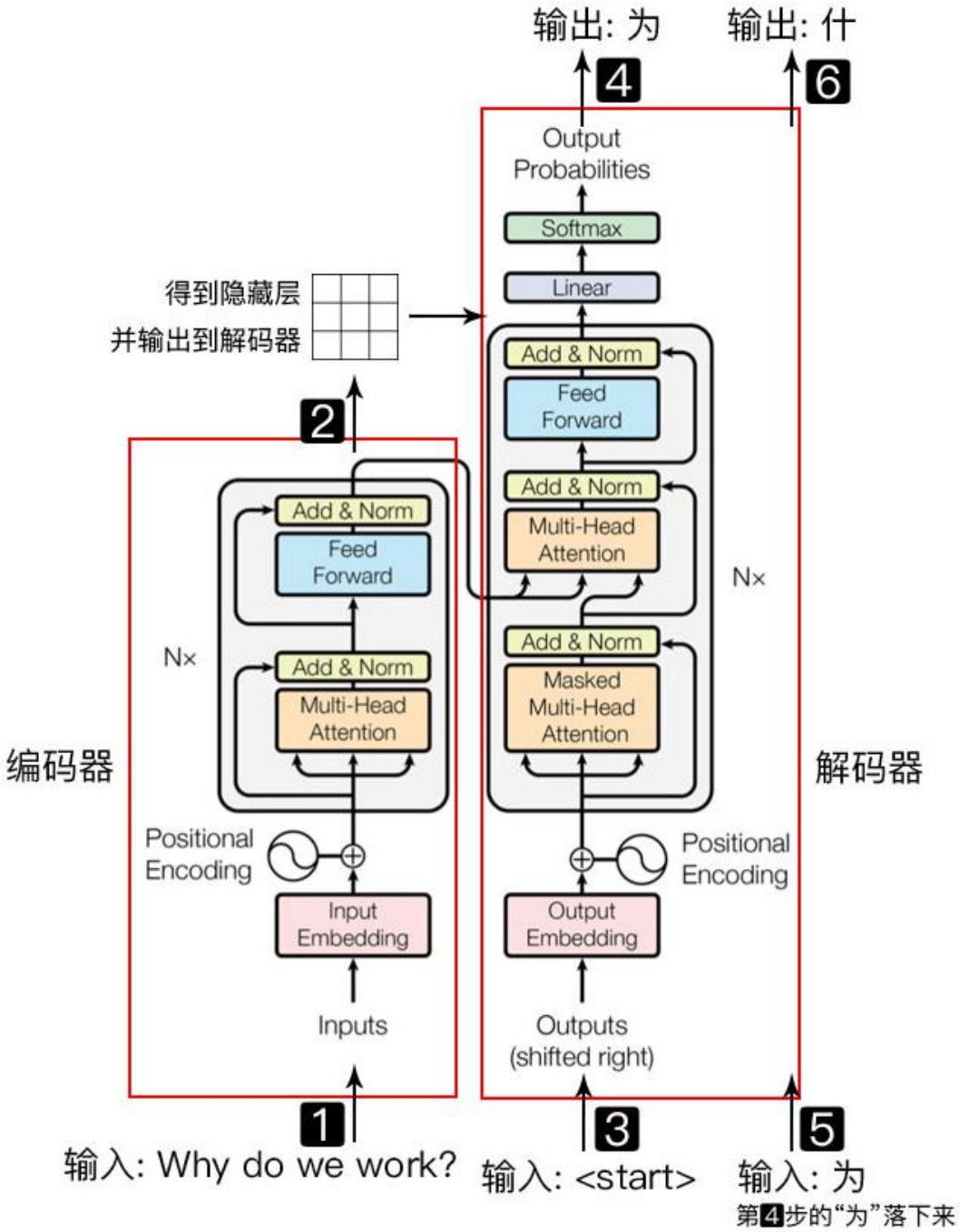

Decoder RNN (target language: French)
l' accord sur la zone économique européenne a été signé en août 1992 . <end>
B B B B B B B B B B B B B B B B
A A A A A A A A A A A A A A A A
the agreement on the European Economic Area was signed in August 1992 . <end>
Encoder RNN (source language: English)


EOS
1
8
9
1
/
Jan
/
1
1
8 1 - 0 1 - 1 1
0.8
0.6
0.4
0.2
0.0


9 7 - 0 9 - 1 7
1 8 - 1 2 - 0 1
8 7 - 1 0 - 2 7
2 3 - 0 3 - 2 0
2 1 - 0 1 - 2 0
8 9 - 1 1 - 1 1
Output
Input
<EOS>
Sep
Dec
Oct
Mar
Jan
Nov

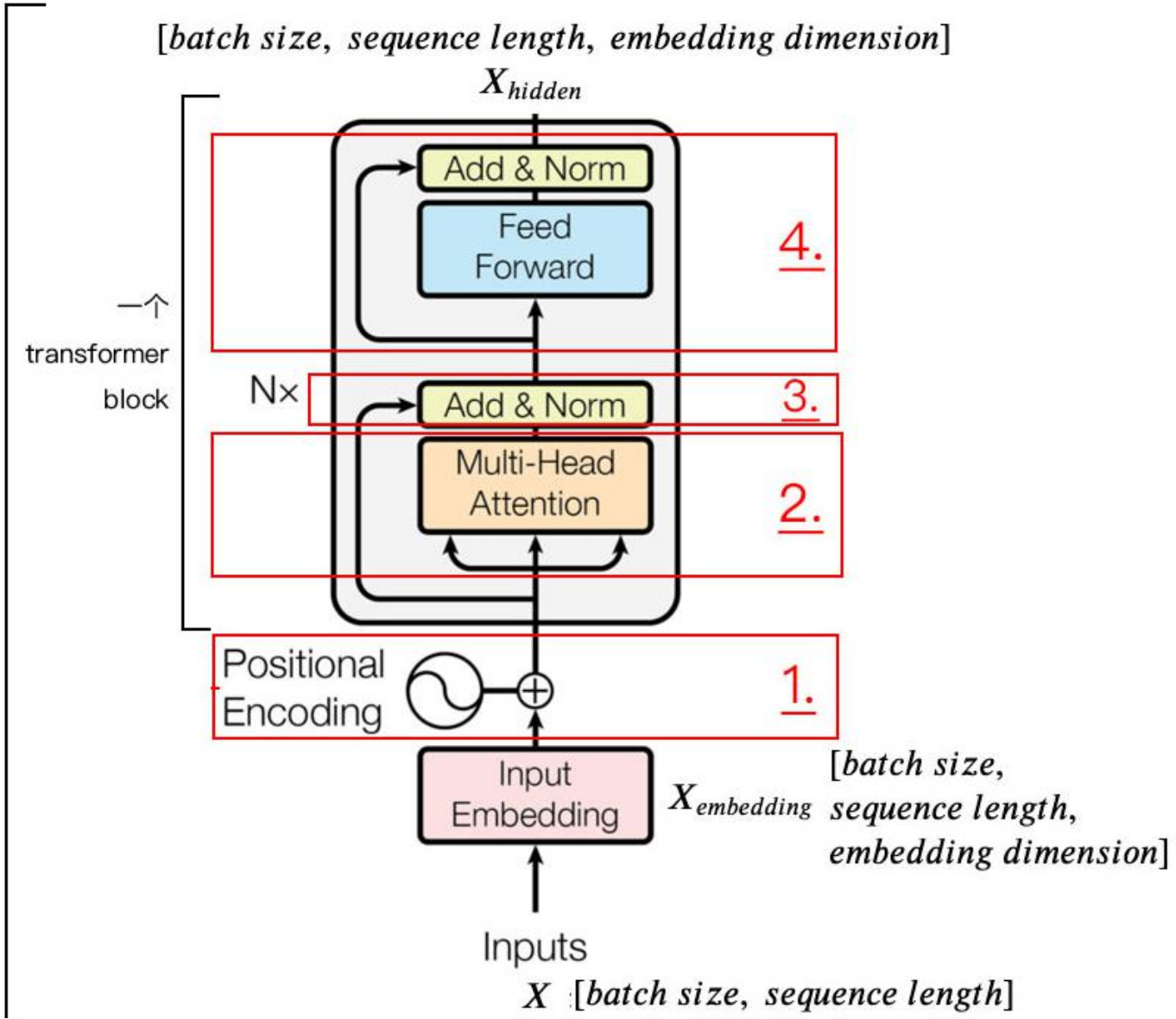
Transformer

transformer整体
=encoder+decoder



encoder

从自然语言序列
经过计算
到隐藏层的过程



```
model Transformer(
  (encoder): Encoder(
    (src_emb): Embedding(6, 512)
    (pos_emb): PositionalEncoding(
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (layers): ModuleList(
      (0): EncoderLayer(
        (enc_self_attn): MultiHeadAttention(
          (W_Q): Linear(in_features=512, out_features=512, bias=False)
          (W_K): Linear(in_features=512, out_features=512, bias=False)
          (W_V): Linear(in_features=512, out_features=512, bias=False)
          (fc): Linear(in_features=512, out_features=512, bias=False)
        )
        (pos_ffn): PoswiseFeedForwardNet(
          (fc): Sequential(
            (0): Linear(in_features=512, out_features=2048, bias=False)
            (1): ReLU()
            (2): Linear(in_features=2048, out_features=512, bias=False)
          )
        )
      )
```

```
  (decoder): Decoder(
    (tgt_emb): Embedding(9, 512)
    (pos_emb): PositionalEncoding(
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (layers): ModuleList(
      (0): DecoderLayer(
        (dec_self_attn): MultiHeadAttention(
          (W_Q): Linear(in_features=512, out_features=512, bias=False)
          (W_K): Linear(in_features=512, out_features=512, bias=False)
          (W_V): Linear(in_features=512, out_features=512, bias=False)
          (fc): Linear(in_features=512, out_features=512, bias=False)
        )
        (dec_enc_attn): MultiHeadAttention(
          (W_Q): Linear(in_features=512, out_features=512, bias=False)
          (W_K): Linear(in_features=512, out_features=512, bias=False)
          (W_V): Linear(in_features=512, out_features=512, bias=False)
          (fc): Linear(in_features=512, out_features=512, bias=False)
        )
        (pos_ffn): PoswiseFeedForwardNet(
          (fc): Sequential(
            (0): Linear(in_features=512, out_features=2048, bias=False)
            (1): ReLU()
            (2): Linear(in_features=2048, out_features=512, bias=False)
          )
        )
      )
```

```
    (projection): Linear(in_features=512, out_features=9, bias=False)
```

encoder

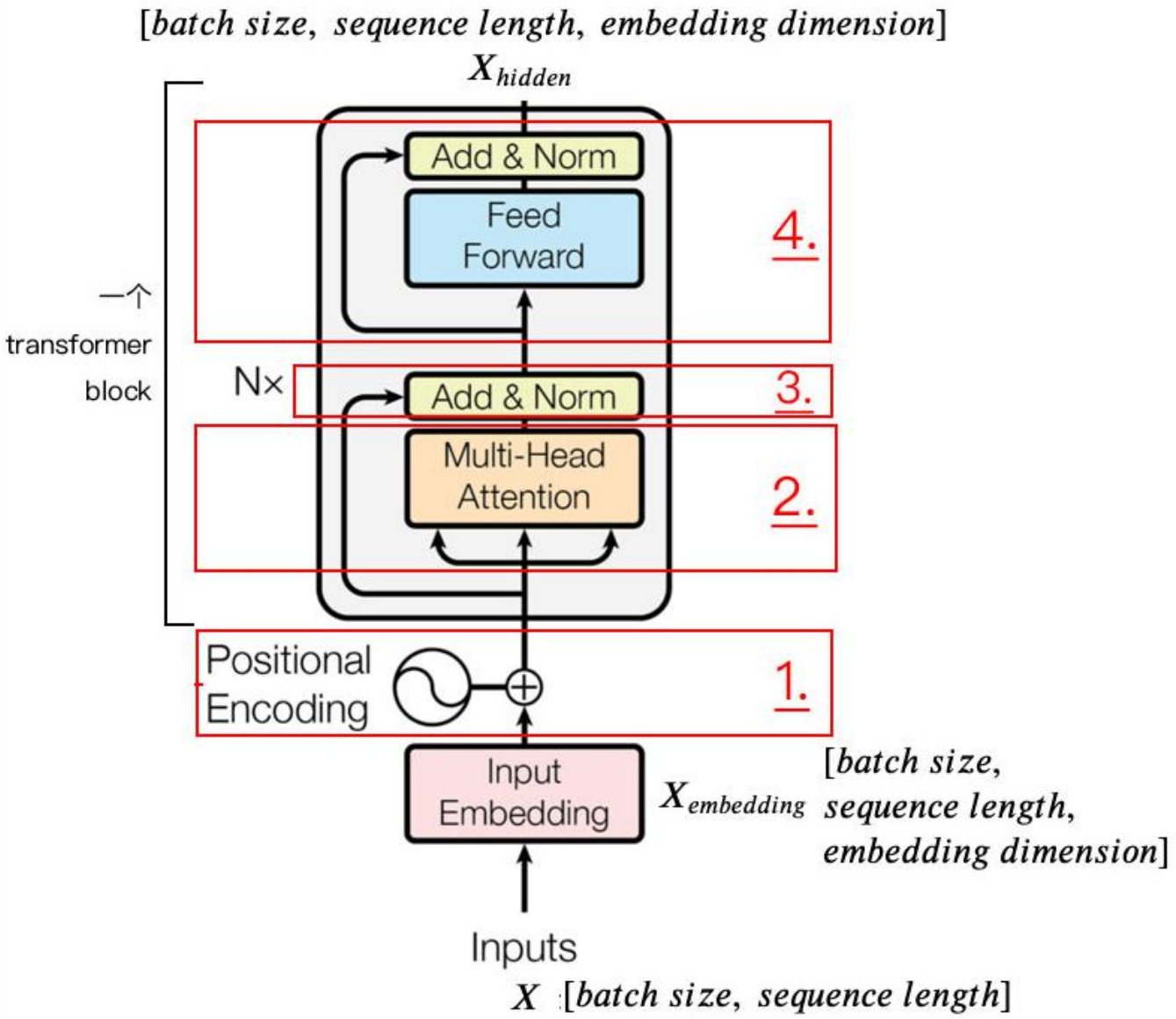
enc_input(编码样本)	dec_input (解码样本)	dec_outputdec_input (标签)
ich mochte ein bier P	S i want a beer .	i want a beer . E
ich mochte ein cola P	S i want a coke .	i want a coke . E
5	6	6

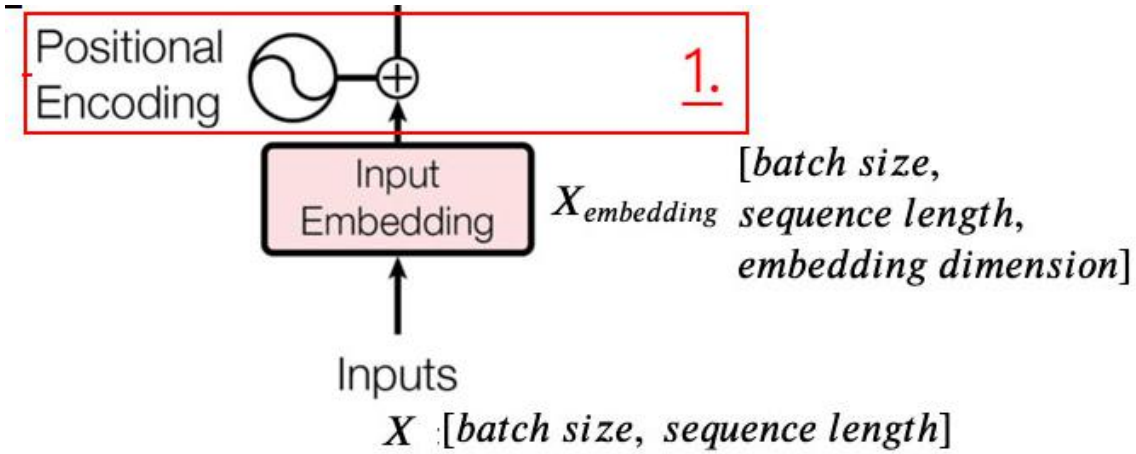
Src_vocab	
0	P
1	ich
2	mochte
3	ein
4	bier
5	cola

tgt_vocab	
0	P
1	i
2	want
3	a
4	beer
5	coke
6	S
7	E
8	.

encoder

从自然语言序列
经过计算
到隐藏层的过程





将位置嵌入和字嵌入相加，然后作为输入

$$\overrightarrow{p_t}^{(i)} = f(t)^{(i)} := \begin{cases} \sin(\omega_k \cdot t), & \text{if } i = 2k \\ \cos(\omega_k \cdot t), & \text{if } i = 2k+1 \end{cases}$$

$$\omega_k = \frac{1}{10000^{2k/d}}$$

t：句中该词序号
d：词向量维度（512）
i：嵌入的维度序号
nn.Dropout：结果放大1/（1-p）即10/9倍

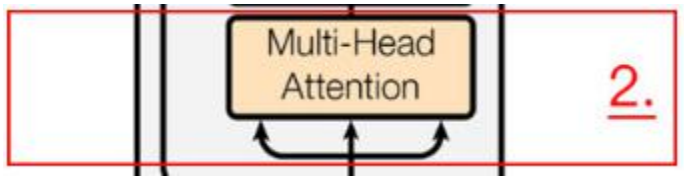
batch1	1	2	3	4	0
batch2	1	2	3	5	0

Word Embedding(1-->512)

batch1	1	2	3	4	0
batch2	1	2	3	5	0

+=Positional Embedded

batch1	1	2	3	4	0
batch2	1	2	3	5	0



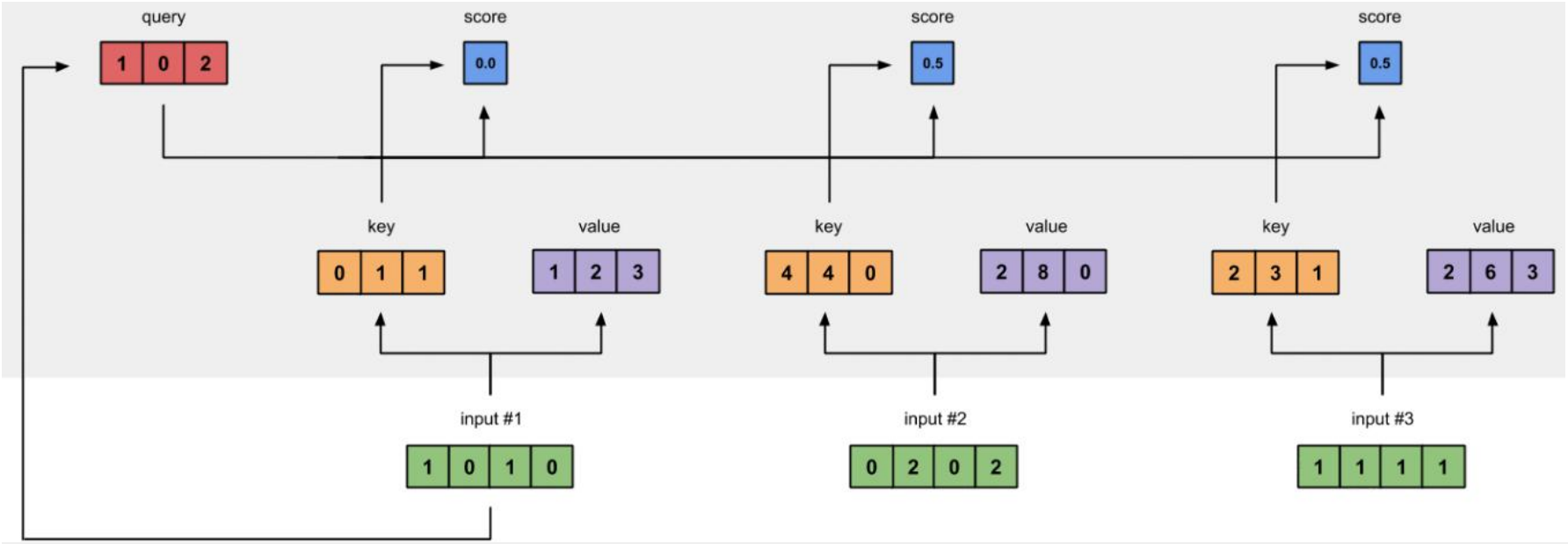
Query向量	当前元素的查询向量
Key向量	所有元素的关键字向量
Value向量	表示所有元素的值向量

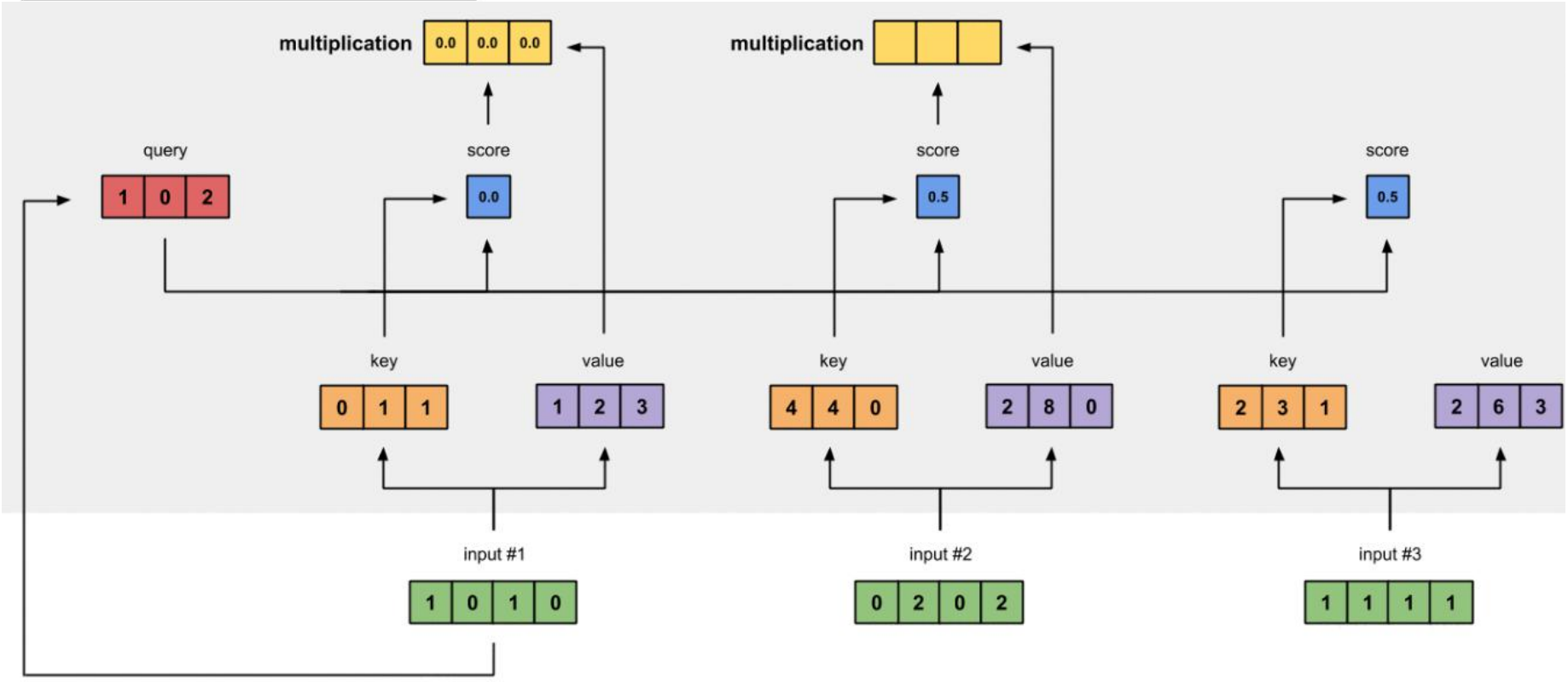
单个词向量分别乘以矩阵Wq，Wk，Wv

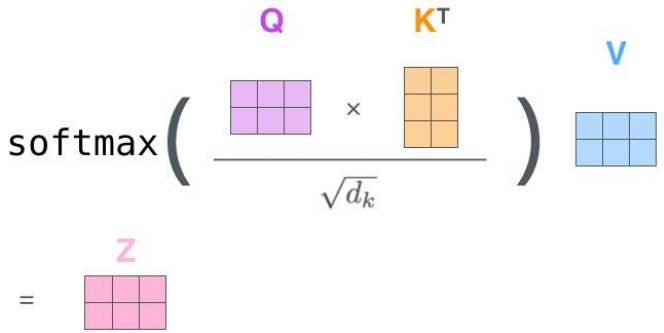
Word Embedding+Positional Embedded					
batch1	1	2	3	4	0
batch2	1	2	3	5	0

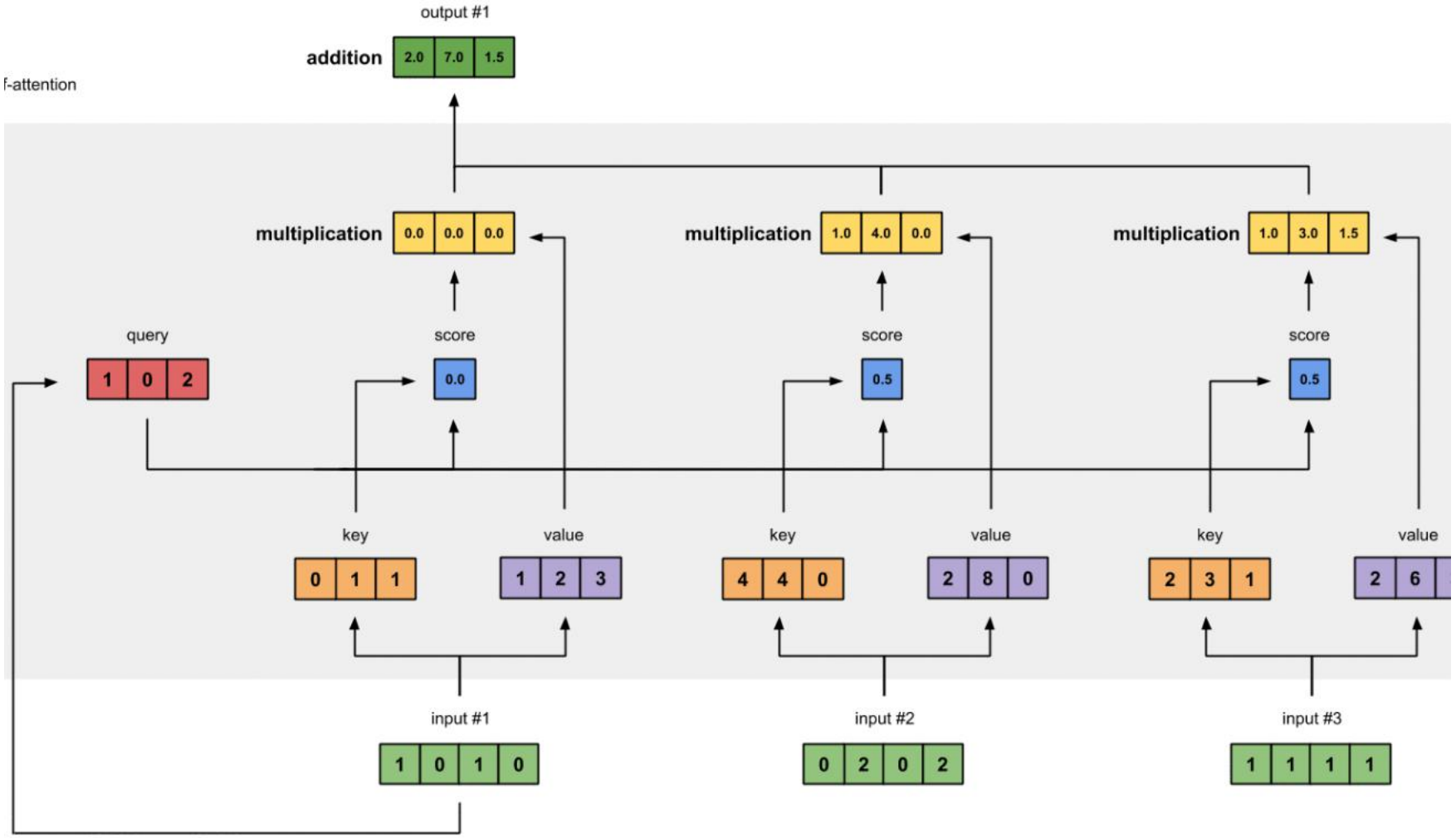
Wq	Wk	Wv
…	…	…
…	…	…

Word Embedding+Positional Embedded					
batch1	1	2	3	4	0
	q k v				
batch2	1	2	3	5	0
	q k v				


query
1 0 2
score
0.0
score
0.5
score
0.5
key
0 1 1
value
1 2 3
key
4 4 0
value
2 8 0
key
2 3 1
value
2 6 3
input #1
1 0 1 0
input #2
0 2 0 2
input #3
1 1 1 1

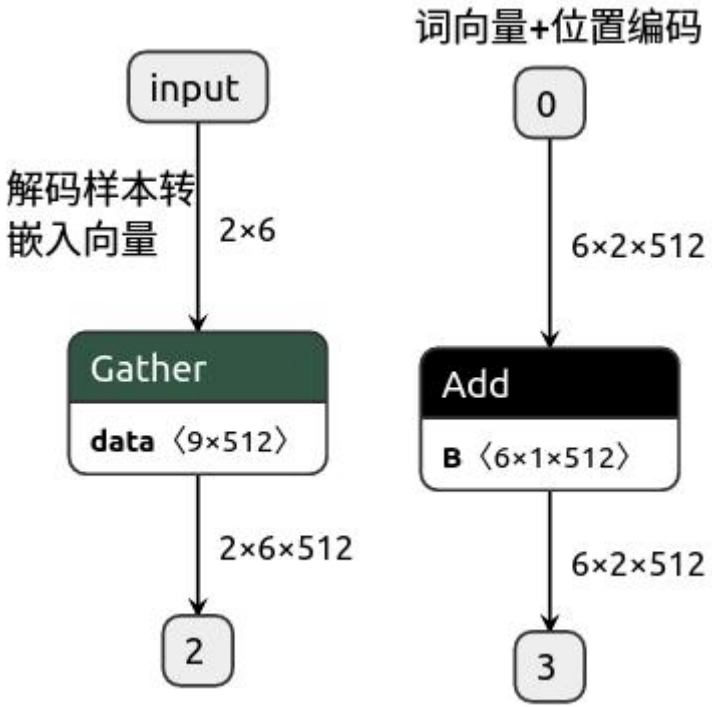

multiplication
0.0 0.0 0.0
multiplication
query
1 0 2
score
0.0
score
0.5
score
0.5
key
0 1 1
value
1 2 3
key
4 4 0
value
2 8 0
key
2 3 1
value
2 6 3
input #1
1 0 1 0
input #2
0 2 0 2
input #3
1 1 1 1

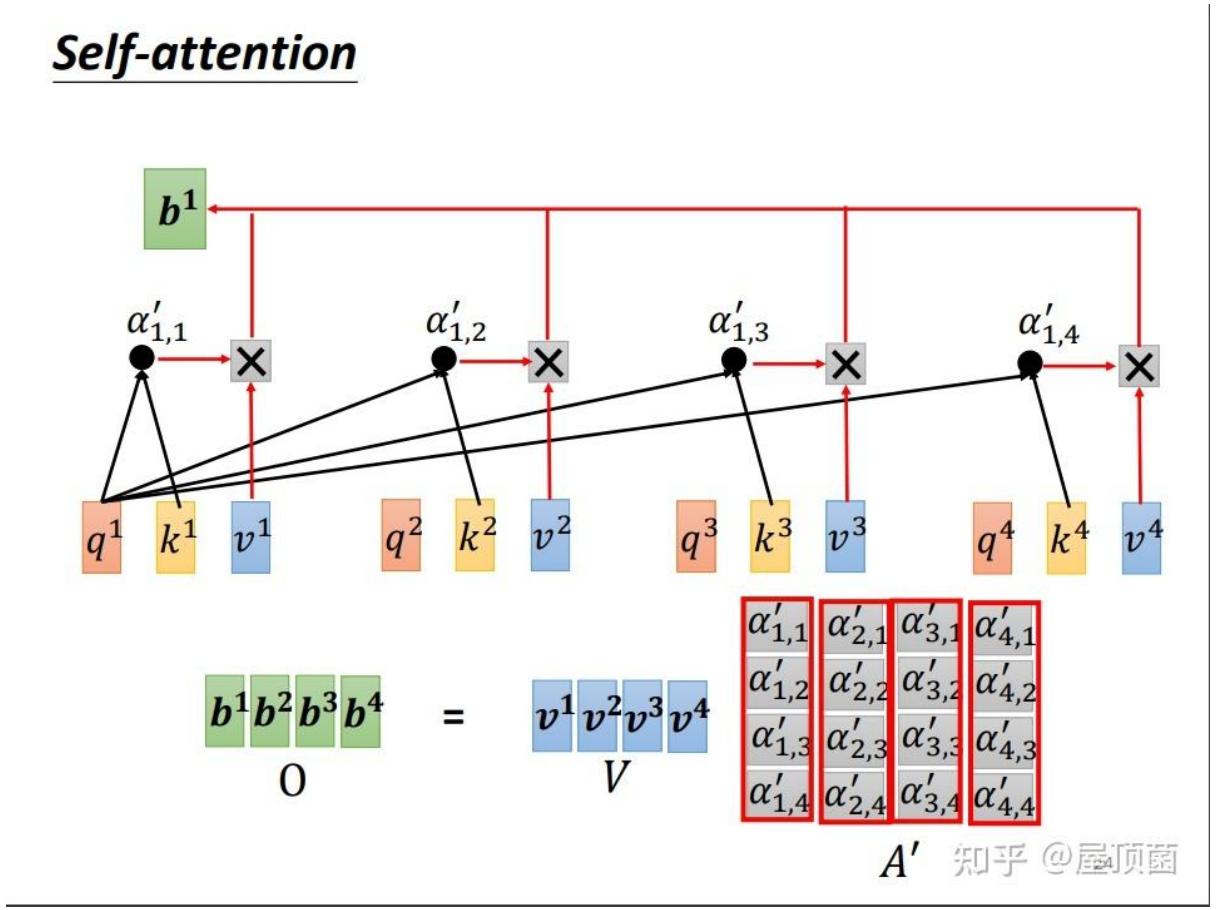

Q
K^T
V
softmax
√d_k
Z
=

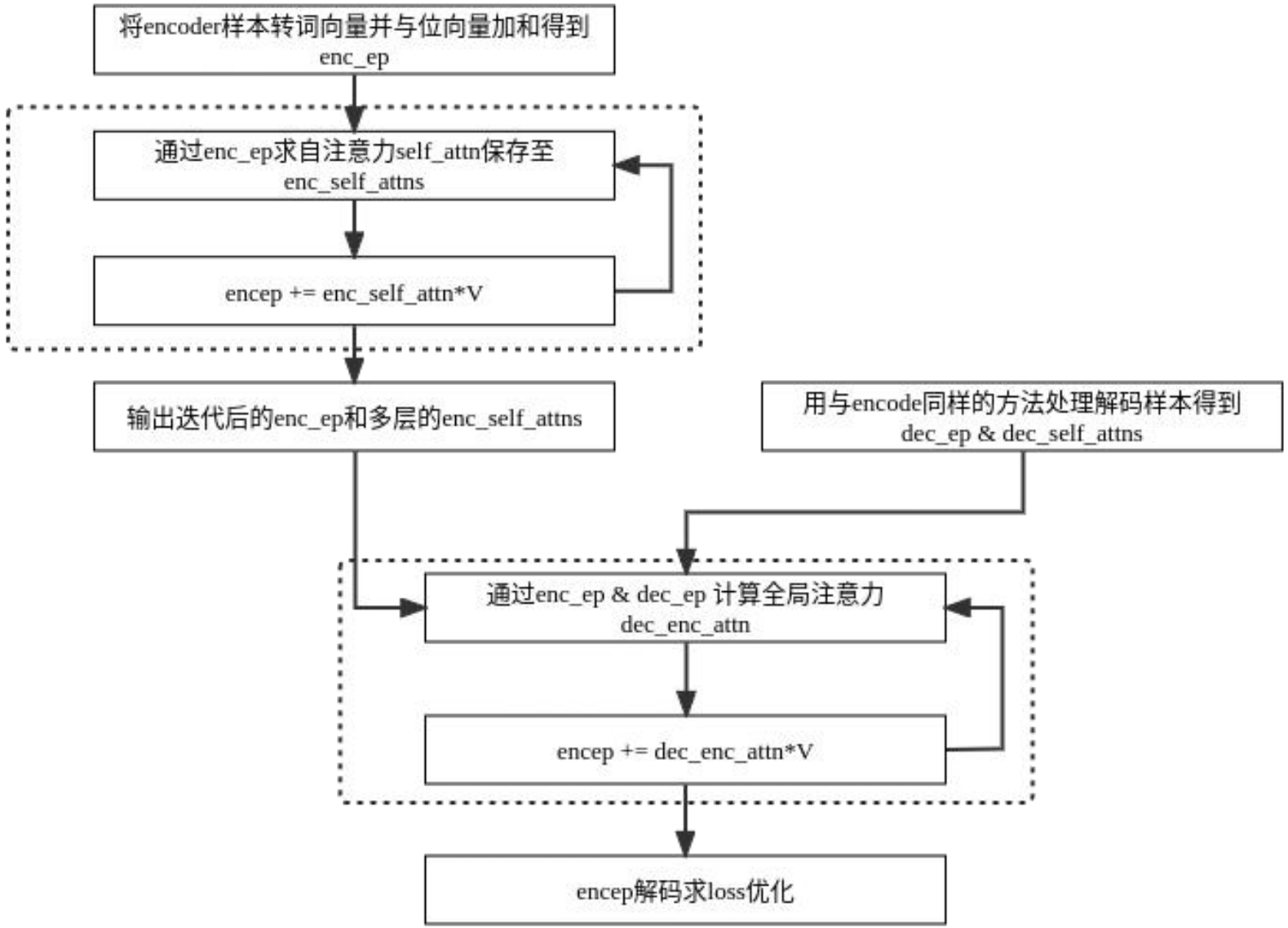


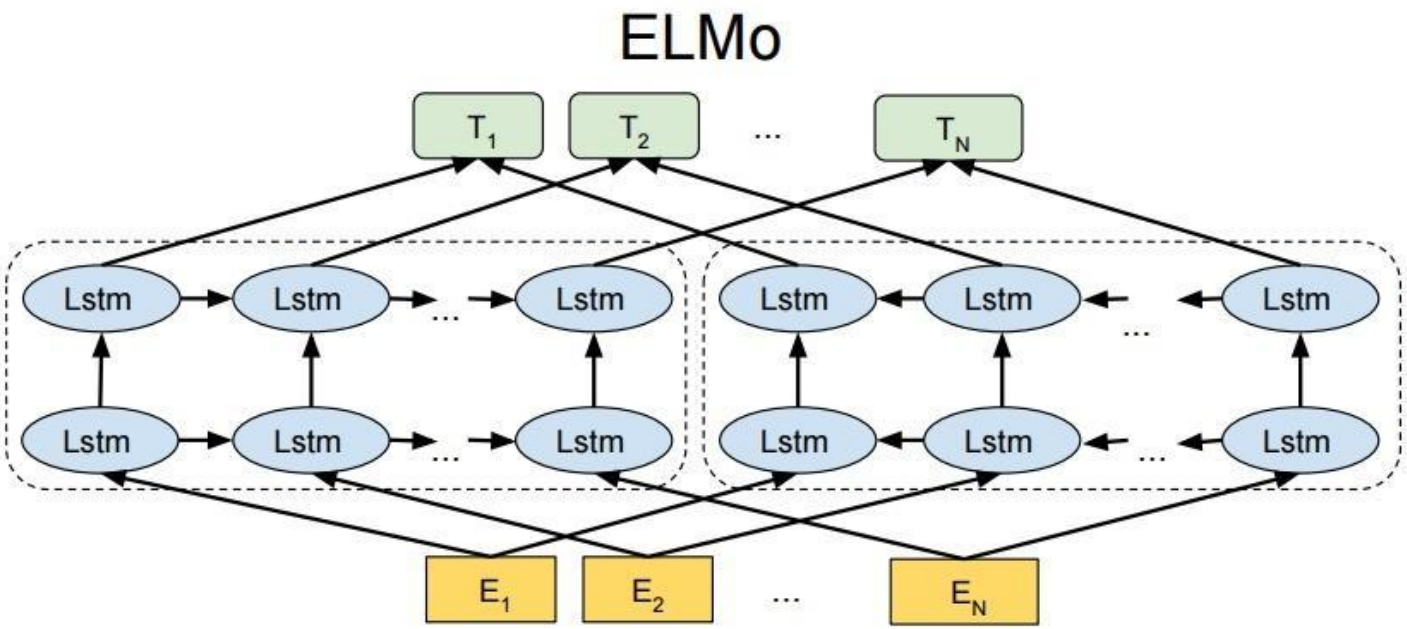
其它的输入向量也执行相同的操作，得到通过 self-attention 后的所有输出
输出结果：Value向量经过attn加权（multiplication） & 自注意力(score集合--互相之间的注意力)

```
# enc_self_attns        list 6 * tensor(2,8,5,5)      # 分层(6)注意力
# enc_outputs           tensor(2,5,512)               # 6层注意力加权结果
```


input
解码样本转
嵌入向量
2×6
Gather
data〈9×512〉
2×6×512
2
词向量+位置编码
0
6×2×512
Add
B〈6×1×512〉
6×2×512
3


Self-attention
b^1
$\alpha'_{1,1}$
$\alpha'_{1,2}$
$\alpha'_{1,3}$
$\alpha'_{1,4}$
q^1 k^1 v^1
q^2 k^2 v^2
q^3 k^3 v^3
q^4 k^4 v^4
$b^1 b^2 b^3 b^4$
O
=
$v^1 v^2 v^3 v^4$
V
A'
知乎 @屋顶菌


将encoder样本转词向量并与位向量加和得到
enc_ep
通过enc_ep求自注意力self_attn保存至
enc_self_attns
encep += enc_self_attn*V
输出迭代后的enc_ep和多层的enc_self_attns
用与encode同样的方法处理解码样本得到
dec_ep & dec_self_attns
通过enc_ep & dec_ep 计算全局注意力
dec_enc_attn
encep += dec_enc_attn*V
encep解码求loss优化



16样本38字符转embedded为
torch.Size([16, 38, 256])

↓

解码样本fxs：embeded前37
解码标签bxs：embeded后37

↓

初始化标签&样本的隐藏层&记忆
h_f,c_f.h_b,c_b

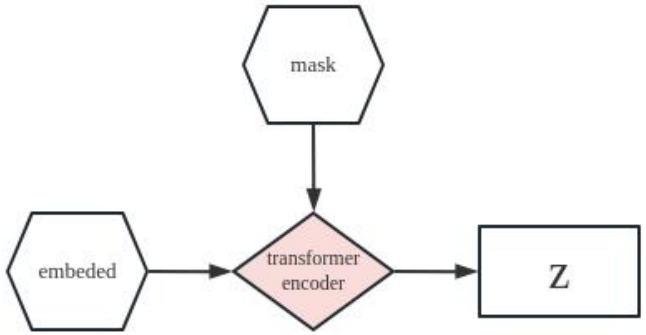
↓

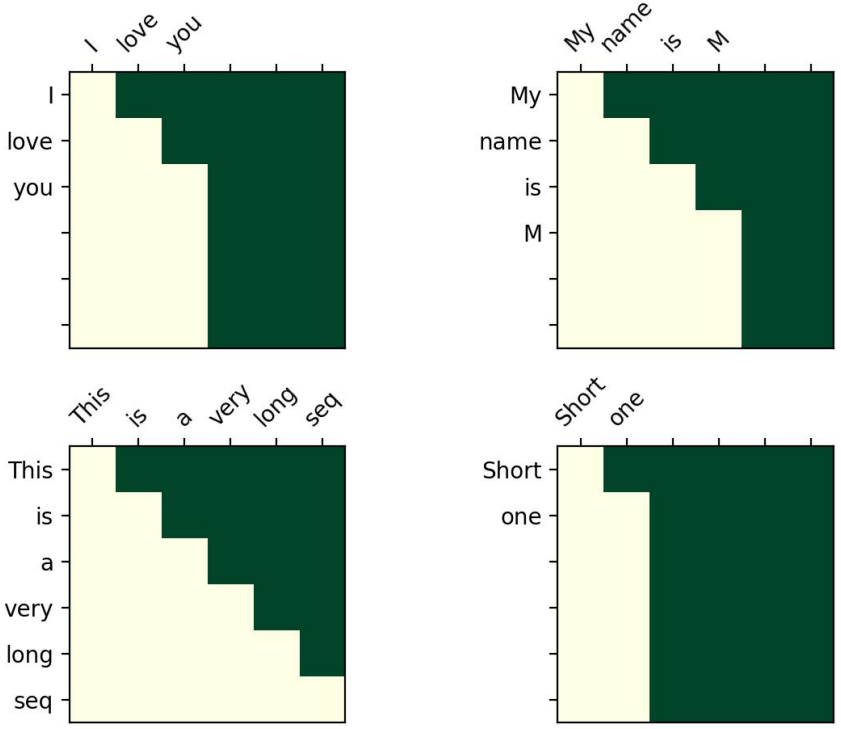
分别正向&反向输入样本到model

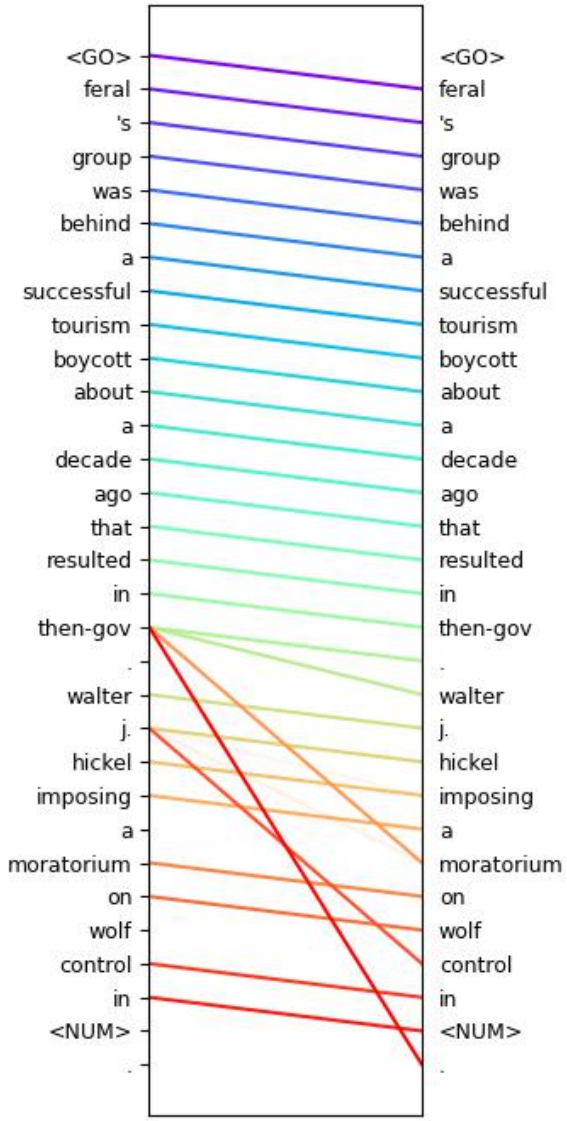
↓

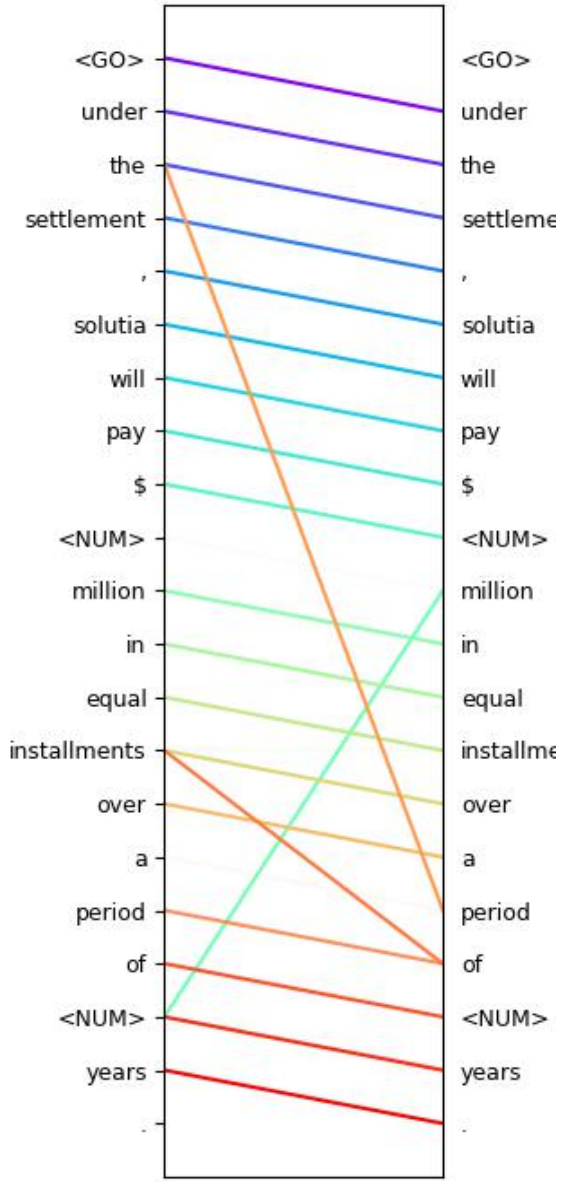
计算输出值与标签的交叉熵

2023.04.23



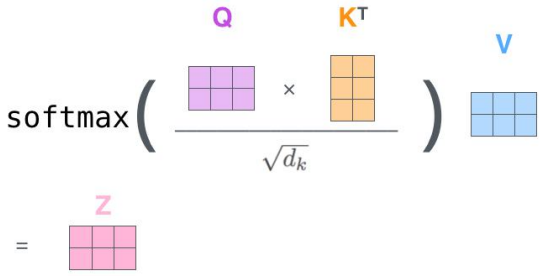


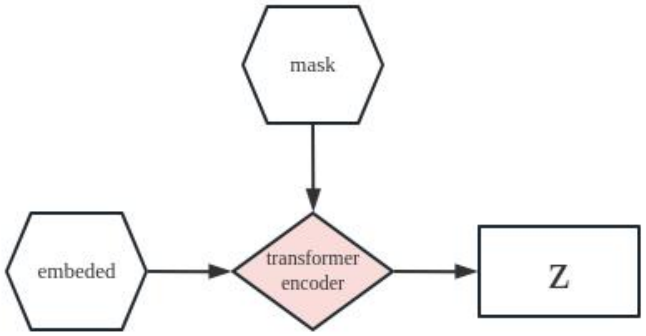


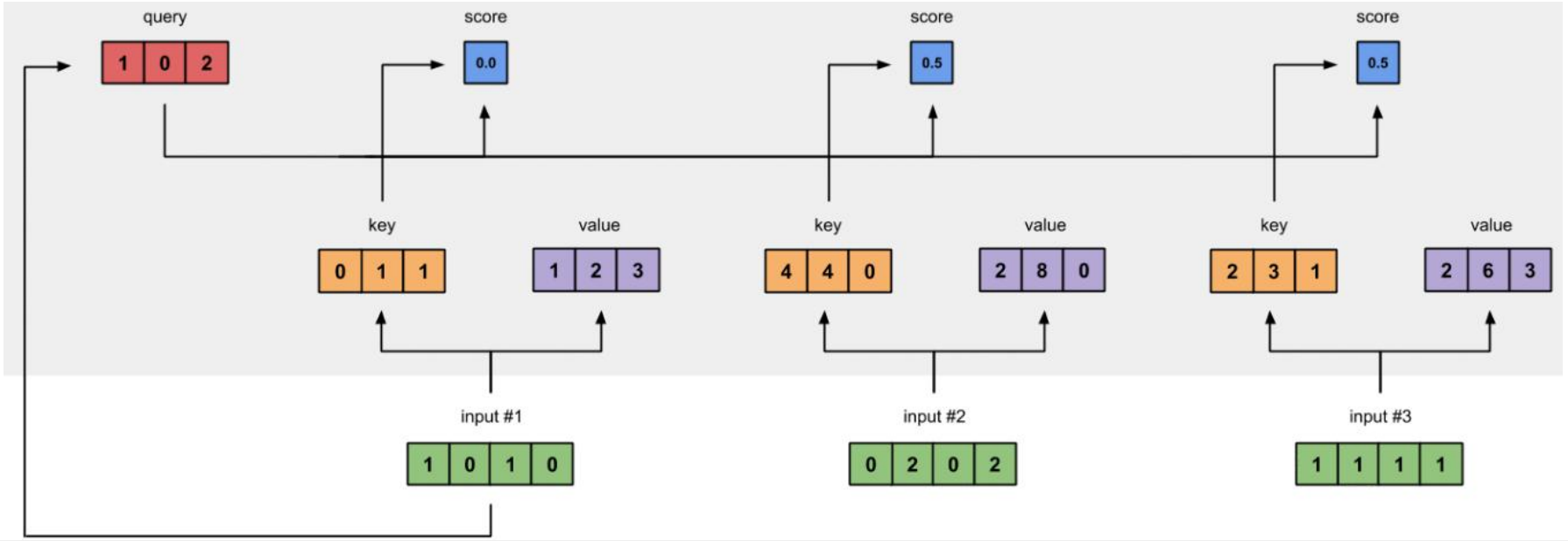


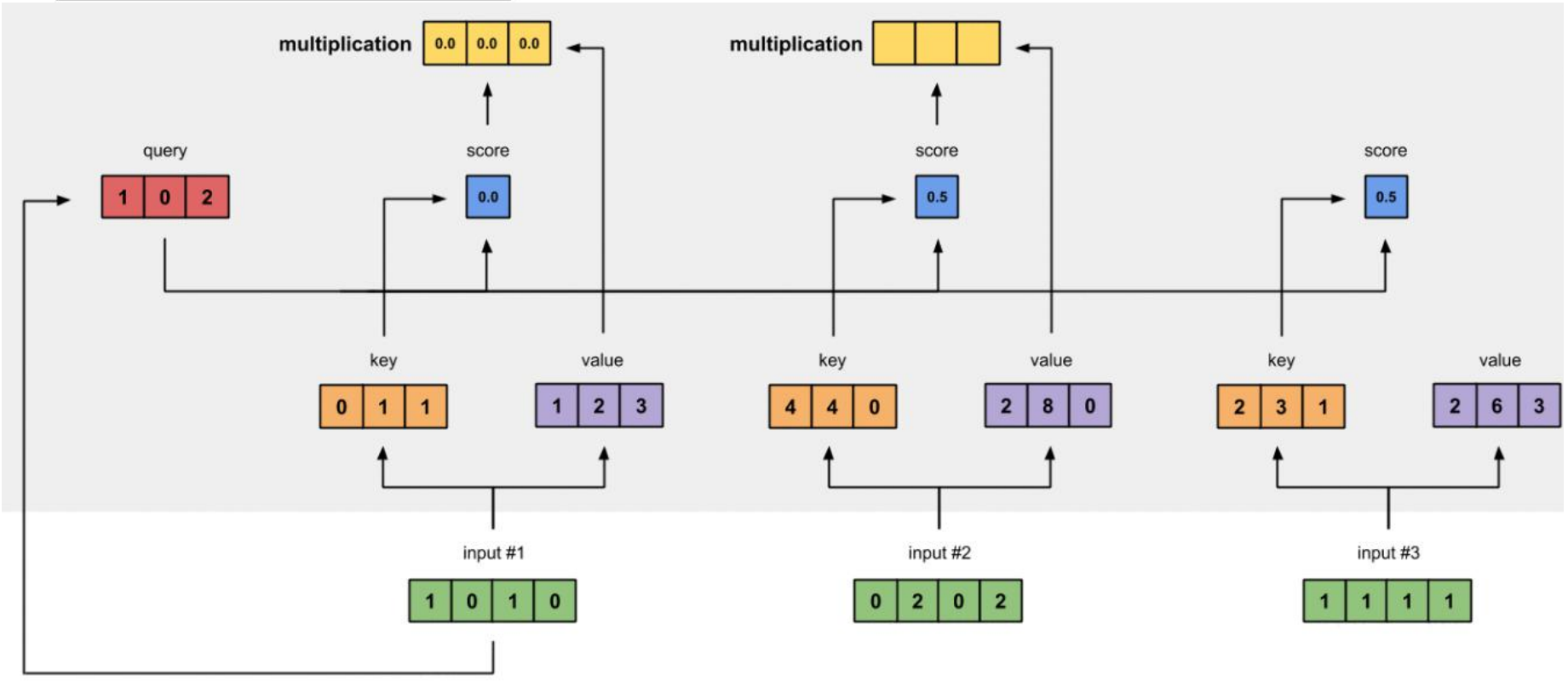
- 用前文的信息预测后文的信息，所以用上了Future Mask。mask将后文遮挡

encoder

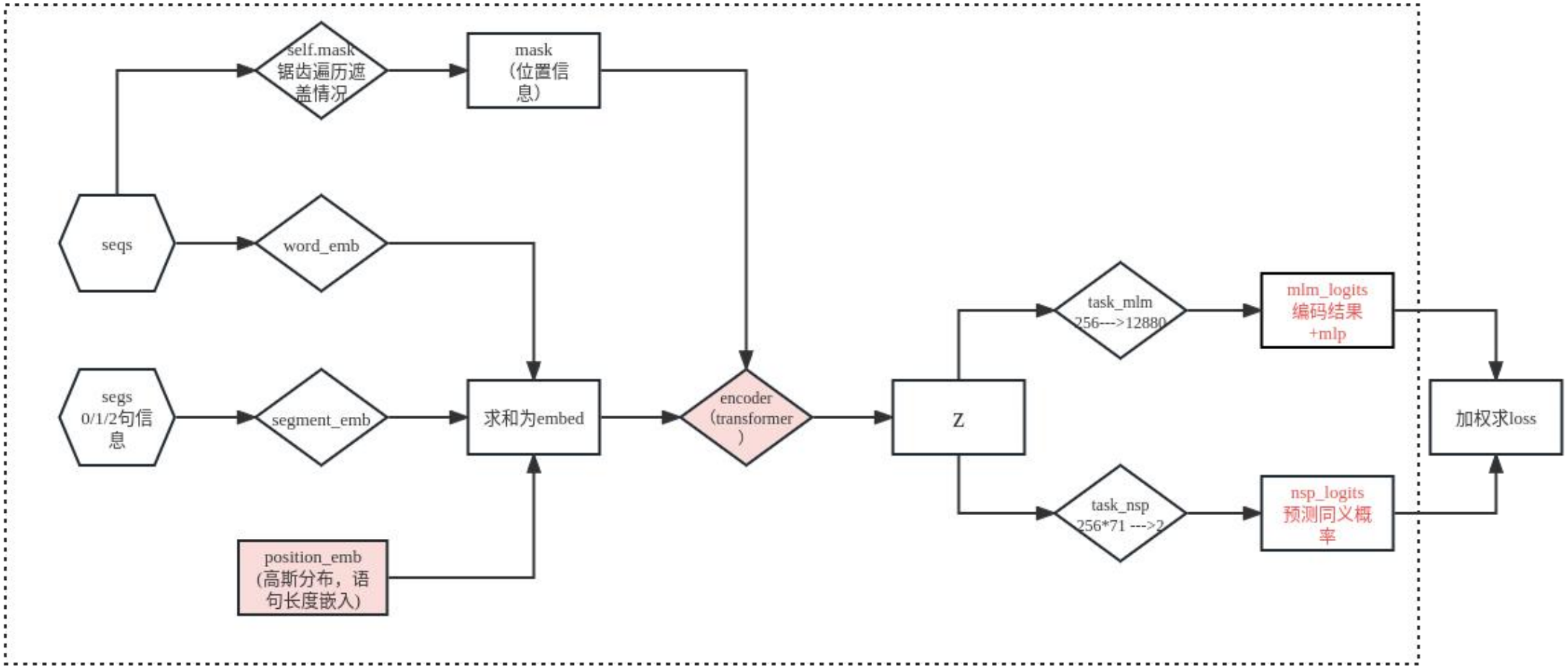

Q
K^T
V
softmax
$\sqrt{d_k}$
Z
=

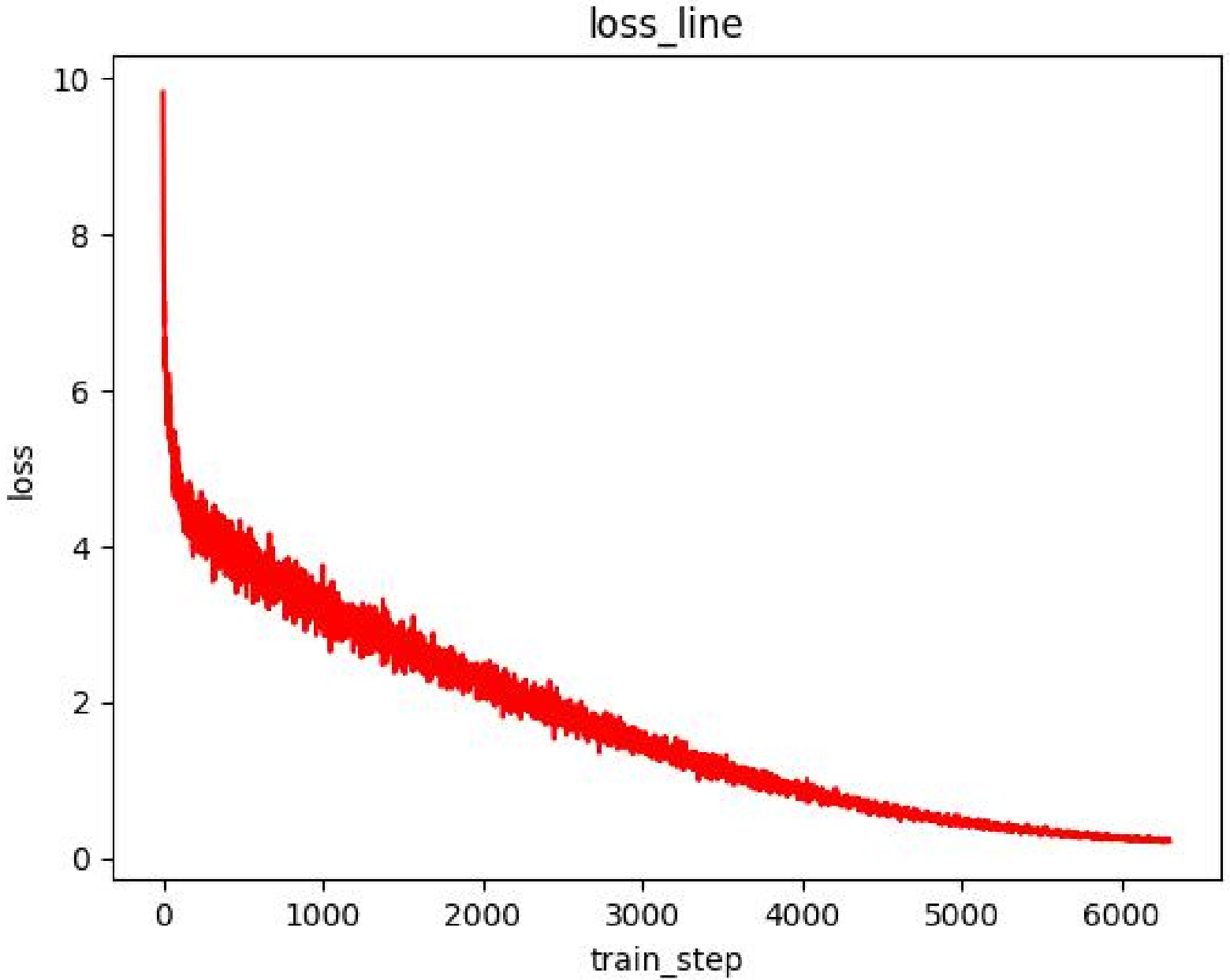

mask
embeded
transformer encoder
Z


query
score
score
score
1 0 2
0.0
0.5
0.5
key
value
key
value
key
value
0 1 1
1 2 3
4 4 0
2 8 0
2 3 1
2 6 3
input #1
input #2
input #3
1 0 1 0
0 2 0 2
1 1 1 1

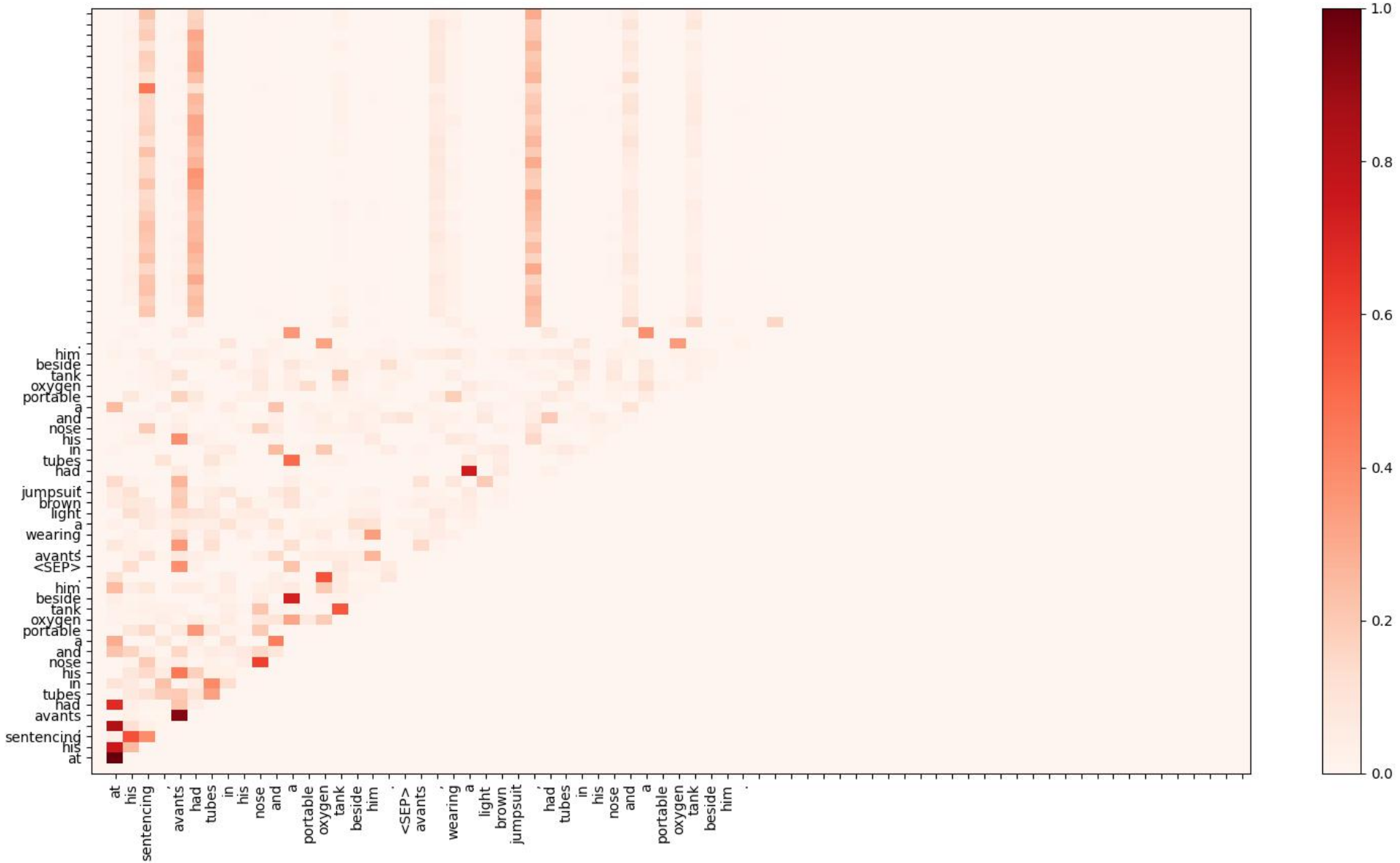

multiplication
multiplication
0.0 0.0 0.0
query
score
score
score
1 0 2
0.0
0.5
0.5
key
value
key
value
key
value
0 1 1
1 2 3
4 4 0
2 8 0
2 3 1
2 6 3
input #1
input #2
input #3
1 0 1 0
0 2 0 2
1 1 1 1

seqs	语句索引	
segs	前后句信息，0前句，1后句，2空符	
xlen	两句话长度	
nsp_labels	标签 array([0]) 0or1	

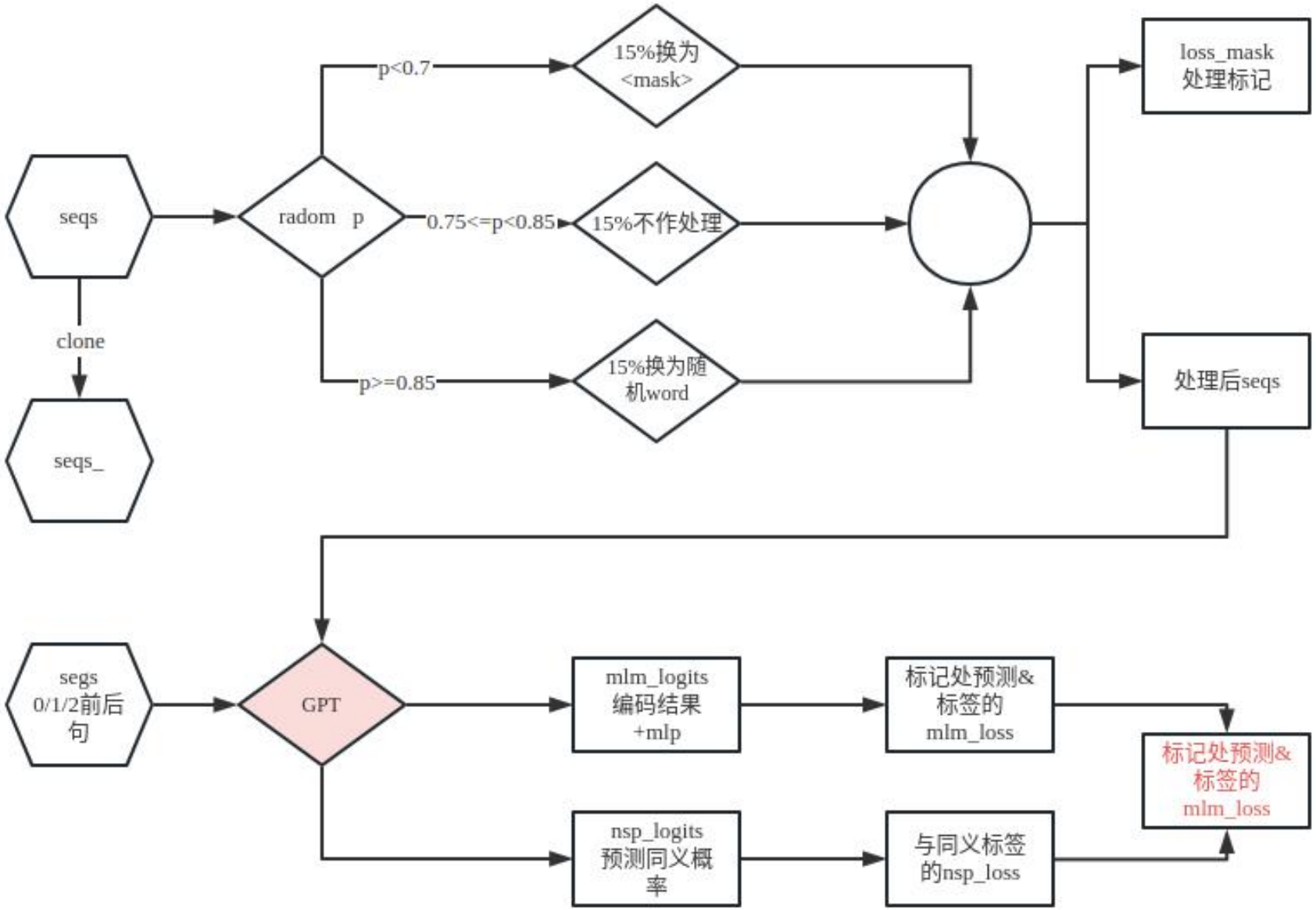


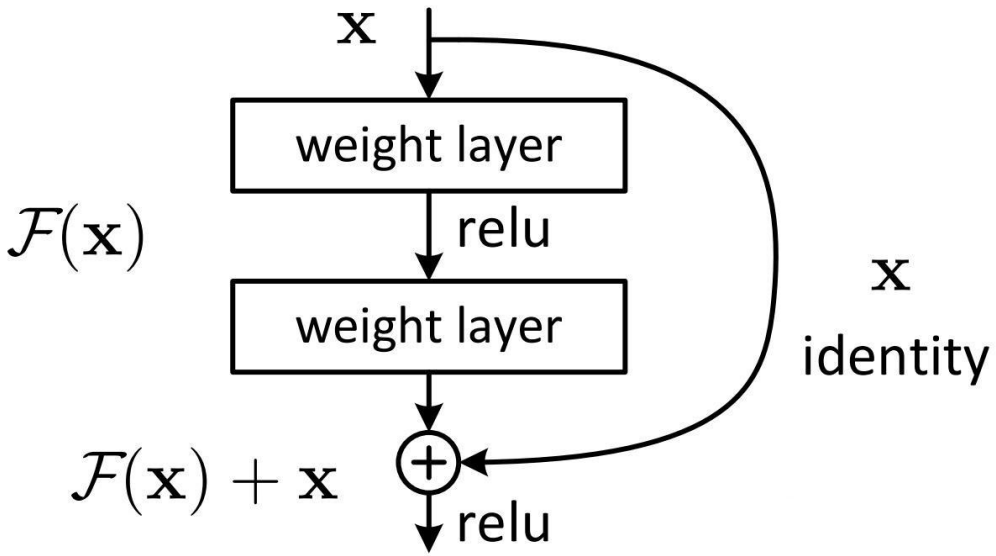

loss_line
10
8
6
4
2
0
loss
0
1000
2000
3000
4000
5000
6000
train_step


1.0
0.8
0.6
0.4
0.2
0.0


at
his
sentencing
,
avants
had
tubes
in
his
nose
and
a
portable
oxygen
tank
beside
him
.
<SEP>
avants
,
wearing
a
light
brown
jumpsuit
,
had
tubes
in
his
nose
and
a
portable
oxygen
tank
beside
him
.
1.0
0.8
0.6
0.4
0.2
0.0

batch	list 1	tensor(32,72)	32样本72字符在词汇总表索引	seqs
	list 2	tensor(32,72)	前后句信息，0前句，1后句，2空符	segs
	list 3	tensor(32,2)	一行两句话各自长度	xlen
	list 4	tensor(32,1)	标签 array([0]) 0or1	nsp_labels


seqs
radom p
p<0.7
15%换为
<mask>
0.75<=p<0.85
15%不作处理
p>=0.85
15%换为随
机word
clone
seqs_
loss_mask
处理标记
处理后seqs
segs
0/1/2前后
句
GPT
mlm_logits
编码结果
+mlp
标记处预测&
标签的
mlm_loss
nsp_logits
预测同义概
率
与同义标签
的nsp_loss
标记处预测&
标签的
mlm_loss



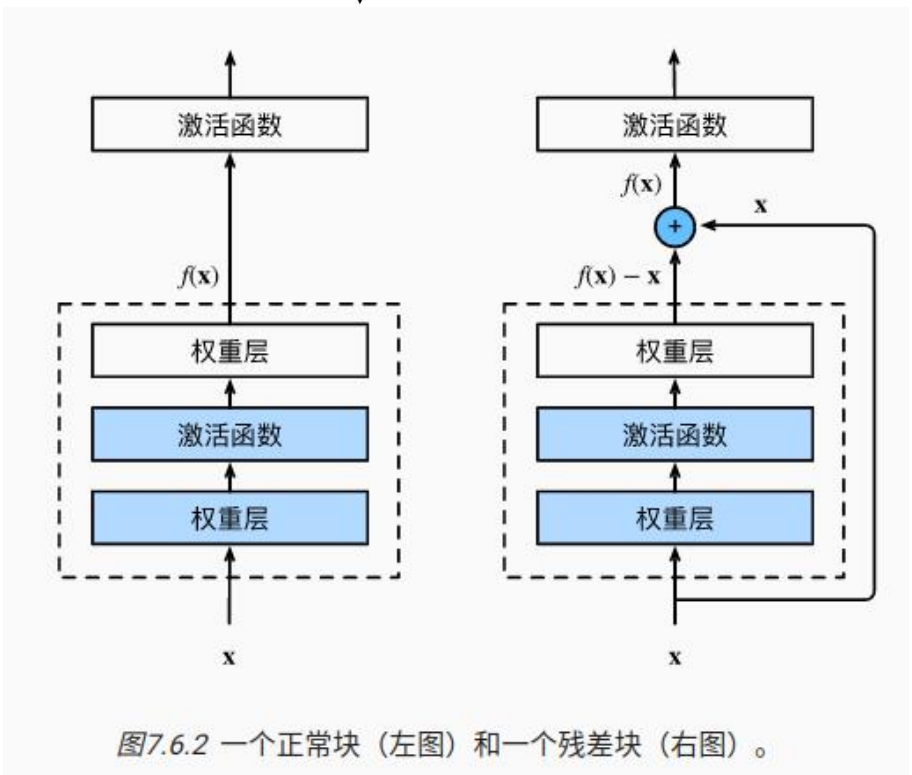
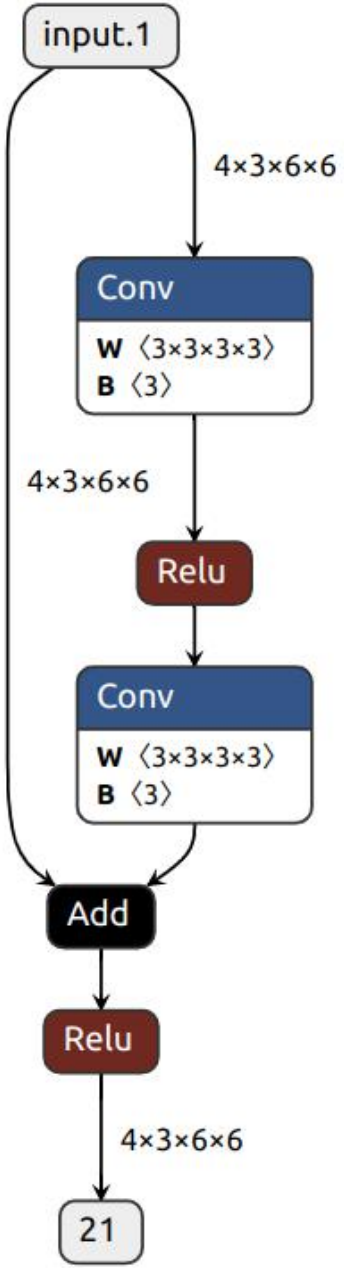
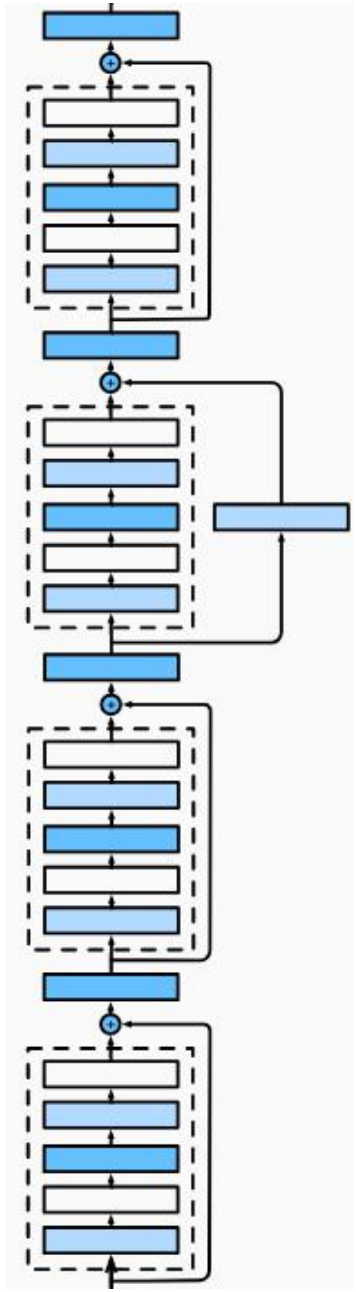


图7.6.2 一个正常块（左图）和一个残差块（右图）。

- F（x）和x的shape不一致时
- pad或者乘以一个权重





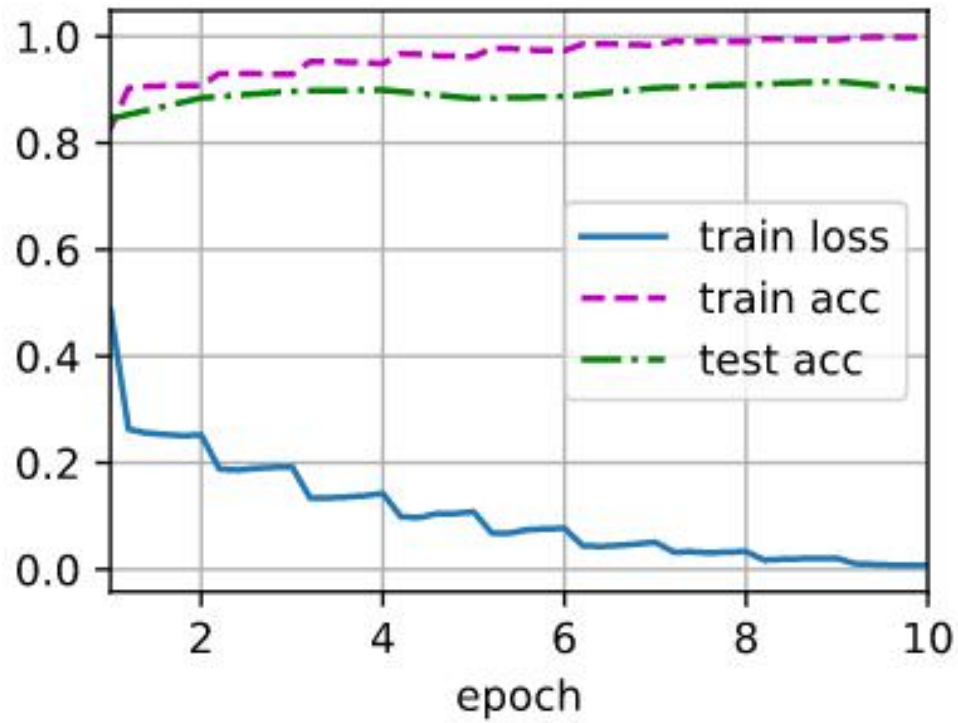
```
Sequential output shape:     torch.Size([1, 64, 56, 56])
Sequential output shape:     torch.Size([1, 64, 56, 56])
Sequential output shape:     torch.Size([1, 128, 28, 28])
Sequential output shape:     torch.Size([1, 256, 14, 14])
Sequential output shape:     torch.Size([1, 512, 7, 7])
AdaptiveAvgPool2d output shape:  torch.Size([1, 512, 1, 1])
Flatten output shape:    torch.Size([1, 512])
Linear output shape:     torch.Size([1, 10])
```

```
loss 0.112, train acc 0.960, test acc 0.902
1613.8 examples/sec on cuda:0
```

设置残差块训练 5 回合结果

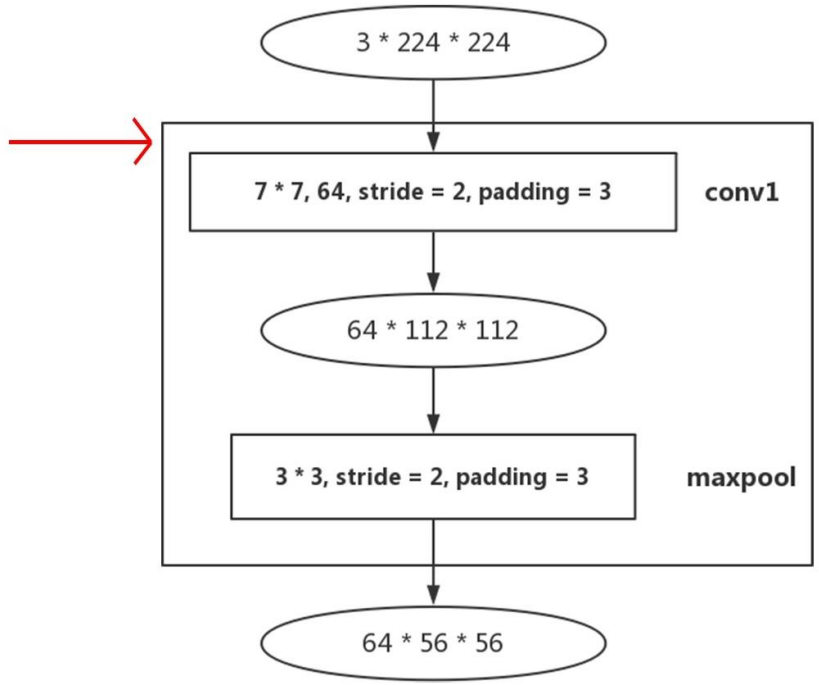
```
loss 0.144, train acc 0.948, test acc 0.861
1751.9 examples/sec on cuda:0
```

未设置残差块训练 5 回合结果

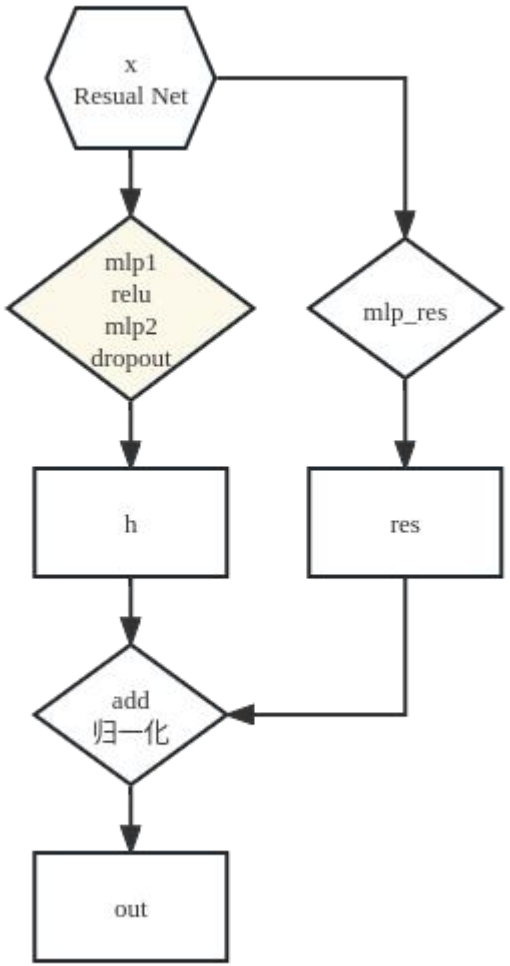


layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

layer name	output size	18-layer
conv1	112×112	7×7, 64, stride 2 3×3 max pool, stride 2
conv2_x	56×56	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$
	1×1	average pool, 1000-d fc, softmax

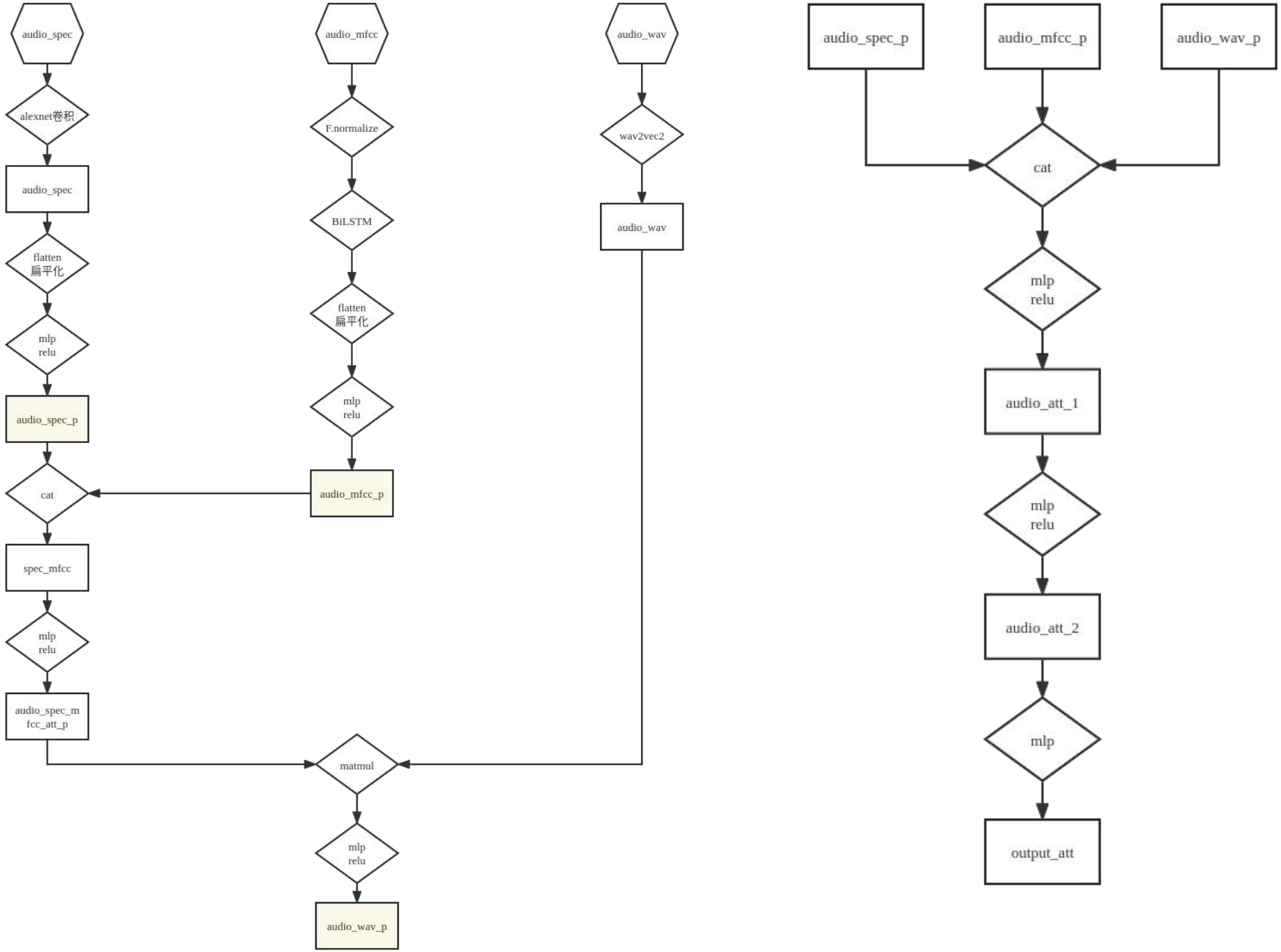


- 数字18 50 101 等分别代表的是卷积层+全连接层的层数
- 50-layer (1+3*3+4*3+6*3+3*3+1) =50层



group name	output size	block type = $B(3,3)$
conv1	32×32	[3×3, 16]
conv2	32×32	[3×3, 16×k; 3×3, 16×k] ×N
conv3	16×16	[3×3, 32×k; 3×3, 32×k] ×N
conv4	8×8	[3×3, 64×k; 3×3, 64×k] ×N
avg-pool	1×1	[8×8]

- WideResNet（WRN）希望使用一种较浅的，并在每个单层上更宽的（维度）模型来提升模型性能。
- 对于卷积层来说，宽度是指输出维度（通道）


audio_spec
alexnet卷积
audio_spec
flatten
扁平化
mlp
relu
audio_spec_p
audio_mfcc
F.normalize
BiLSTM
flatten
扁平化
mlp
relu
audio_mfcc_p
audio_wav
wav2vec2
audio_wav
cat
spec_mfcc
mlp
relu
audio_spec_m
fcc_att_p
matmul
mlp
relu
audio_wav_p
audio_spec_p
audio_mfcc_p
audio_wav_p
cat
mlp
relu
audio_att_1
mlp
relu
audio_att_2
mlp
output_att

- Weather 收集自2010-2013四年间美国1600个地点的每小时气候数据。包括预测目标湿球温度以及11个气候指标。数据集/验证集/测试集分别为28/10/10个月。

- ETT(Electricity Transformer Temperature) 收集自中国两个县的电力变压器温度数据集，有2小时、1小时、15分钟三种粒度，每个数据点包括油温以及6个电力负载指标。数据集/验证集/测试集分别为12/4/4个月。

- Traffic2是加利福尼亚州交通部每小时数据的集合，描述了旧金山湾地区高速公路上不同传感器测量的道路占用率。

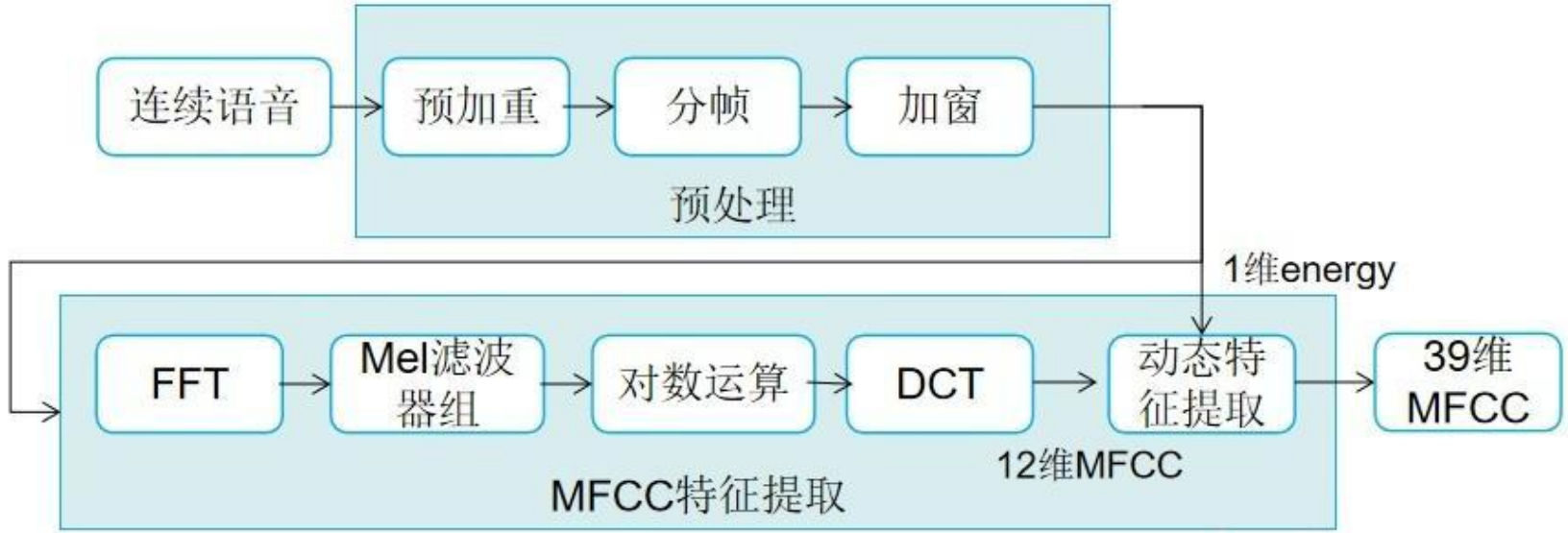
MSE（均方误差）
MAE(平均绝对误差)

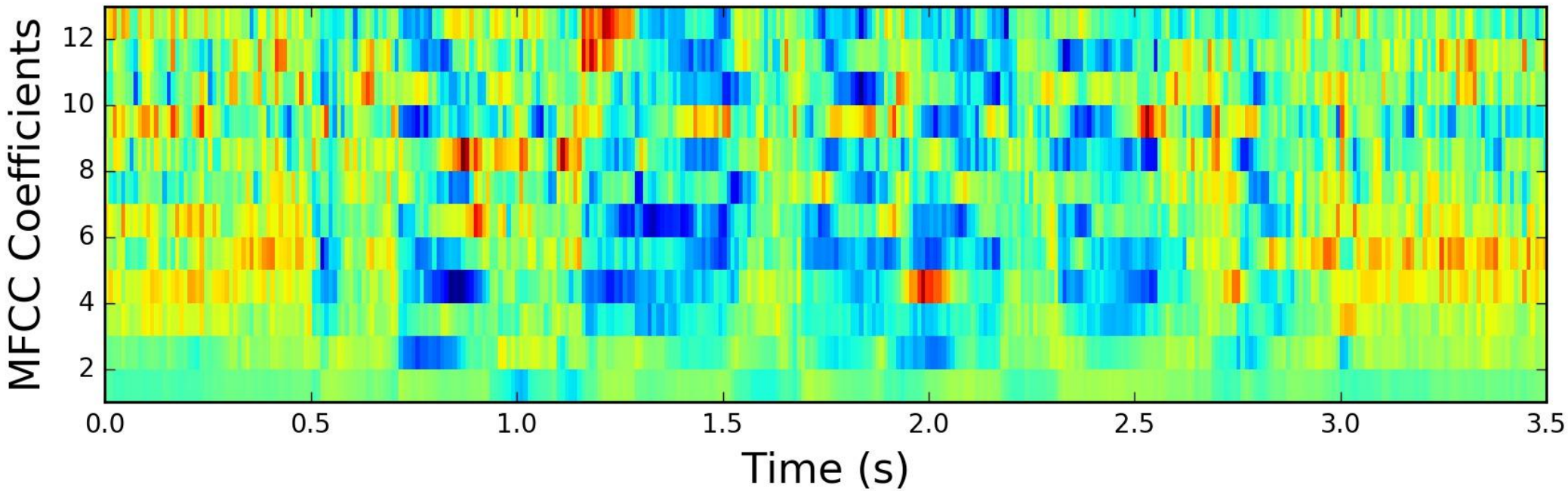
Models		TiDE		PatchTST/64		DLinear		FEDformer		Autoformer		Informer		Pyraformer		LogTrans	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Weather	96	0.166	0.222	**0.149**	**0.198**	0.176	0.237	0.238	0.314	0.249	0.329	0.354	0.405	0.896	0.556	0.458	0.490
	192	0.209	0.263	**0.194**	**0.241**	0.220	0.282	0.275	0.329	0.325	0.370	0.419	0.434	0.622	0.624	0.658	0.589
	336	0.254	0.301	**0.245**	**0.282**	0.265	0.319	0.339	0.377	0.351	0.391	0.583	0.543	0.739	0.753	0.797	0.652
	720	**0.313**	0.340	**0.314**	**0.334**	0.323	0.362	0.389	0.409	0.415	0.426	0.916	0.705	1.004	0.934	0.869	0.675
Traffic	96	**0.336**	0.253	0.360	**0.249**	0.410	0.282	0.576	0.359	0.597	0.371	0.733	0.410	2.085	0.468	0.684	0.384
	192	**0.346**	**0.257**	0.379	**0.256**	0.423	0.287	0.610	0.380	0.607	0.382	0.777	0.435	0.867	0.467	0.685	0.390
	336	**0.355**	**0.260**	0.392	0.264	0.436	0.296	0.608	0.375	0.623	0.387	0.776	0.434	0.869	0.469	0.734	0.408
	720	**0.386**	**0.273**	0.432	0.286	0.466	0.315	0.621	0.375	0.639	0.395	0.827	0.466	0.881	0.473	0.717	0.396
Electricity	96	**0.132**	0.229	**0.129**	**0.222**	0.140	0.237	0.186	0.302	0.196	0.313	0.304	0.393	0.386	0.449	0.258	0.357
	192	**0.147**	0.243	**0.147**	**0.240**	0.153	0.249	0.197	0.311	0.211	0.324	0.327	0.417	0.386	0.443	0.266	0.368
	336	**0.161**	0.261	0.163	**0.259**	0.169	0.267	0.213	0.328	0.214	0.327	0.333	0.422	0.378	0.443	0.280	0.380
	720	**0.196**	0.294	**0.197**	**0.290**	0.203	0.301	0.233	0.344	0.236	0.342	0.351	0.427	0.376	0.445	0.283	0.376
ETTh1	96	**0.375**	**0.398**	0.379	0.401	0.375	**0.399**	0.376	0.415	0.435	0.446	0.941	0.769	0.664	0.612	0.878	0.740
	192	**0.412**	0.422	0.413	0.429	**0.412**	**0.420**	0.423	0.446	0.456	0.457	1.007	0.786	0.790	0.681	1.037	0.824
	336	**0.435**	**0.433**	**0.435**	0.436	0.439	0.443	0.444	0.462	0.486	0.487	1.038	0.784	0.891	0.738	1.238	0.932
	720	0.454	**0.465**	**0.446**	**0.464**	0.472	0.490	0.469	0.492	0.515	0.517	1.144	0.857	0.963	0.782	1.135	0.852
ETTh2	96	**0.270**	**0.336**	0.274	**0.337**	0.289	0.353	0.332	0.374	0.332	0.368	1.549	0.952	0.645	0.597	2.116	1.197
	192	**0.332**	0.380	0.338	**0.376**	0.383	0.418	0.407	0.446	0.426	0.434	3.792	1.542	0.788	0.683	4.315	1.635
	336	**0.360**	0.407	0.363	**0.397**	0.448	0.465	0.400	0.447	0.477	0.479	4.215	1.642	0.907	0.747	1.124	1.604
	720	0.419	0.451	**0.393**	**0.430**	0.605	0.551	0.412	0.469	0.453	0.490	3.656	1.619	0.963	0.783	3.188	1.540
ETTm1	96	0.306	0.349	**0.293**	**0.346**	0.299	0.343	0.326	0.390	0.510	0.492	0.626	0.560	0.543	0.510	0.600	0.546
	192	**0.335**	**0.366**	**0.333**	0.370	0.335	**0.365**	0.365	0.415	0.514	0.495	0.725	0.619	0.557	0.537	0.837	0.700
	336	**0.364**	**0.384**	0.369	0.392	0.369	0.386	0.392	0.425	0.510	0.492	1.005	0.741	0.754	0.655	1.124	0.832
	720	**0.413**	**0.413**	0.416	0.420	0.425	0.421	0.446	0.458	0.527	0.493	1.133	0.845	0.908	0.724	1.153	0.820
ETTm2	96	**0.161**	**0.251**	0.166	0.256	0.167	0.260	0.180	0.271	0.205	0.293	0.355	0.462	0.435	0.507	0.768	0.642
	192	**0.215**	**0.289**	0.223	0.296	0.224	0.303	0.252	0.318	0.278	0.336	0.595	0.586	0.730	0.673	0.989	0.757
	336	**0.267**	**0.326**	0.274	0.329	0.281	0.342	0.324	0.364	0.343	0.379	1.270	0.871	1.201	0.845	1.334	0.872
	720	**0.352**	**0.383**	0.362	**0.385**	0.397	0.421	0.410	0.420	0.414	0.419	3.001	1.267	3.625	1.451	3.048	1.328

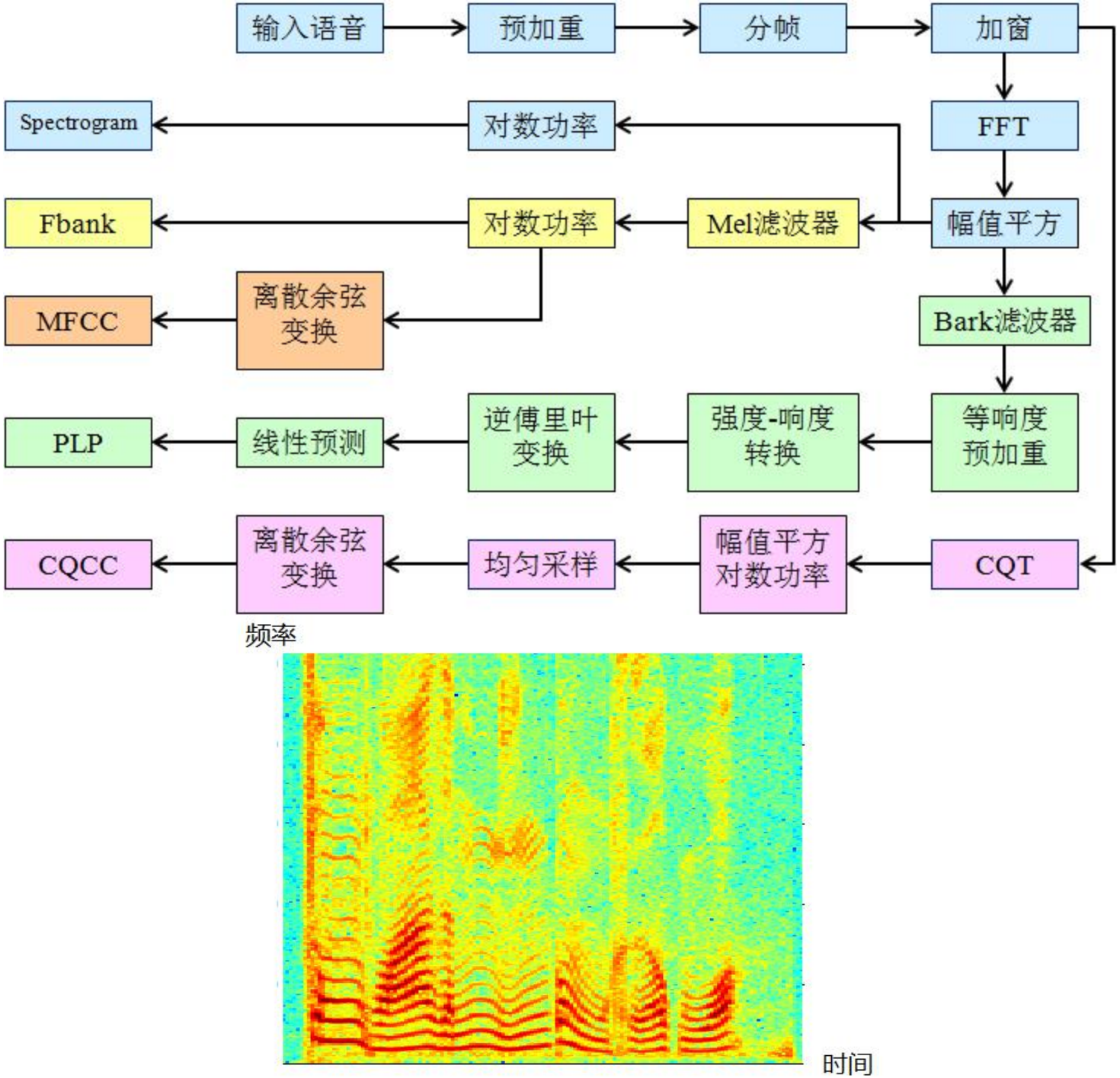
https://github.com/yuqinie98/PatchTST/issues/7

Models		**Autoformer**		Informer[48]		LogTrans[26]		Reformer[23]		LSTNet[25]		LSTM[17]		TCN[4]	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETT*	96	**0.255**	**0.339**	0.365	0.453	0.768	0.642	0.658	0.619	3.142	1.365	2.041	1.073	3.041	1.330
	192	**0.281**	**0.340**	0.533	0.563	0.989	0.757	1.078	0.827	3.154	1.369	2.249	1.112	3.072	1.339
	336	**0.339**	**0.372**	1.363	0.887	1.334	0.872	1.549	0.972	3.160	1.369	2.568	1.238	3.105	1.348
	720	**0.422**	**0.419**	3.379	1.388	3.048	1.328	2.631	1.242	3.171	1.368	2.720	1.287	3.135	1.354
Electricity	96	**0.201**	**0.317**	0.274	0.368	0.258	0.357	0.312	0.402	0.680	0.645	0.375	0.437	0.985	0.813
	192	**0.222**	**0.334**	0.296	0.386	0.266	0.368	0.348	0.433	0.725	0.676	0.442	0.473	0.996	0.821
	336	**0.231**	**0.338**	0.300	0.394	0.280	0.380	0.350	0.433	0.828	0.727	0.439	0.473	1.000	0.824
	720	**0.254**	**0.361**	0.373	0.439	0.283	0.376	0.340	0.420	0.957	0.811	0.980	0.814	1.438	0.784
Exchange	96	**0.197**	**0.323**	0.847	0.752	0.968	0.812	1.065	0.829	1.551	1.058	1.453	1.049	3.004	1.432
	192	**0.300**	**0.369**	1.204	0.895	1.040	0.851	1.188	0.906	1.477	1.028	1.846	1.179	3.048	1.444
	336	**0.509**	**0.524**	1.672	1.036	1.659	1.081	1.357	0.976	1.507	1.031	2.136	1.231	3.113	1.459
	720	**1.447**	**0.941**	2.478	1.310	1.941	1.127	1.510	1.016	2.285	1.243	2.984	1.427	3.150	1.458
Traffic	96	**0.613**	**0.388**	0.719	0.391	0.684	0.384	0.732	0.423	1.107	0.685	0.843	0.453	1.438	0.784
	192	**0.616**	**0.382**	0.696	0.379	0.685	0.390	0.733	0.420	1.157	0.706	0.847	0.453	1.463	0.794
	336	**0.622**	**0.337**	0.777	0.420	0.733	0.408	0.742	0.420	1.216	0.730	0.853	0.455	1.479	0.799
	720	**0.660**	**0.408**	0.864	0.472	0.717	0.396	0.755	0.423	1.481	0.805	1.500	0.805	1.499	0.804
Weather	96	**0.266**	**0.336**	0.300	0.384	0.458	0.490	0.689	0.596	0.594	0.587	0.369	0.406	0.615	0.589
	192	**0.307**	**0.367**	0.598	0.544	0.658	0.589	0.752	0.638	0.560	0.565	0.416	0.435	0.629	0.600
	336	**0.359**	**0.395**	0.578	0.523	0.797	0.652	0.639	0.596	0.597	0.587	0.455	0.454	0.639	0.608
	720	**0.419**	**0.428**	1.059	0.741	0.869	0.675	1.130	0.792	0.618	0.599	0.535	0.520	0.639	0.610
ILI	24	**3.483**	**1.287**	5.764	1.677	4.480	1.444	4.400	1.382	6.026	1.770	5.914	1.734	6.624	1.830
	36	**3.103**	**1.148**	4.755	1.467	4.799	1.467	4.783	1.448	5.340	1.668	6.631	1.845	6.858	1.879
	48	**2.669**	**1.085**	4.763	1.469	4.800	1.468	4.832	1.465	6.080	1.787	6.736	1.857	6.968	1.892
	60	**2.770**	**1.125**	5.264	1.564	5.278	1.560	4.882	1.483	5.548	1.720	6.870	1.879	7.127	1.918

* *ETT* means the ETTm2. See Appendix A for the **full benchmark** of ETTh1, ETTh2, ETTm1.


连续语音
预加重
分帧
加窗
预处理
1维energy
FFT
Mel滤波器组
对数运算
DCT
动态特征提取
39维MFCC
12维MFCC
MFCC特征提取


MFCC Coefficients
12
10
8
6
4
2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Time (s)

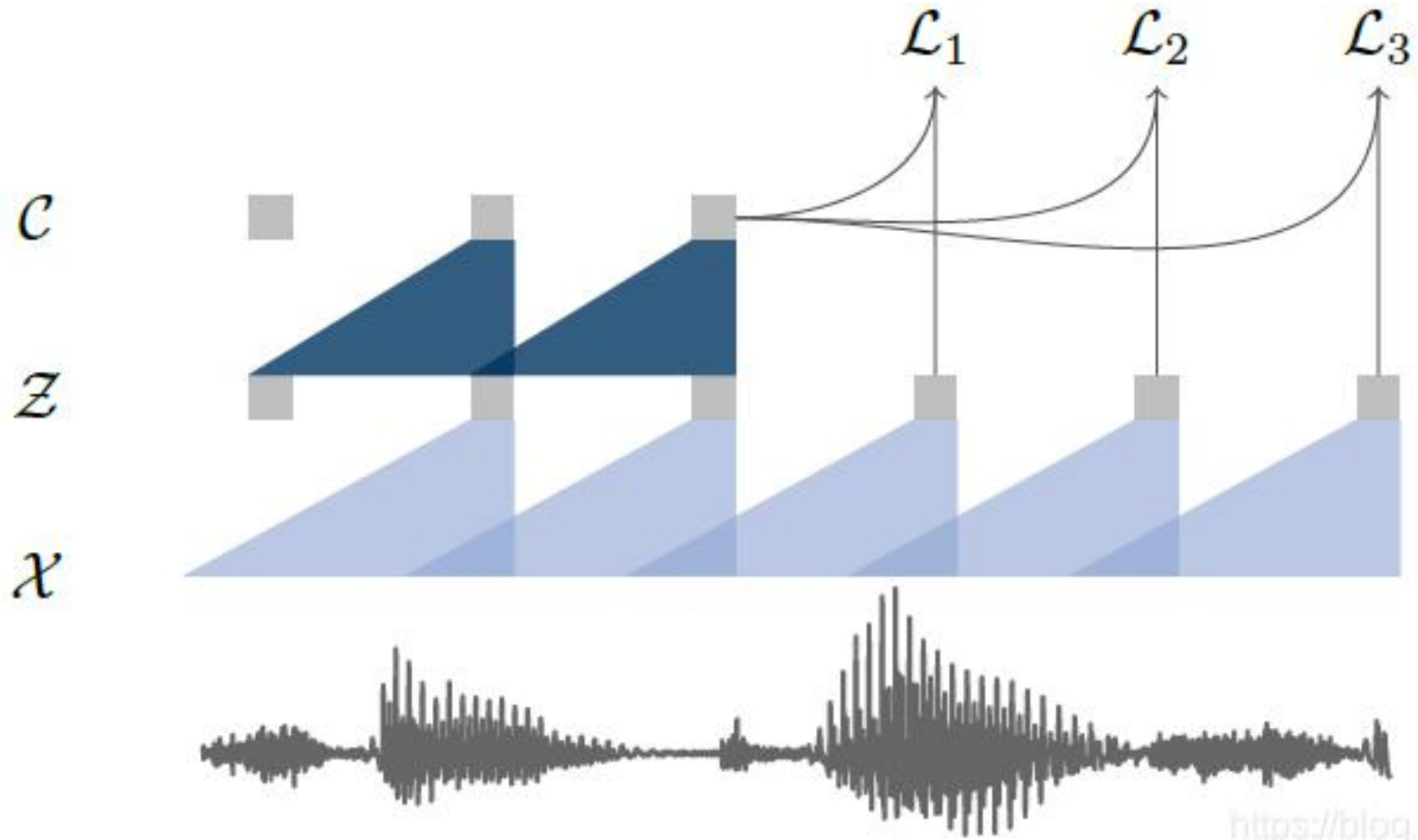


语谱图(spectrogram)

- 声谱图的横坐标是时间
- 纵坐标是频率
- 坐标点值为语音数据能量

由于是采用二维平面表达三维信息，所以能量值的大小是通过颜色来表示的，颜色深，表示该点的语音能量越强。

将原始音频x编码为潜在空间z的 encoder network（5层卷积），和将z转换为contextualized representation的context network（9层卷积），最终特征维度为512x帧数。目标是在特征层面使用当前帧预测未来帧。



MFCC

针对数据集1M进行训练&测试

github原代码
1e-05　0.3476　1.3293　61.94%　65.89%　60.07%　66.39%

MFCC+BiLSTM
1e-05　1.0440　0.9828　60.50%　60.32%　60.62%　62.47%

		训练集loss	测试集loss		测试集最优		训练集最优
github原代码	1e-05	0.3476	1.3293	61.94%	65.89%	60.07%	66.39%
MFCC+BiLSTM	1e-05	1.0440	0.9828	60.50%	60.32%	60.62%	62.47%
MFCC+DLinear	1e-05	1.0739	1.0352	55.83%	56.91%	57.79%	58.28%
MFCC+TiDE	1e-05	1.1036	1.1508	45.27%	46.21%	49.40%	49.74%
Spec+Alexnet	1e-05	0.5312	1.0506	63.55%	64.81%	62.15%	66.82%
Spec+BiLstm	1e-05	0.2206	1.6786	55.48%	61.40%	56.50%	62.50%
wav2vec	1e-05	0.2117	1.7233	70.74%	71.99%	73.74%	74.20%

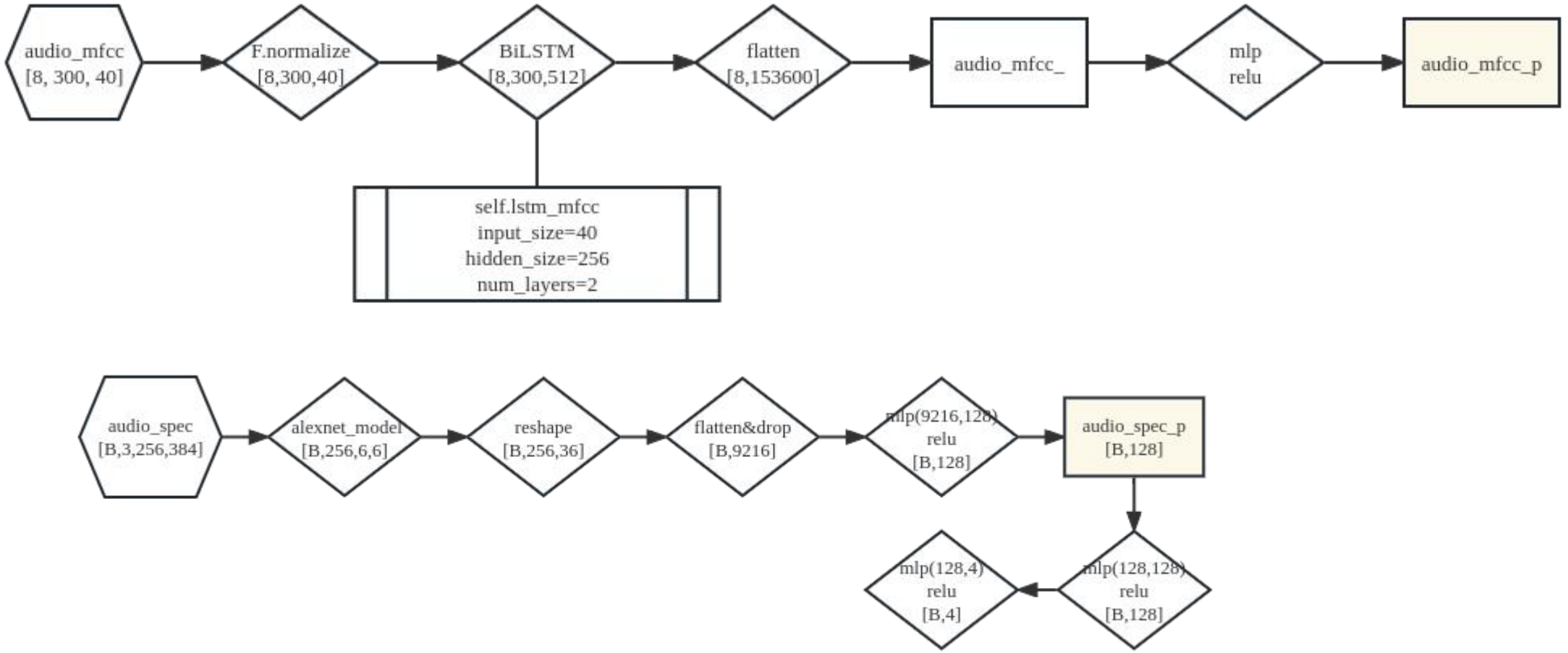
```
  (lstm_spec): LSTM(384, 512, num_layers=2, batch_first=True, dropout=0.5, bidirectional=True)
  (spec_linear): Linear(in_features=786432, out_features=128, bias=True)
  (spec_linear_out): Linear(in_features=128, out_features=4, bias=True)
)
Number of trainable parameters: 285281765
```

```
  (spec_linear_out): Linear(in_features=128, out_features=4, bias=True)
  (out_res): Linear(in_features=2048, out_features=4, bias=True)
)
Number of trainable parameters: 10332176


Start Training!!!
 13%|                                                  | 79/614 [00:16<01:47,  4.96it/s]
```

/home/windsky/Data/point_cloud/CA-github4_spec3_bilstm2

mfcc&spec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F.normalize
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BiLSTM
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flatten
[8,153600]
audio_mfcc_
mlp
relu
audio_mfcc_p
self.lstm_mfcc
input_size=40
hidden_size=256
num_layers=2
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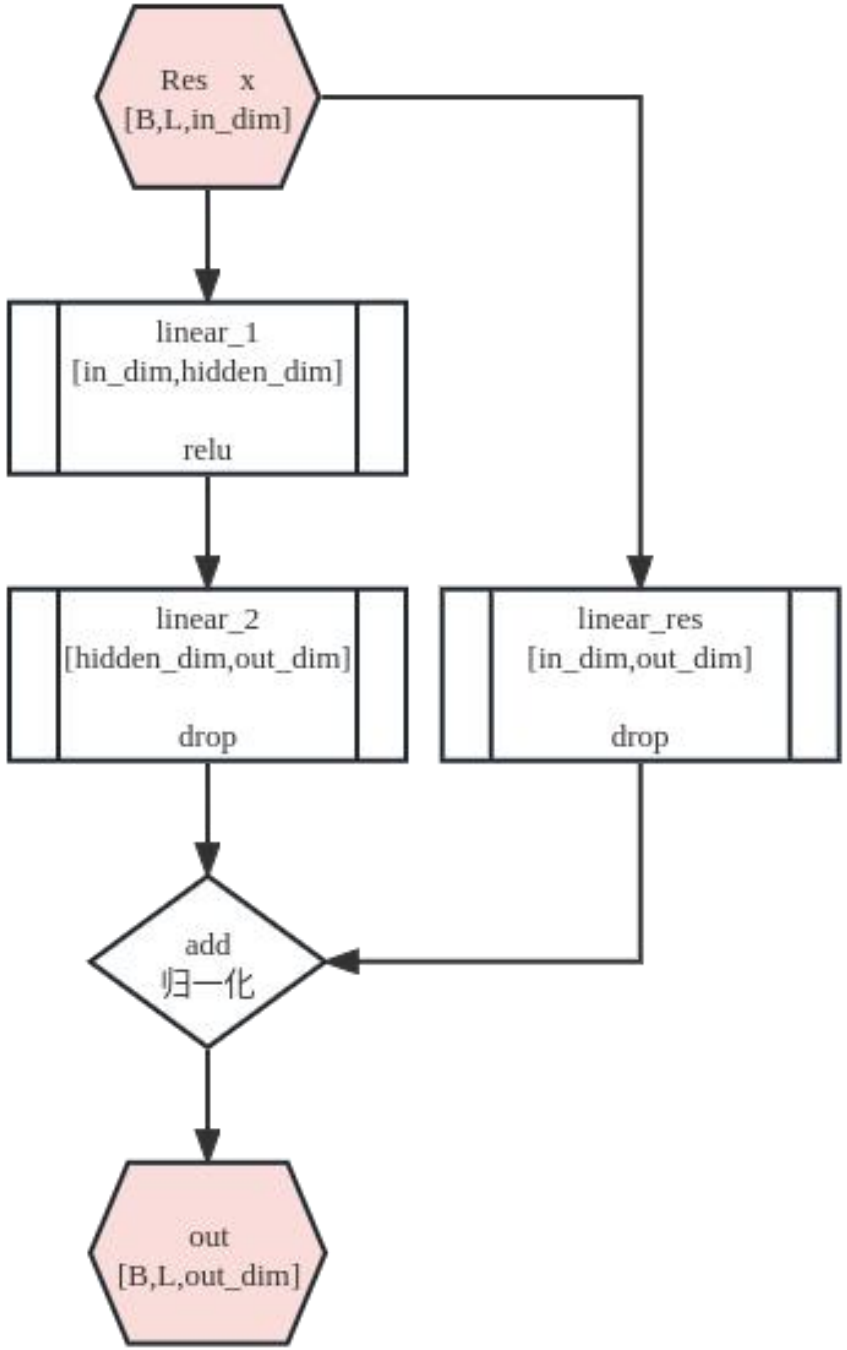
alexnet_model
[B,256,6,6]
reshape
[B,256,36]
flatten&drop
[B,9216]
mlp(9216,128)
relu
[B,128]
audio_spec_p
[B,128]
mlp(128,128)
relu
[B,128]
mlp(128,4)
relu
[B,4]

作者	数据库	模态	特征	分类器	使用数据	验证方法	UA /%	WA /%
Cai et al.[64]	IEMOCAP	Speech	log power-spectrum	CNN LSTM MAML	Angry, Happy, Sad, Neutral	5-fold cross validation	70.32	76.64
Liu et al. [65]	IEMOCAP	Speech	MFCCs	CNN Bi-LSTM	Angry, Happy +(excited), Sad, Neutral	5-fold cross validation	78.30	79.52
Peng et al.[66]	RECOLA SEWA	Speech	MMCG	LSTM	Arousal, Valence	5-fold cross validation		
Zhu et al.[67]	EmoDB CREMA-D	Speech	Log-mel spectro-gram	ACRNN	Angry, Happy +(excited), Sad, Neutral, Disgust, Fear, Boredom	10-fold cross validation	77.20 73.40	77.90 79.60
Latif et al.[68]	IEMOCAP MSP-IMPROV	Speech	Emobase2010	Mixup GAN	Angry, Happy +(excited), Sad, Neutral			46.60
Liu et al. [69]	IEMOCAP	Speech Text	Mel-spectrogram MFCCs Glove	CNN Bi-LSTM Attention	Angry, Happy, Sad, Neutral	5-fold cross validation	70.08	72.39
Chen et al. [70]	IEMOCAP	Speech Text	Log-mel spectrogram	CNN Bi-LSTM Attention ALBERT	Angry, Happy +(excited), Sad, Neutral	5-fold cross validation	72.05	71.06
Li et al.[71]	IEMOCAP RAVDESS	Speech Text	Filterbank GloVe	Bi-LSTM	Angry, Happy, Sad, Neutral	5-fold cross validation	73.50 70.80	72.70 72.00
Kumar et al.[72]	IEMOCAP	Speech Text	Mel-spectrogram MFCC Chroma	GRU Attention BERT	Angry, Happy +(excited), Sad, Neutral	5-fold cross validation	75.00	71.70
Wang et al.[34]	IEMOCAP	Speech Text	MFCCs One-hot embedding	Transformer Attention	Angry, Happy, Sad, Neutral	5-fold cross validation	77.10	76.80
Hou [73]	IEMOCAP MELD	Speech Text	LMBFs BERT	GRU Attention	Angry, Happy +(excited), Sad, Neutral	5-fold cross validation	77.60	75.60 64.90
Lian et al.[74]	IEMOCAP	Speech Text	IS13-ComparE ELMo	GRU Multi-Head-Attention	Angry, Happy +(excited), Sad, Neutral	session 5 validation		82.68
Pan et al.[75]	IEMOCAP	Speech Visual Text	IS13-ComparE Word2vec	LSTM Attention	Angry, Happy +(excited), Sad, Neutral, Frustrated			73.94
Khare et al.[76]	CMU-MOSEI	Speech Visial Text	LFBE GloVe	Transformer Multi-Head-Attention	Angry, Happy, Sad, Disgust, Surprise, Fear			66.40

Table 2. Ablation study on the proposed model.

Model	WA	UA
MFCC	57.60	58.09
Spectrogram	62.13	62.25
W2E	64.03	65.67
MFCC+W2E (w/o co-att)	64.62	65.93
Spectrogram+W2E (w/o co-att)	66.20	67.22
MFCC+Spectrogram+W2E (w/o co-att)	67.22	67.81
W2E (w/ co-att)	67.55	68.65
MFCC+W2E (w/ co-att)	69.11	70.30
Spectrogram+W2E (w/ co-att)	70.05	71.30
MFCC+Spectrogram+W2E (w/ co-att)	**71.64**	**72.70**


Res x
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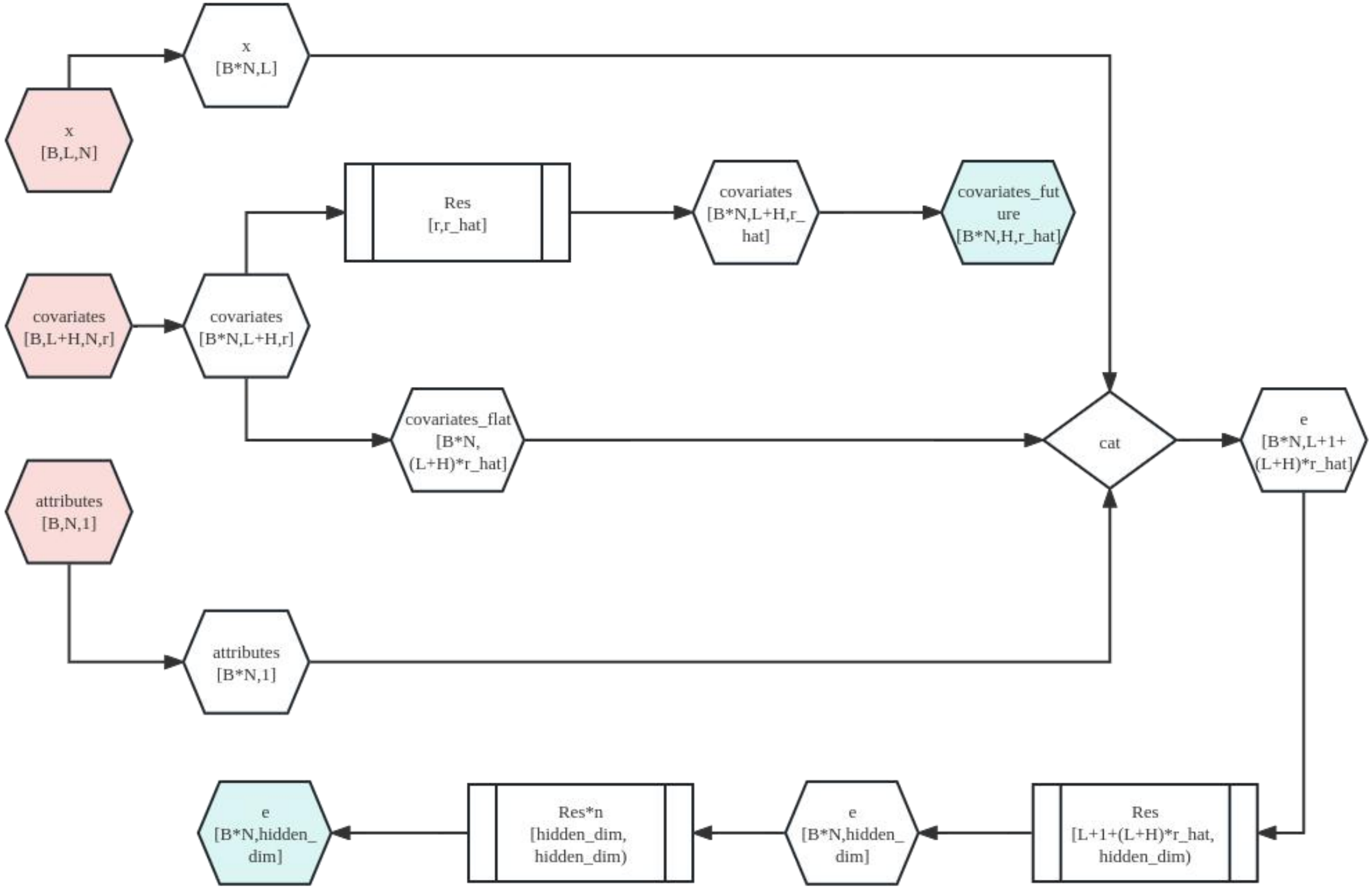
relu
linear_2
[hidden_dim,out_dim]
drop
linear_res
[in_dim,out_dim]
drop
add
归一化
out
[B,L,out_dim]


x
[B,L,N]
x
[B*N,L]
Res
[r,r_hat]
covariates
[B*N,L+H,r_
hat]
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ure
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covariates
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covariates
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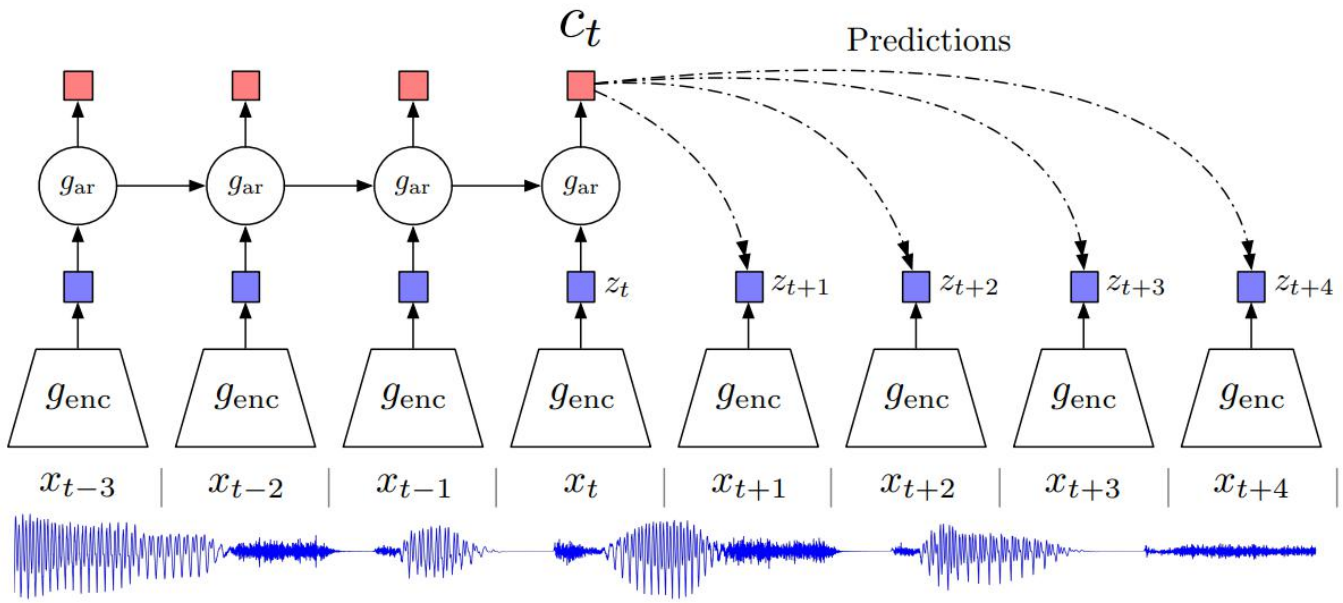
covariates_flat
[B*N,
(L+H)*r_hat]
cat
e
[B*N,L+1+
(L+H)*r_hat]
attributes
[B,N,1]
attributes
[B*N,1]
e
[B*N,hidden_
dim]
Res*n
[hidden_dim,
hidden_dim)
e
[B*N,hidden_
dim]
Res
[L+1+(L+H)*r_hat,
hidden_dim)

Contrastive Predictive Coding (对比预测编码)

- target (future) x：预测目标
- context (present) c ：已知上下文



$$I(x,c) = \sum_{x,c} p(x,c)\ log\frac{p(x|c)}{p(x)}$$
$$= \sum_{x,c} p(x,c)\ log\frac{p(x|c)p(c)}{p(x)p(c)}$$
$$= \sum_{x,c} p(x,c)\ log\, p(x|c) - \sum_{x,c} p(x,c)\ log\, p(x)$$
$$= \sum_{c} p(c) \sum_{x} p(x|c)\ log\, p(x|c) - \sum_{x} \sum_{c} p(x,c)\ log\, p(x$$
$$= H(x) - H(x|c)$$

网络的目标是最大化x和c之间的互信息

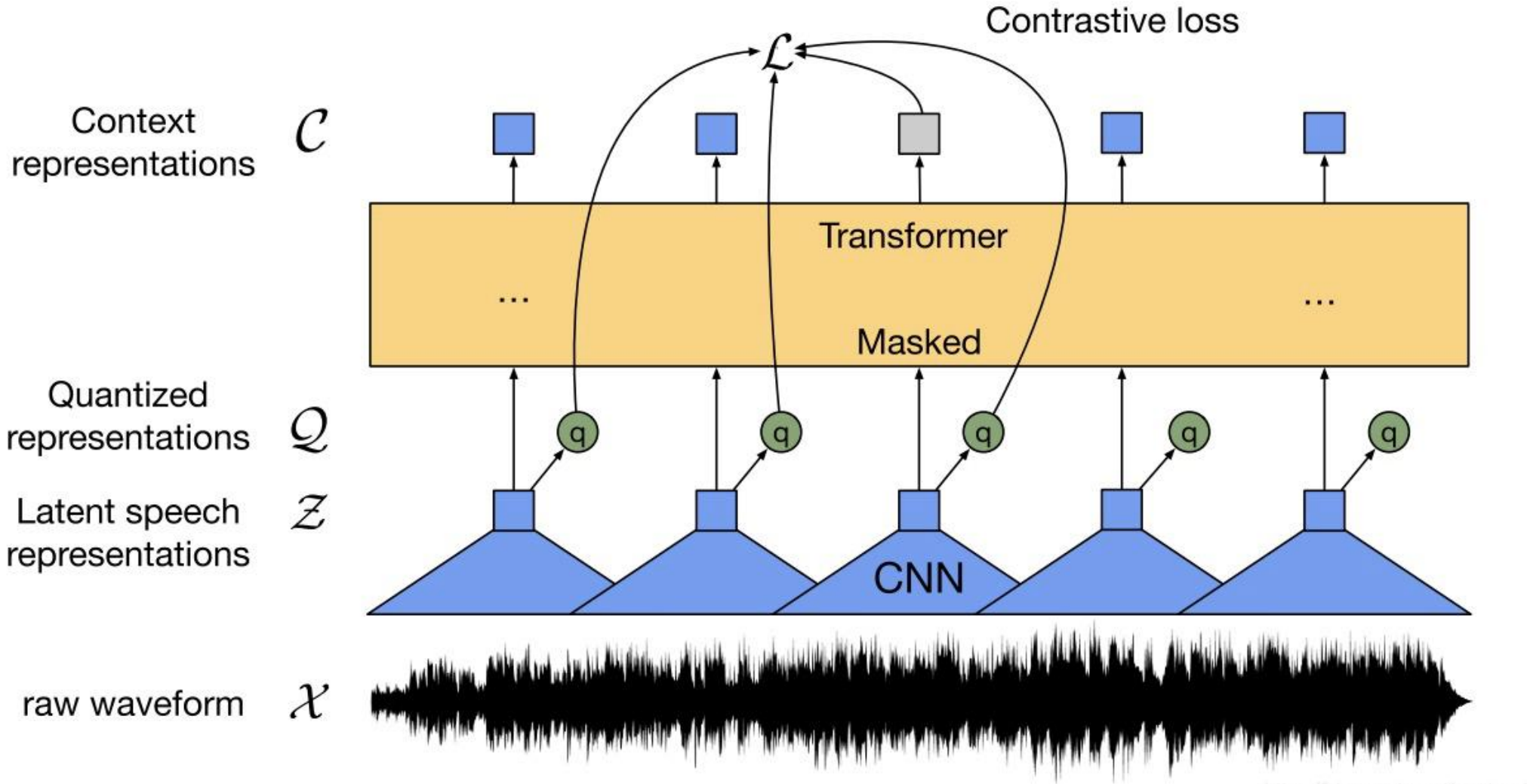
$$I(x;c) = \sum_{x,c} p(x,c) \log \frac{p(x|c)}{p(x)}$$

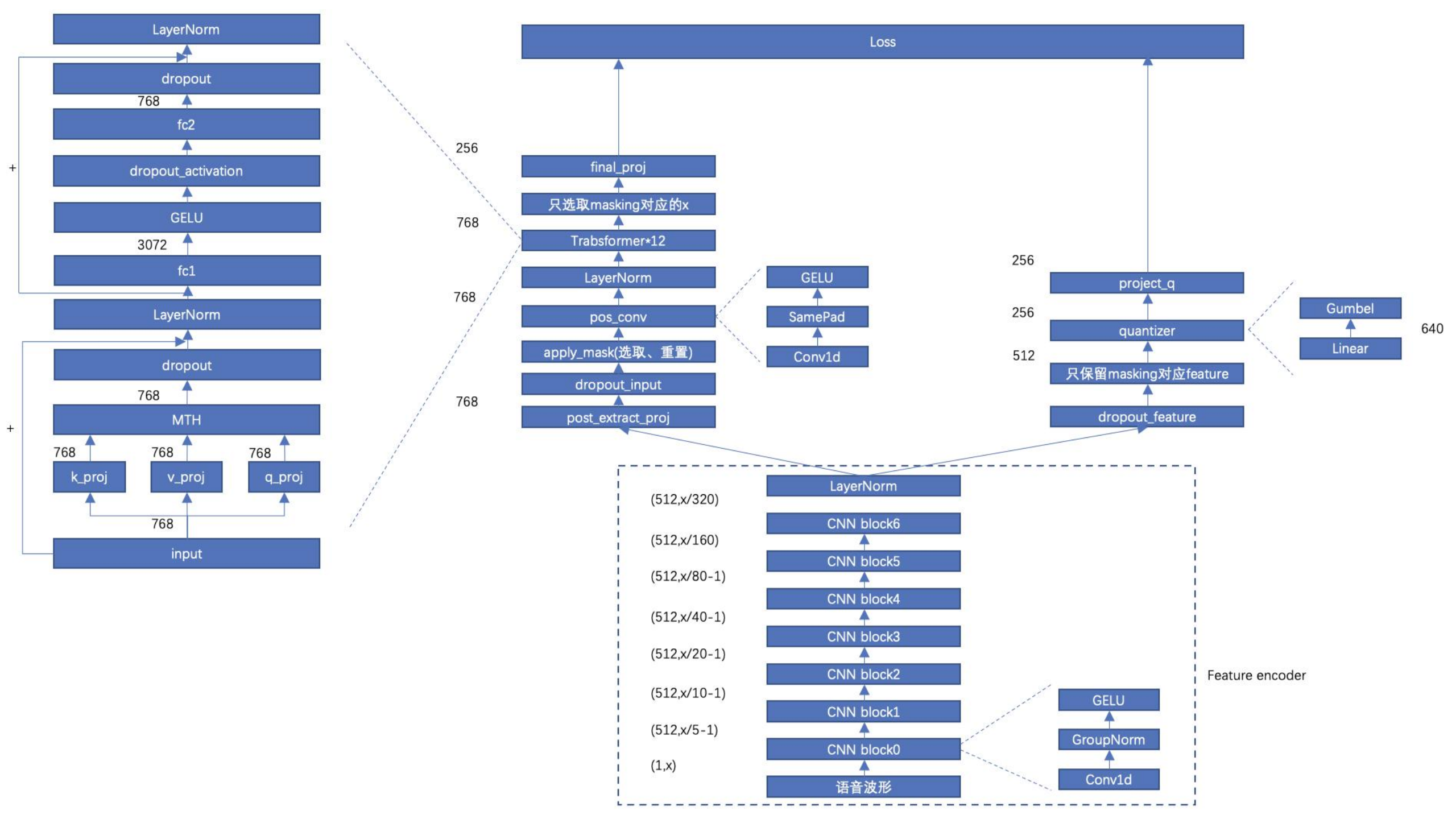
给定c，找到最契合的x，而不是随机采样

- 样本：X ={x1, … , xN}
- x_{t+k}为正样本 (positive sample)，其余N-1个是负样本。
- 在用ct预测xt+k，尽可能预测到正样本

$$\mathcal{L}_N = -\mathop{\mathbb{E}}_{X}\left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)}\right]$$

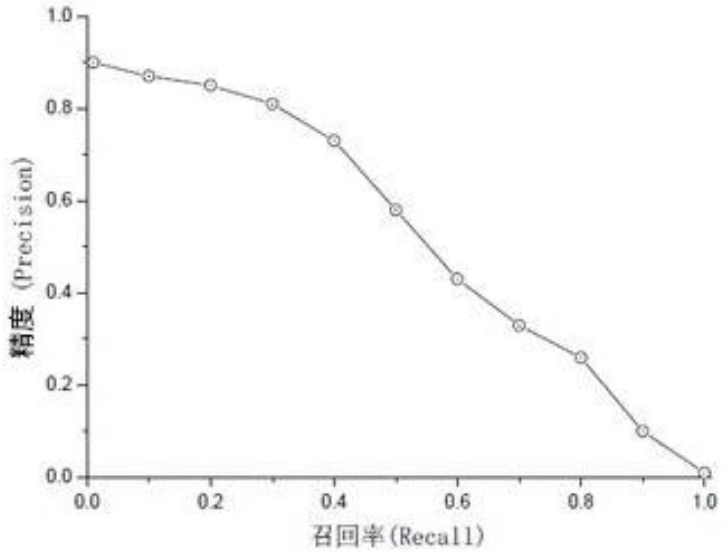
信息熵 $H(X) = -\sum_{i=1}^{n} p(x_i) \log p(x_i)$




LayerNorm
dropout
768
fc2
dropout_activation
GELU
3072
fc1
LayerNorm
dropout
768
MTH
768
768
768
k_proj
v_proj
q_proj
768
input
+
+
Loss
256
final_proj
768
只选取masking对应的x
Trabsformer*12
768
LayerNorm
pos_conv
apply_mask(选取、重置)
768
dropout_input
post_extract_proj
GELU
SamePad
Conv1d
256
project_q
256
quantizer
512
只保留masking对应feature
dropout_feature
Gumbel
Linear
640
(512,x/320)
(512,x/160)
(512,x/80-1)
(512,x/40-1)
(512,x/20-1)
(512,x/10-1)
(512,x/5-1)
(1,x)
LayerNorm
CNN block6
CNN block5
CNN block4
CNN block3
CNN block2
CNN block1
CNN block0
语音波形
GELU
GroupNorm
Conv1d
Feature encoder

准确/召回 率
评价指标 roc auc

真阳性	true positive	正确地检测到阳性结果
假阳性	false positive	错误地检测到阳性结果
真阴性	true negative	正确地检测到阴性结果
假阴性	false negative	错误地检测到阴性结果



$$AP = \int_0^1 p(r)dr$$

准确率 (Accuracy)	全部预测中，正确预测结果占比
精度 (Precision)	全部阳性预测中，正确预测结果占比
召回率 (Recall)	全部阳性事件中，正确预测结果占的比例
误报率	全部阳性预测中，错误预测结果占比

$$a = \frac{TP + TN}{TP + TN + FP + FN}$$

$$p = \frac{TP}{TP + FP}$$

$$r = \frac{TP}{TP + FN} = \frac{TP}{P}$$

$$q = \frac{FP}{TP + FP} = 1 - p$$

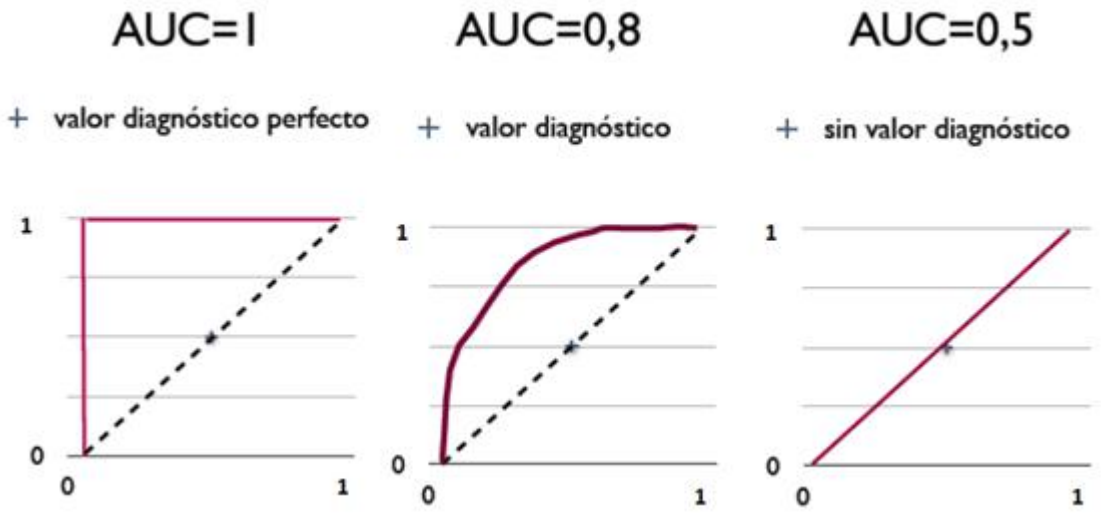
- FP越小越好
- FN越小越好

存疑结果
预测为
阴or阳

受试者工作特征曲线
（ROC曲线）

$$AP = \int_0^1 p(r)dr$$

AUC (Area under Curve)：ROC 曲线下的面积，介于 0.1 和 1 之间，作为数值可以直观的评价分类器的好坏，值越大越好。

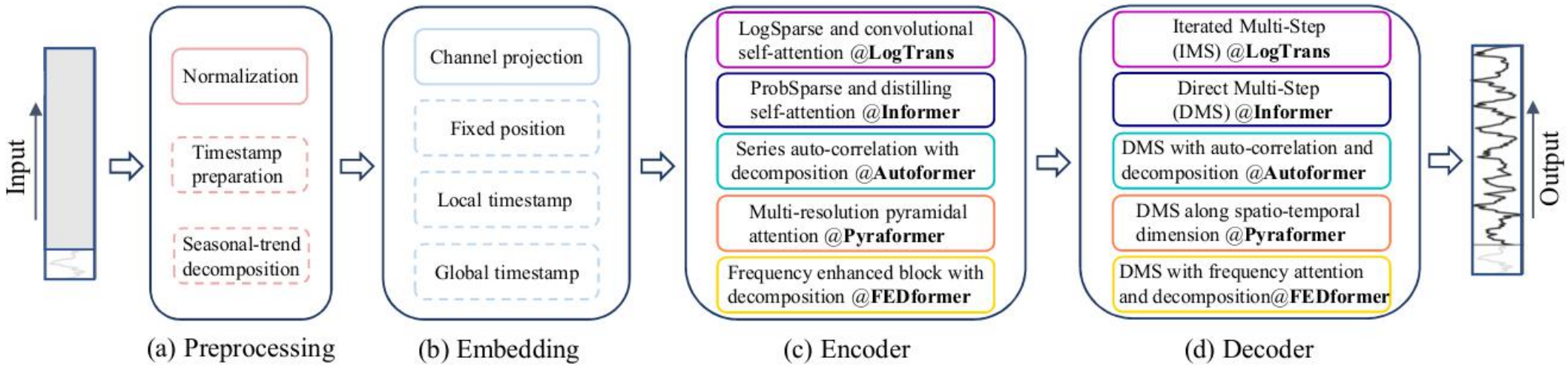


AUC = 1	完美分类器，采用这个预测模型时，存在至少一个阈值能得出完美预测。。绝大多数预测的场合，不存在完美分类器。
0.5 < AUC < 1	优于随机猜测。这个分类器（模型）妥善设定阈值的话，能有预测价值。
AUC = 0.5	跟随机猜测一样（例：丢铜板），模型没有预测价值。
AUC < 0.5	比随机猜测还差；但只要总是反预测而行，就优于随机猜测。

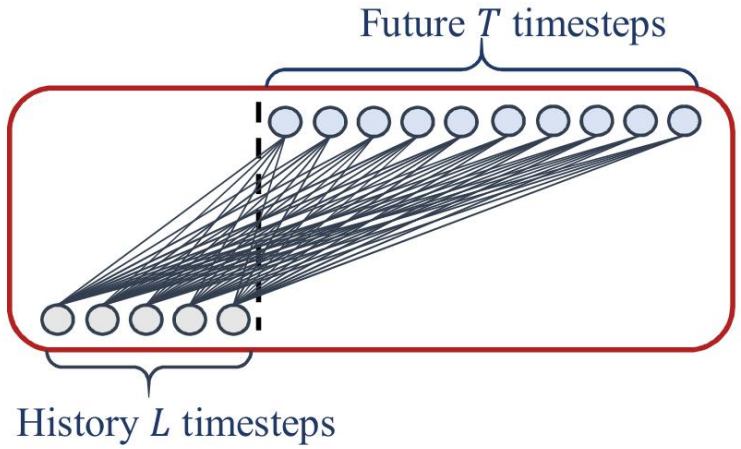
sample_width(eg:8,16,24,32):每个样本的位数(分辨率)
sample_rate（采样率）:音频中每秒钟采样的样本数。
num_frames（帧数）:音频信号中的采样帧的总数。
num_channels（声道数）:音频信号中的独立声道数量

model	算法	DLinear	[Autoformer, Informer, Transformer]
假阳性	false positive		错误地检测到阳性结果
真阴性	true negative		正确地检测到阴性结果
假阴性	false negative		错误地检测到阴性结果

2023.6.3


Input
Normalization
Timestamp preparation
Seasonal-trend decomposition
Channel projection
Fixed position
Local timestamp
Global timestamp
LogSparse and convolutional self-attention @LogTrans
ProbSparse and distilling self-attention @Informer
Series auto-correlation with decomposition @Autoformer
Multi-resolution pyramidal attention @Pyraformer
Frequency enhanced block with decomposition @FEDformer
Iterated Multi-Step (IMS) @LogTrans
Direct Multi-Step (DMS) @Informer
DMS with auto-correlation and decomposition @Autoformer
DMS along spatio-temporal dimension @Pyraformer
DMS with frequency attention and decomposition@FEDformer
Output
(a) Preprocessing
(b) Embedding
(c) Encoder
(d) Decoder

D_linear



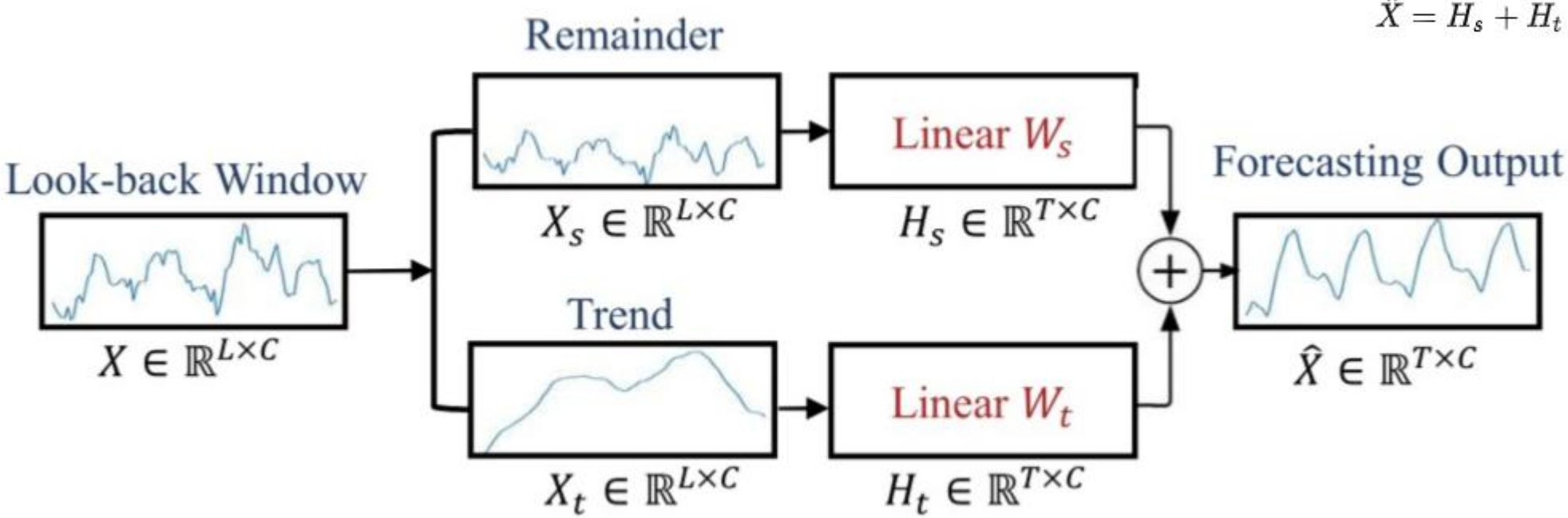
- 历史时间序列数据
- 趋势(Trend)数据
- 剩余(Remainder)数据

$$H_s = W_s X_s \in \mathbb{R}^{T\times C}, W_s \in \mathbb{R}^{T\times L}$$

$$H_t = W_t X_t \in \mathbb{R}^{T\times C}, W_t \in \mathbb{R}^{T\times L}$$

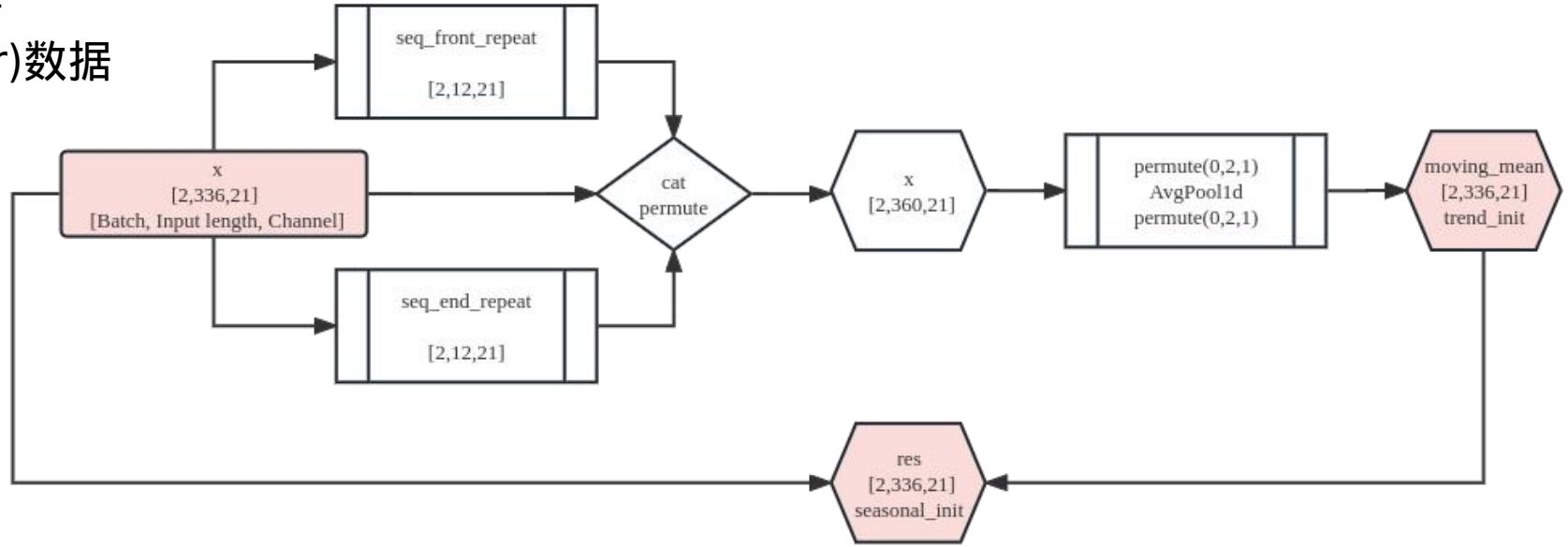
Figure 2. Illustration of the basic linear model.

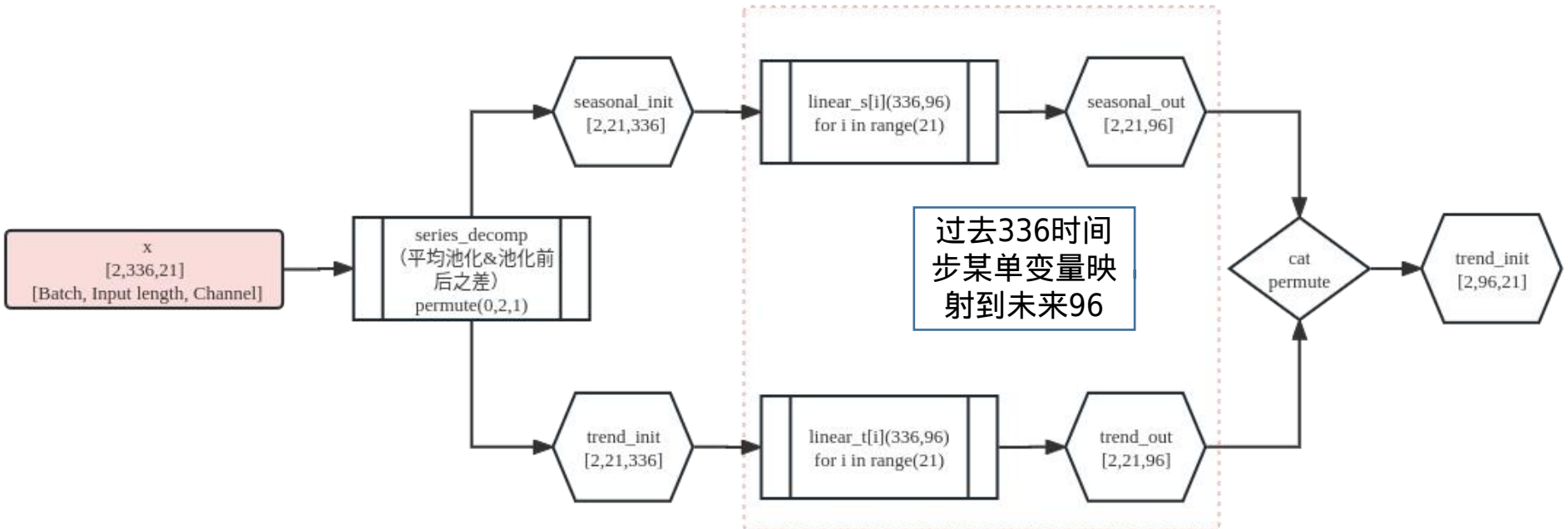
$$\hat{X} = H_s + H_t$$



(a) The whole structure of *DLinear*

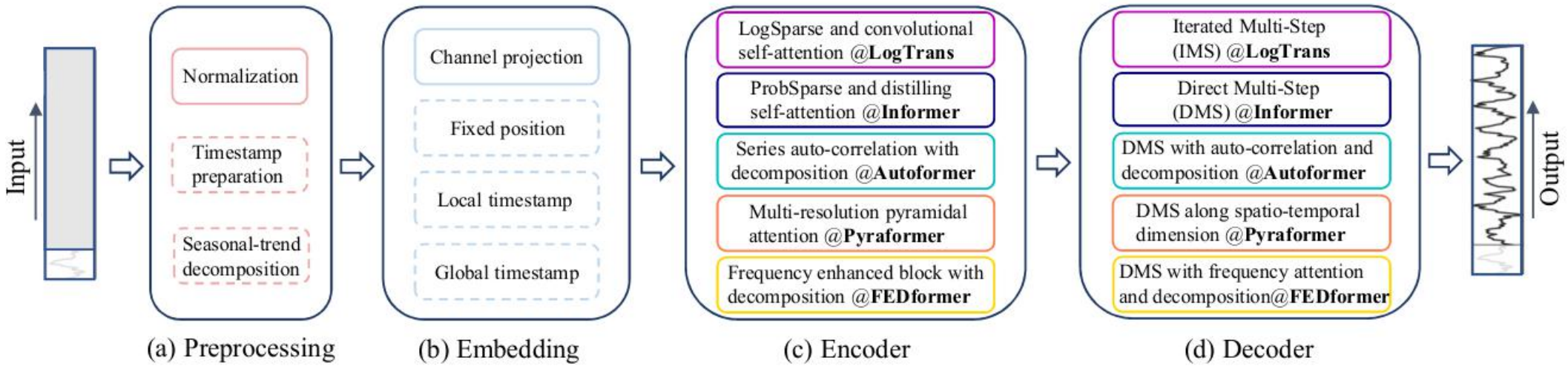
- 历史时间序列数据
- 趋势(Trend)数据
- 剩余(Remainder)数据





Methods	WA	UA
Speech-only		
LSTM+Attn (our implementation)	63.4	57.4
LSTM+Attn (Mirsamadi *et al.*, 2017)	63.5	58.8
CNN+LSTM (Satt *et al.*, 2018)	68	59.4
TDNN+LSTM (Sarma *et al.*, 2018)	70.1	60.7
Text-only (ASR text)		
LSTM+Attn	60.3	54.8
Multimodal (ASR text)		
Concat (our implementation)	68.1	66.0
Concat (Yoon *et al.*, 2018)	69.1	-
Proposed	**70.4**	**69.5**

2023.6.6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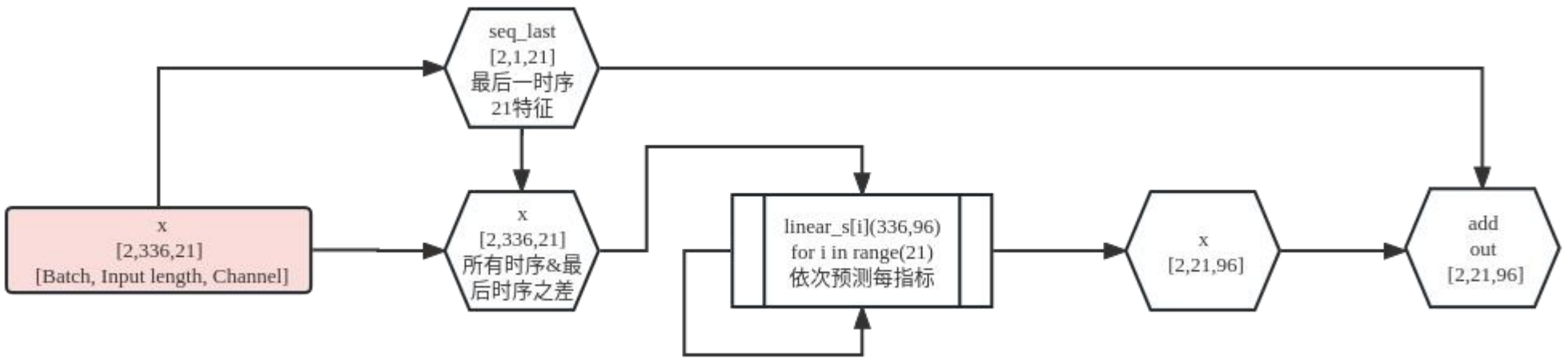
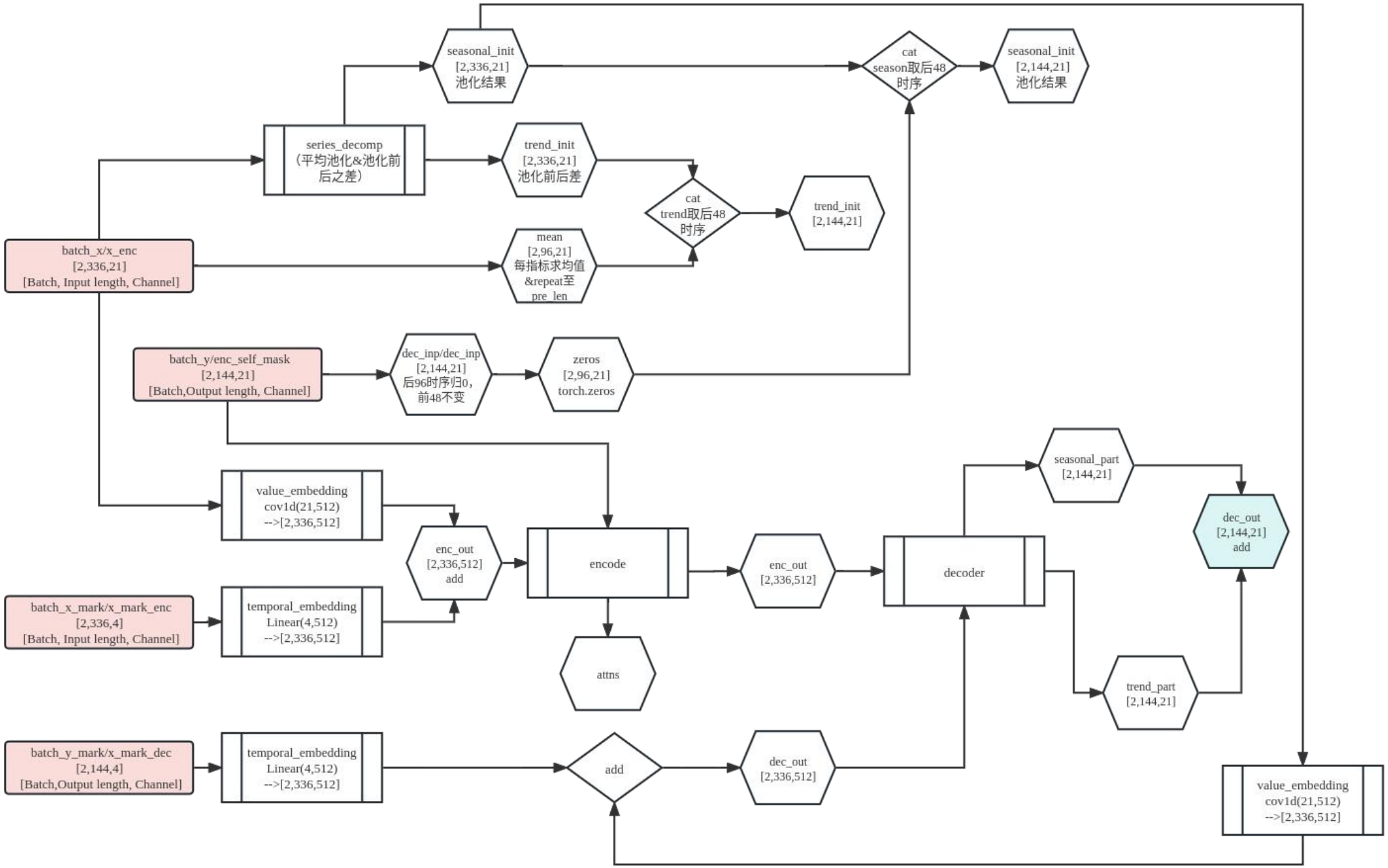
DMS with auto-correlation and decomposition @Autoformer
DMS along spatio-temporal dimension @Pyraformer
DMS with frequency attention and decomposition@FEDformer
Output
(a) Preprocessing
(b) Embedding
(c) Encoder
(d) Decoder

- NLinear将当前输入减去上一时刻的值，然后经过一个线性层，然后在做最终的预测前将减去的部分加回来。对数据的加减是对输入序列的一种简单标准化(normalizaiton)




seasonal_init
[2,336,21]
池化结果
cat
season取后48
时序
seasonal_init
[2,144,21]
池化结果
series_decomp
（平均池化&池化前
后之差）
trend_init
[2,336,21]
池化前后差
cat
trend取后48
时序
trend_init
[2,144,21]
batch_x/x_enc
[2,336,21]
[Batch, Input length, Channel]
mean
[2,96,21]
每指标求均值
&repeat至
pre_len
batch_y/enc_self_mask
[2,144,21]
[Batch,Output length, Channel]
dec_inp/dec_inp
[2,144,21]
后96时序归0，
前48不变
zeros
[2,96,21]
torch.zeros
value_embedding
cov1d(21,512)
-->[2,336,512]
enc_out
[2,336,512]
add
encode
enc_out
[2,336,512]
decoder
seasonal_part
[2,144,21]
dec_out
[2,144,21]
add
batch_x_mark/x_mark_enc
[2,336,4]
[Batch, Input length, Channel]
temporal_embedding
Linear(4,512)
-->[2,336,512]
attns
trend_part
[2,144,21]
batch_y_mark/x_mark_dec
[2,144,4]
[Batch,Output length, Channel]
temporal_embedding
Linear(4,512)
-->[2,336,512]
add
dec_out
[2,336,512]
value_embedding
cov1d(21,512)
-->[2,336,512]

2023.6.9

name	shape
audio_spec	[batch, 3, 256, 384]
audio_mfcc	[batch, 300, 40]
audio_wav	[batch, 48000]

D_linear

```
####spec+dlinear####
Ser_Model(
  (Linear_Seasonal): ModuleList(
    (0-127): 128 x Linear(in_features=768, out_features=128, bias=True)
  )
  (Linear_Trend): ModuleList(
    (0-127): 128 x Linear(in_features=768, out_features=128, bias=True)
  )
  (post_spec_dropout): Dropout(p=0.1, inplace=False)
  (decompsition): series_decomp(
    (moving_avg): moving_avg(
      (avg): AvgPool1d(kernel_size=(25,), stride=(1,), padding=(0,))
    )
  )
  (dlinear1): Linear(in_features=49152, out_features=128, bias=True)
  (dlinear2): Linear(in_features=128, out_features=4, bias=True)
)
```

```
####spec+dlinear####
Ser_Model(
  (decompsition): series_decomp(
    (moving_avg): moving_avg(
      (avg): AvgPool1d(kernel_size=(25,), stride=(1,), padding=(0,))
    )
  )
  (Linear_Seasonal): ModuleList(
    (0-39): 40 x Linear(in_features=300, out_features=16, bias=True)
  )
  (Linear_Trend): ModuleList(
    (0-39): 40 x Linear(in_features=300, out_features=16, bias=True)
  )
  (dlinear1): Linear(in_features=640, out_features=128, bias=True)
  (dlinear2): Linear(in_features=128, out_features=4, bias=True)
  (dropout): Dropout(p=0.01, inplace=False)
)
```

训练集loss	测试集loss	训练准确率	测试准确率
0.8837	1.1109	58.34%	56.91%

训练集loss	测试集loss	训练准确率	测试准确率
0.8097	1.1729	58.28%	57.09%

MFCC—Nlinear			
训练集loss	测试集loss	训练准确率	测试准确率
1.1280	1.1193	55.56%	54.94%

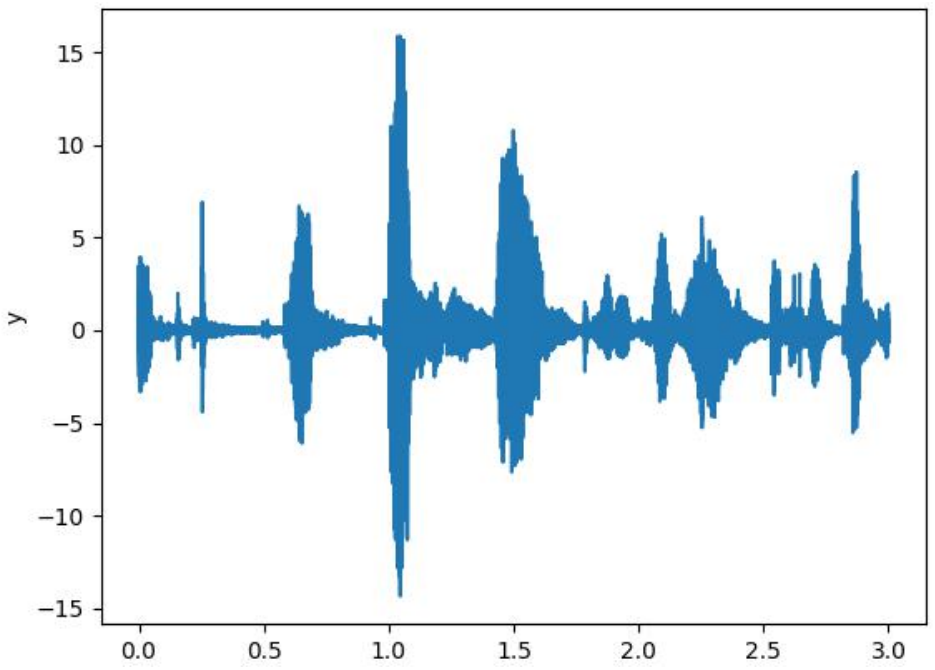
2023.6.10

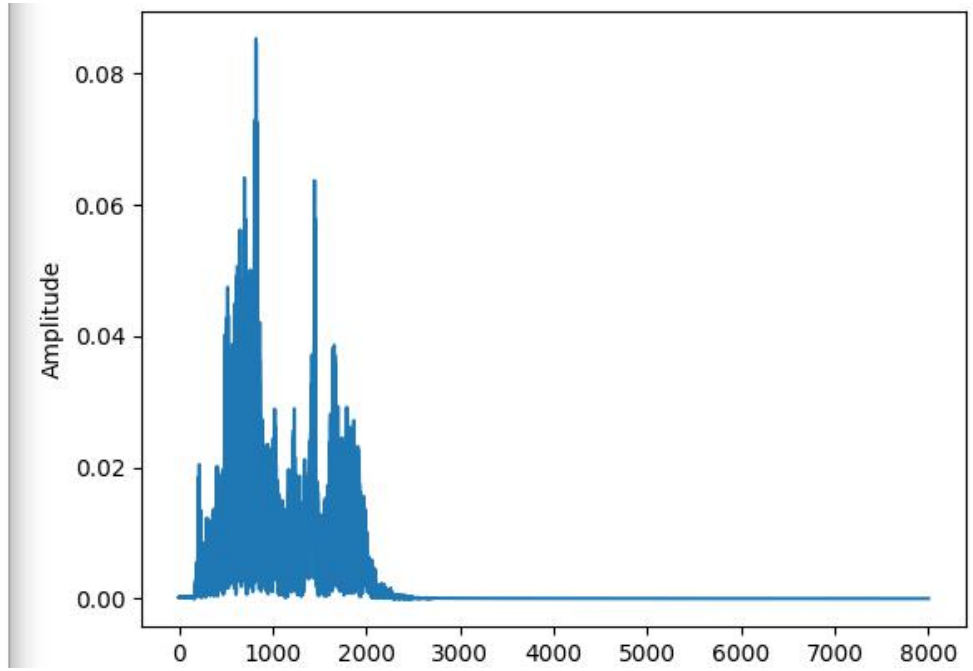
spec_Dlinear
wav2vec
cat

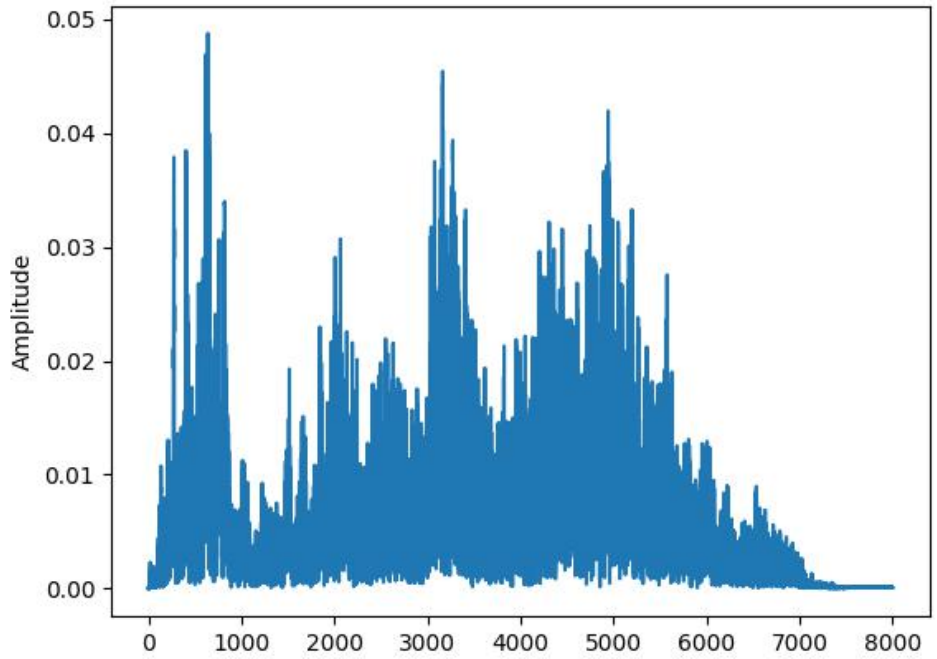
```
CA-github5_wav2vac1测试开始时间 2023-06-10 17:51:31
2023-06-10 17:51:31 v_id,t_id 为    1M1M
Number of trainable parameters: {140510340}
0   1e-05   1.3535  1.3068  39.50%  39.50%  32.95%  32.95%
1   1e-05   1.1907  1.1163  56.19%  56.19%  60.56%  60.56%
2   1e-05   0.9828  1.0227  59.07%  59.07%  64.88%  64.88%
3   1e-05   0.8792  0.9472  67.15%  67.15%  69.82%  69.82%
```

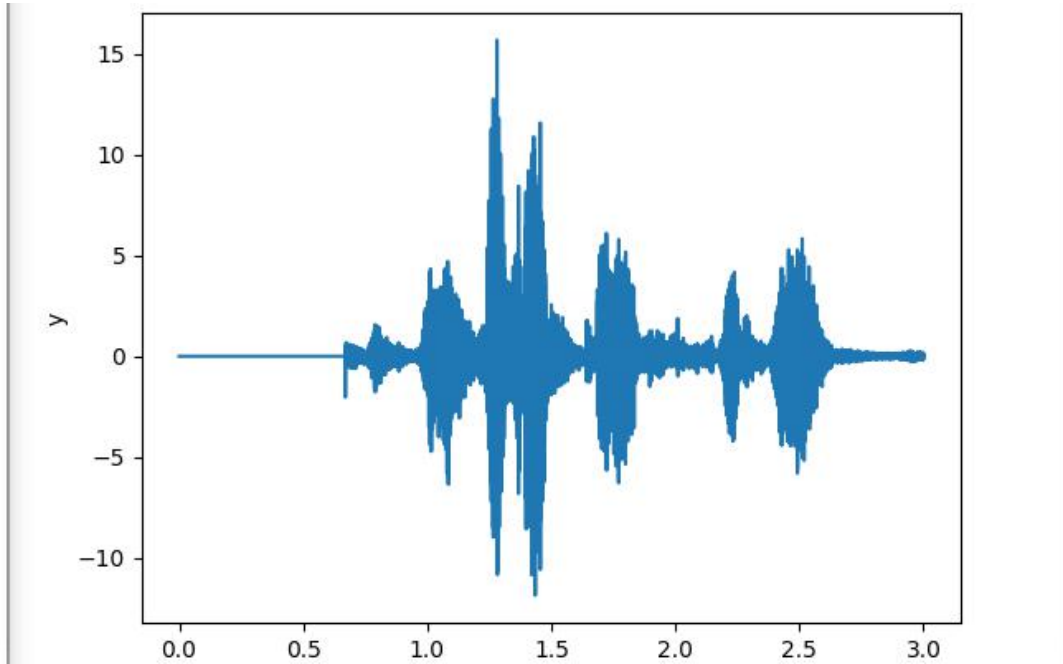
```
CA-github5_wav2vac1测试开始时间 2023-06-10 16:24:24
2023-06-10 16:24:24 v_id,t_id 为    1M1M
Number of trainable parameters: {140510340}
0   1e-05   1.3535  1.3068  39.50%  39.50%  32.95%  32.95%
1   1e-05   1.1907  1.1163  56.19%  56.19%  60.56%  60.56%
2   1e-05   0.9828  1.0227  59.07%  59.07%  64.88%  64.88%
3   1e-05   0.8792  0.9472  67.15%  67.15%  69.82%  69.82%
```

由于dropout设置较少，两次训练雷同


y
15
10
5
0
−5
−10
−15
0.0
0.5
1.0
1.5
2.0
2.5
3.0


Amplitude
0.08
0.06
0.04
0.02
0.00
0
1000
2000
3000
4000
5000
6000
7000
8000


Amplitude
0.05
0.04
0.03
0.02
0.01
0.00
0
1000
2000
3000
4000
5000
6000
7000
8000


y
15
10
5
0
−5
−10
0.0
0.5
1.0
1.5
2.0
2.5
3.0

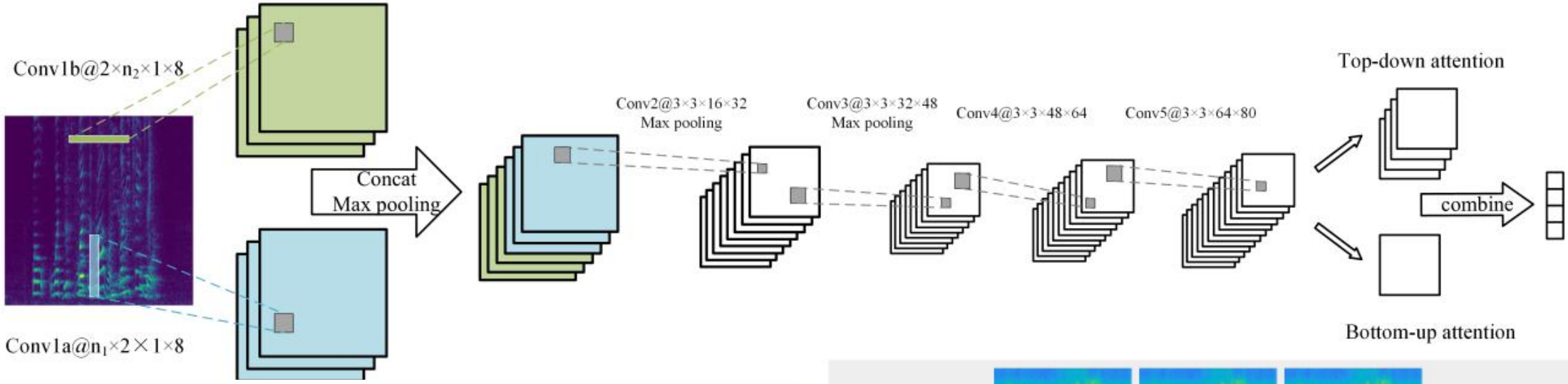
$$J^2(w,b) = J(w,b) + \frac{\lambda}{2m}\Omega(w)$$

式中， $\frac{\lambda}{2m}$ 是一个常数， m 为样本个数， λ 是一个超参数，用于控制正则化程度。

- 目的：防止模型过拟合
- 原理：在损失函数上加上某些规则（限制），缩小解空间，从而减少求出过拟合解的可能性

- 0范数，向量中非零元素的个数。
- 1范数，为绝对值之和。
- 2范数，就是通常意义上的模。
- +∞范数，即求向量元素绝对值中的最值

2023/06/20



Input: spectrogram 200x300 (4000Hz x 3 sec)

Convolution 1: 16 filters of 12x16 (240Hz x 160msec)

Max-pooling 2:1 → 100x150

Convolution 2: 24 filters of 8x12 (320Hz x 240 msec)

Max-pooling 2:1 → 50x75

Convolution 3: 32 filters of 5x7 (400Hz x 280 msec)

Max-pooling 2:1 → 25x37

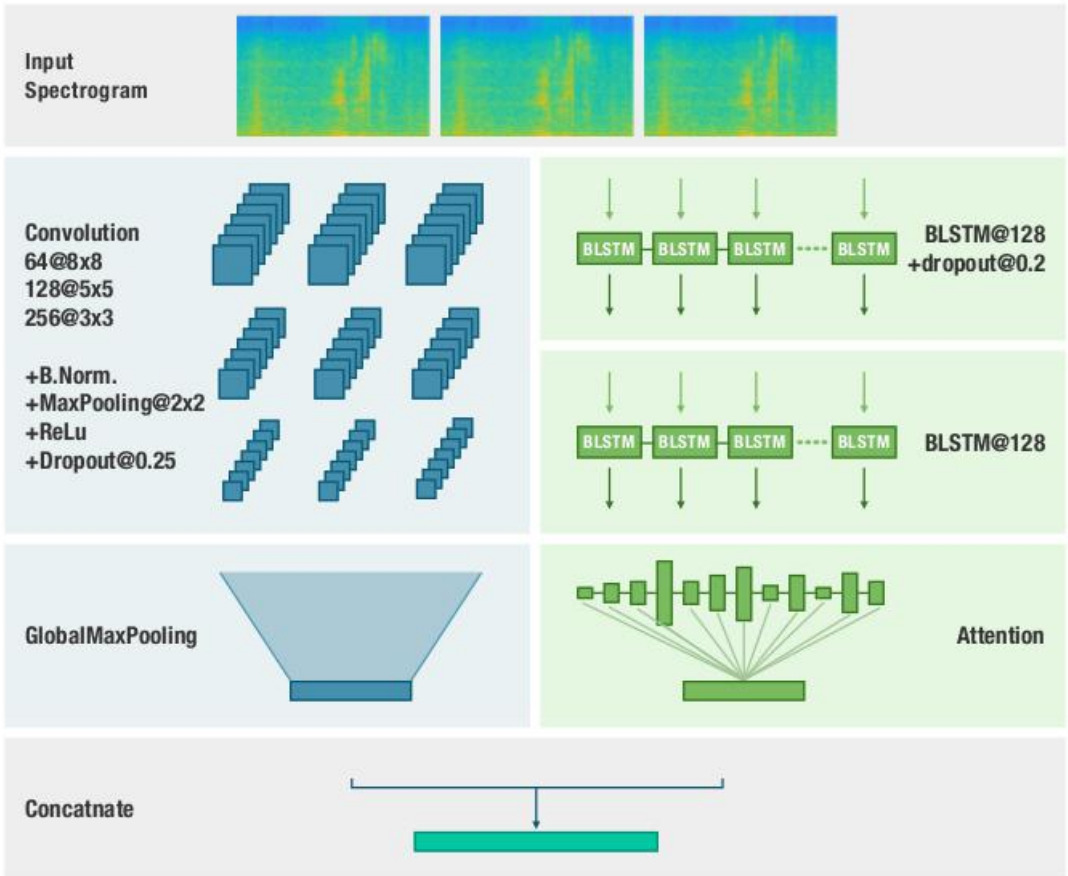
LSTM: bi-directional, 128x2

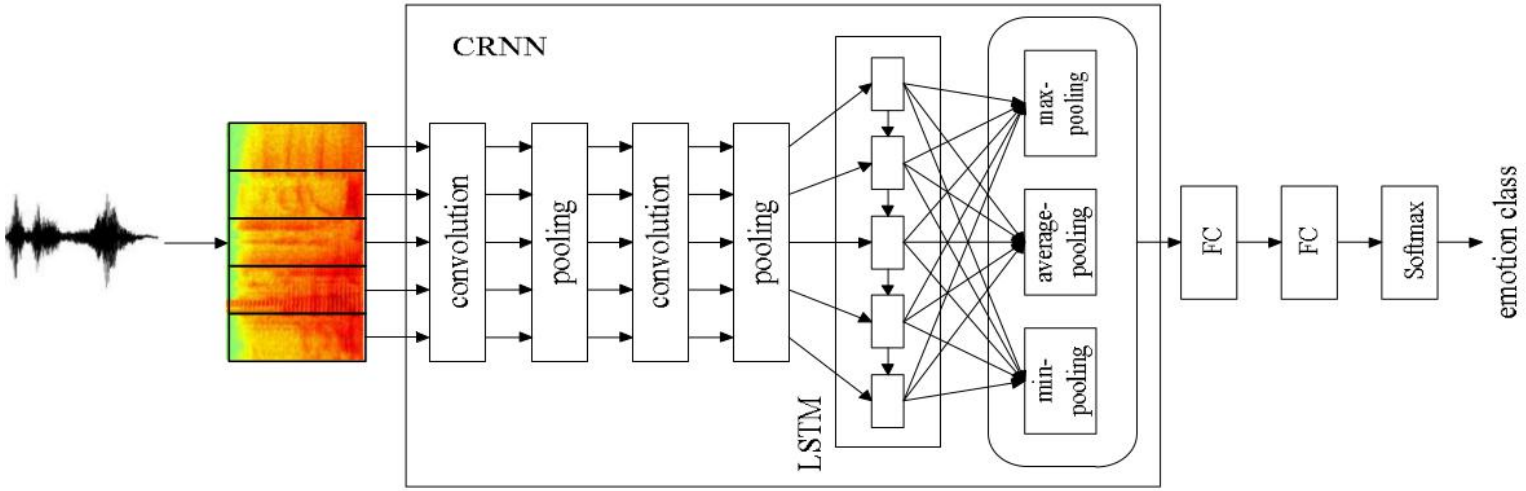
Dense layer: length 64

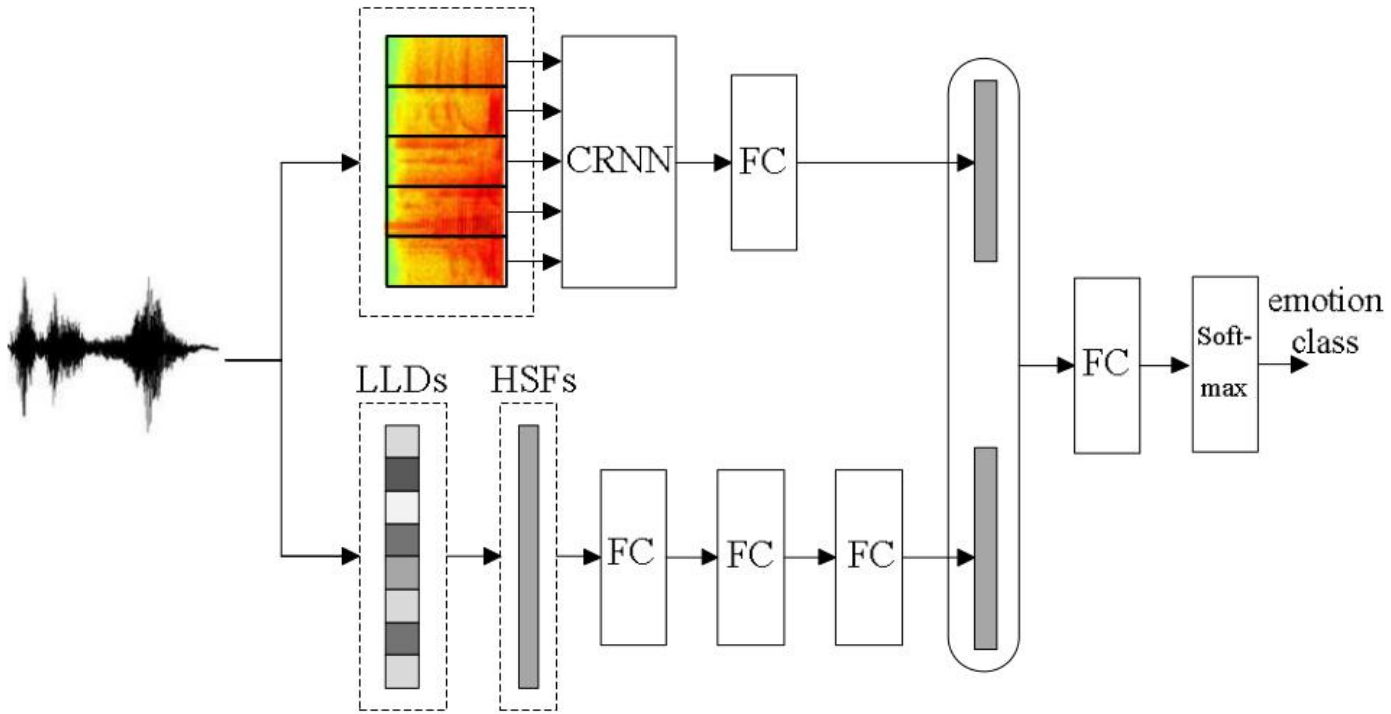
Dropout: length 64

Soft-max: length 4

Output: 4 posterior probabilities



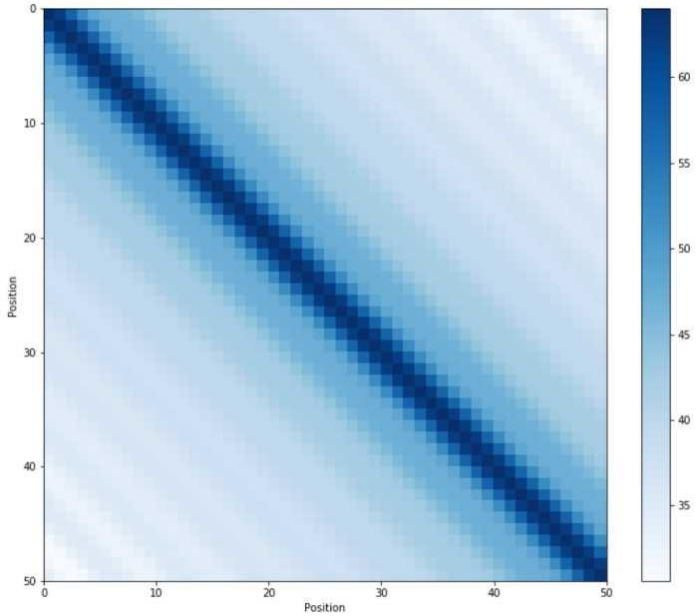

CRNN
convolution
pooling
convolution
pooling
LSTM
max-pooling
average-pooling
min-pooling
FC
FC
Softmax
emotion class

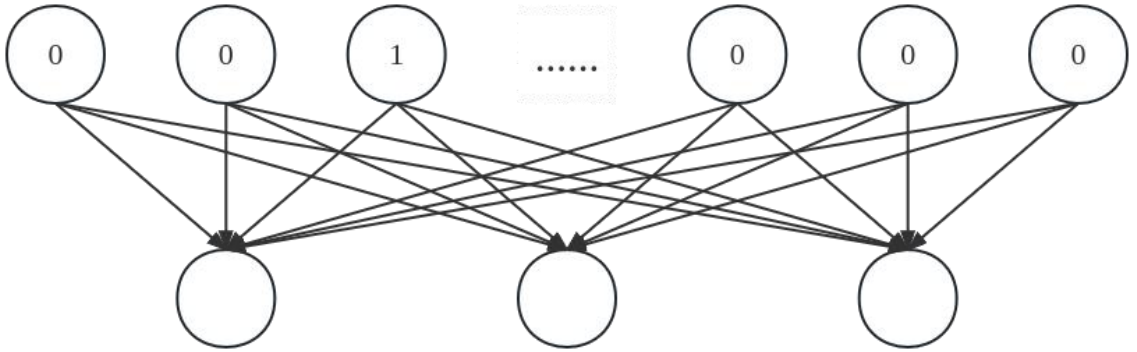

CRNN
FC
LLDs
HSFs
FC
FC
FC
FC
Soft-max
emotion class

2023/06/26

Bert

- BERT 的创建者:当使用超过 512 个令牌的文档时，性能会显着下降。





$$\overrightarrow{p_t}^{(i)} = f(t)^{(i)} := \begin{cases} \sin(\omega_k \cdot t), & \text{if } i = 2k \\ \cos(\omega_k \cdot t), & \text{if } i = 2k+1 \end{cases}$$

$$\omega_k = \frac{1}{10000^{2k/d}}$$

- t：句中该词序号
- d：词向量维度（512）
- i：嵌入的维度序号

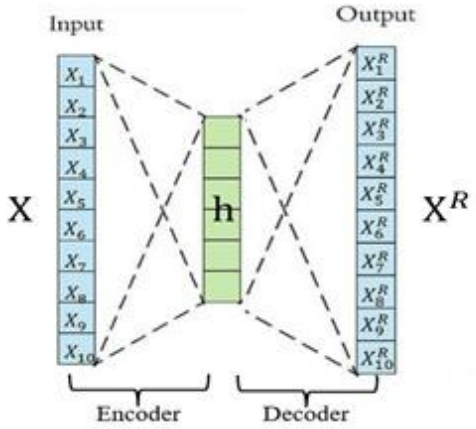
Clipping（截断法）
Pooling（池化法）
划窗法
RNN（循环法）

1		原代码		3任务B64-cu		3任务/4-b16		3任务/4-b16
2	1M	74.33%	74.01%	72.17%	73.25%	73.79%	74.37%	71.81% 74.30%
3	1F	68.37%	71.54%	68.75%	71.93%	68.56%	70.82%	68.56% 71.75%
4	2M	74.91%	78.37%	74.72%	78.64%	75.65%	80.31%	74.91% 76.83%
5	2F	78.17%	78.87%	75.26%	76.33%	75.68%	74.59%	78.59% 78.02%
6	3M	72.97%	73.14%	68.04%	68.59%	69.00%	69.89%	68.52% 68.91%
7	3F	71.84%	71.77%			70.50%	70.10%	69.54% 68.68%
8	4M	65.81%	68.06%			67.00%	68.42%	66.00% 68.24%
9	4F	72.73%	68.48%			75.00%	70.11%	73.67% 73.31%
10	5M	69.43%	70.19%			70.51%	69.60%	70.35% 70.51%
11	5F	71.86%	74.08%			73.05%	74.38%	73.73% 74.45%
12		72.04(t)	72.85(t)			71.87	72.26	71.57 72.95
13		71.64(p)	72.70(p)					

Table 1. Performance comparison with 5-fold leave-one-session-out [12,7,13] and 10-fold leave-one-speaker-out [14,15,16,17,18,5] cross-validation strategy on IEMOCAP.

Model	WA	UA
CNN-ELM+STC attention[12]	61.32	60.43
Audio_{25}[7]	60.64±1.96	61.32±2.26
IS09 - classification [13]	68.1	63.8
Ours	**69.80**	**71.05**
RNN(prop.)-ELM[14]	62.85	63.89
3D ACRNN[15]	-	64.74±5.44
BLSTM-CTC-CA[16]	69.0	67.0
CNN GRU-SeqCap[17]	72.73	59.71
CNN_TF_Att.pooling[18]	71.75	68.06
HNSD[5]	70.5	72.5
Ours	71.64	**72.70**

AutoEncoder
栈式自编码(SAE)



层数深，低层网络梯度弥散

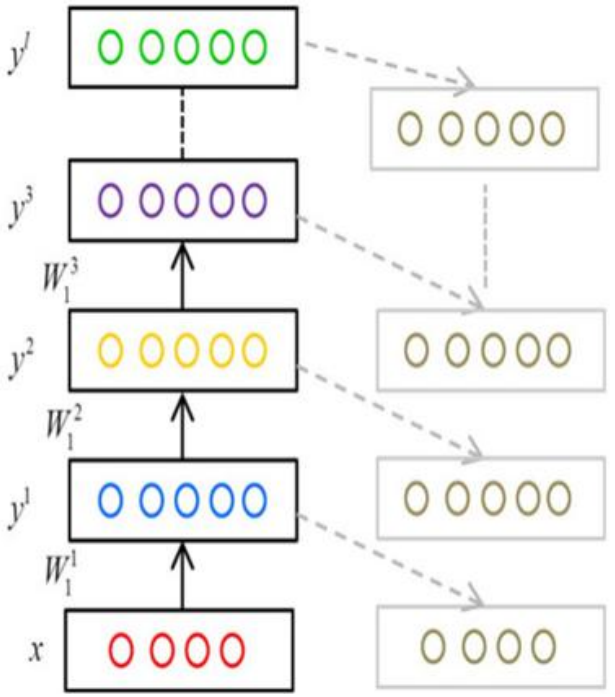
栈式自编码(SAE)

- 参数预训练，每一层进行参数初始化
- 对每层网络的进行逐层无监督训练
- 无监督训练完毕后，用有标签的数据对整个网络的参数继续进行梯度下降调整。

无监督预训练
有监督微调

```
AE(
  (encoder): Sequential(
    (0): Linear(in_features=784, out_features=256, bias=True)
    (1): ReLU()
    (2): Linear(in_features=256, out_features=20, bias=True)
    (3): ReLU()
  )
  (decoder): Sequential(
    (0): Linear(in_features=20, out_features=256, bias=True)
    (1): ReLU()
    (2): Linear(in_features=256, out_features=784, bias=True)
    (3): Sigmoid()
  )
)
```

优化目标函数 $MinimizeLoss = dist(X, X^R)$



变分自编码器（VAE）

优点：泛化性强，无监督，不需数据标注
缺点：针对异常识别场景，训练数据需要为正常数据。

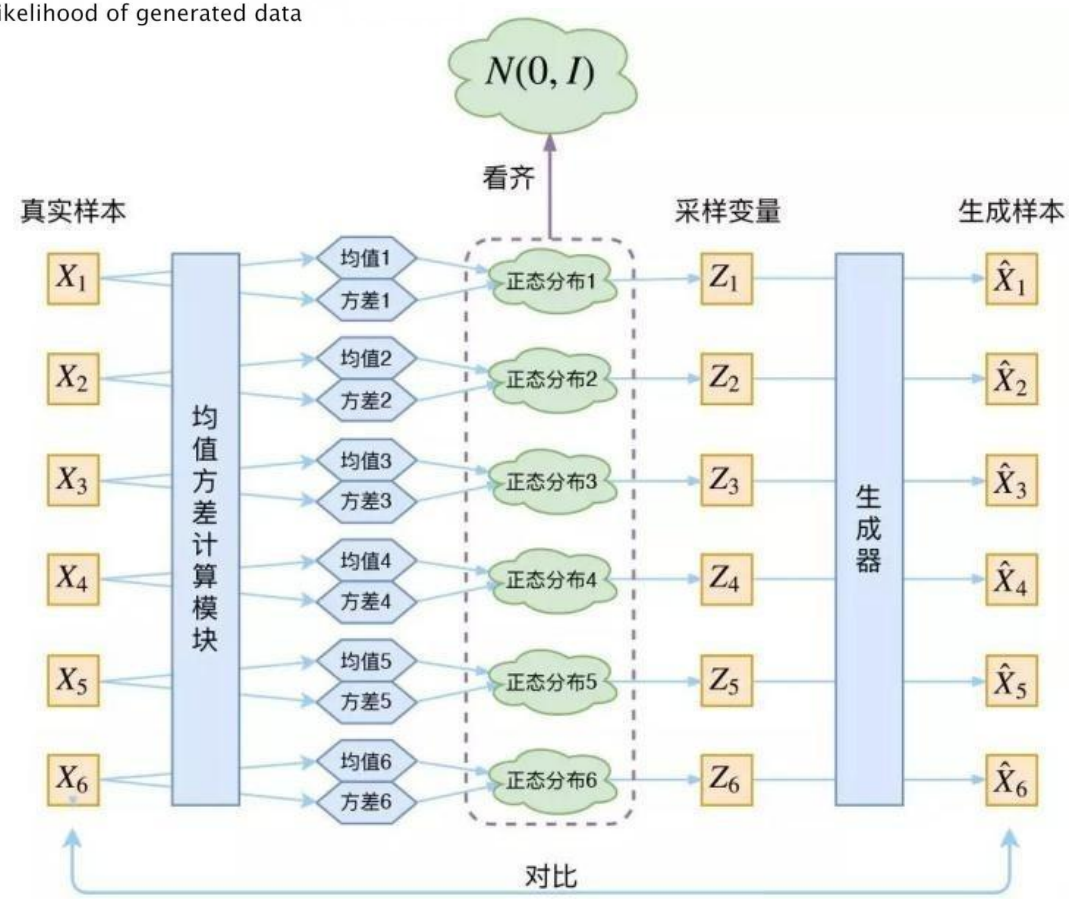
VAE

Hidden Variables

$$p(z|x) = \frac{p(x|z)p(z)}{p(x)}$$

Likelihood of generated data

P（z）就是隐层变量Z的分布又称先验分布



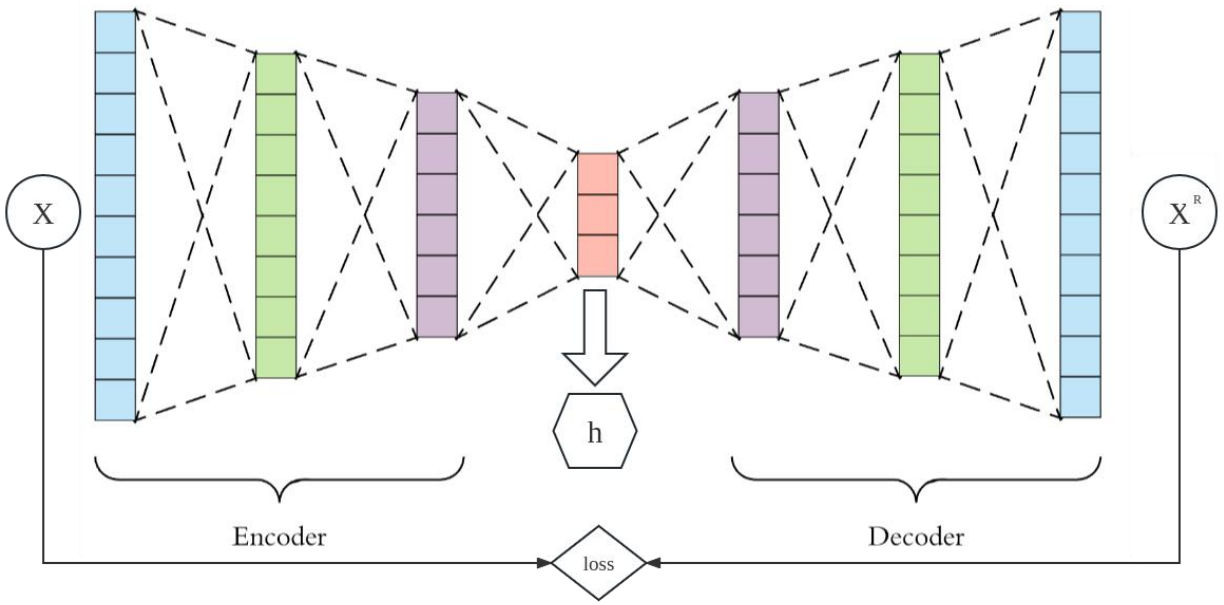
2023/06/26

- AutoEncoder 自编码器
- Variational AutoEncoder变分自编码器

AutoEncoder
自编码器

自编码是一种非线性降维方式

- 目标：使隐藏层h包含高维输入X尽可能多的重要信息。



自编码器特点

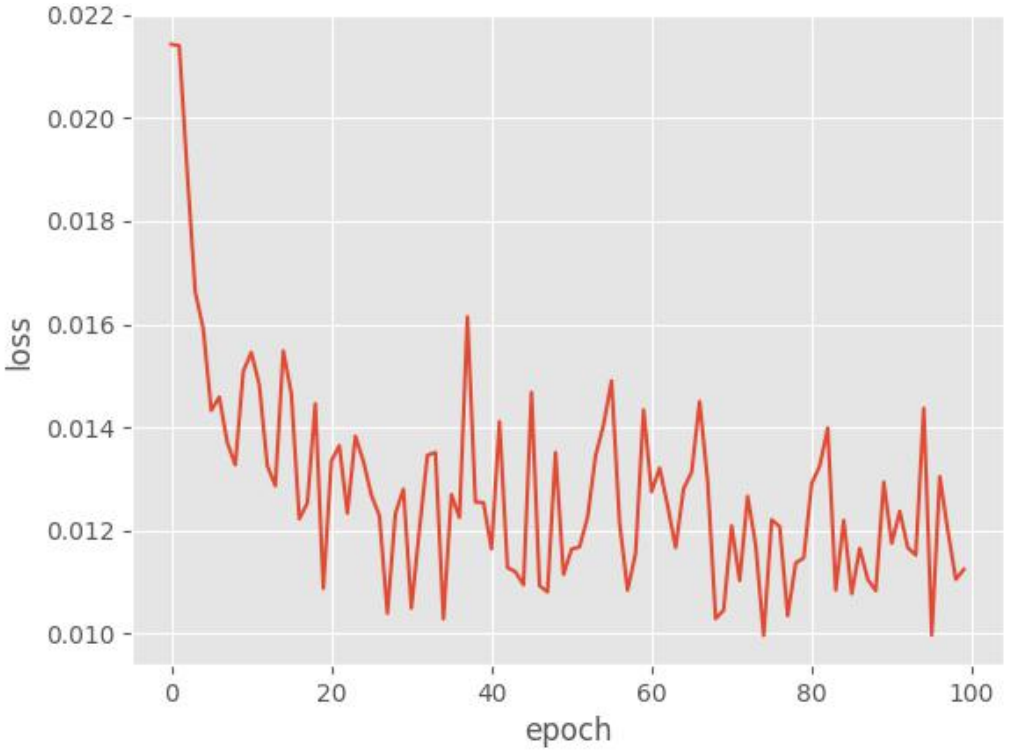
- 前馈神经网络：输出层的数据以及准确率
- 自编码器：中间隐层的结果。

层数深时低层网络梯度弥散时改进方案

- 参数预训练
- 逐层无监督训练
- 用有标签的数据调整

AutoEncoder
自编码器

- 实验：手写字母编码与重建
- 数据集：Minst
- 优化目标：原图和重建图均方误差

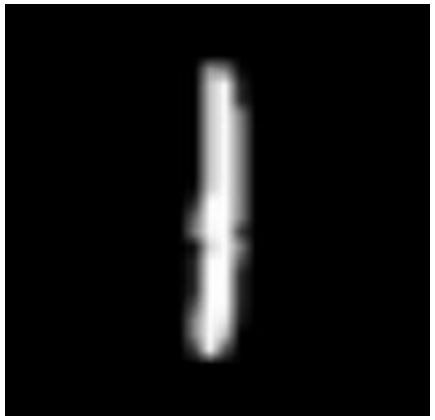


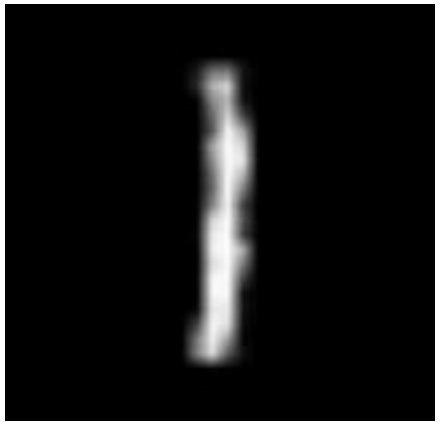
input

output

epoch 0

epoch90

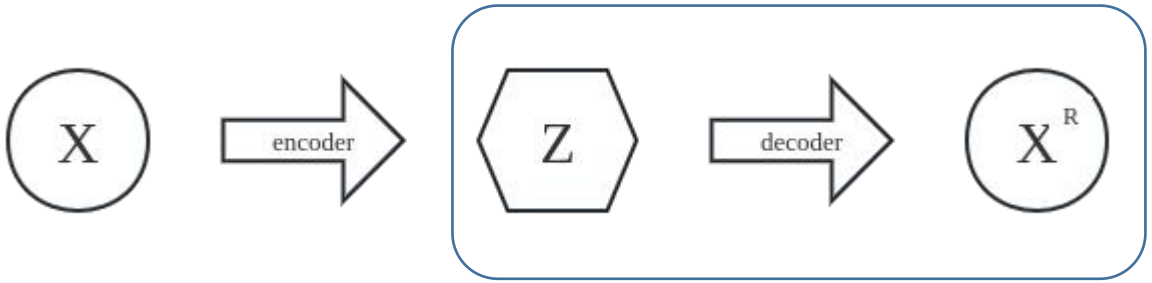


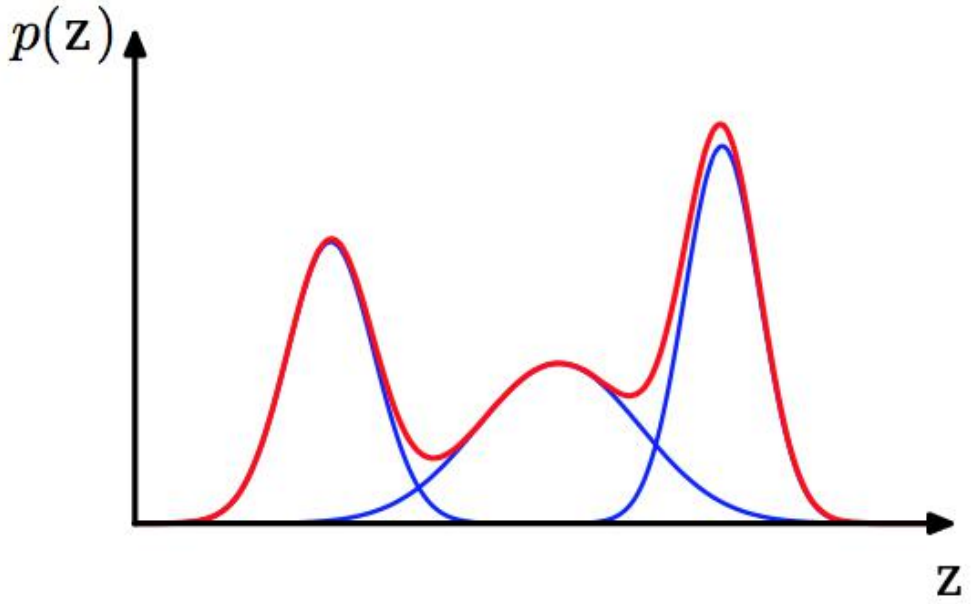


变分自编码器
VAE

主要目标：构建从隐变量 Z生成目标数据 X 的模型

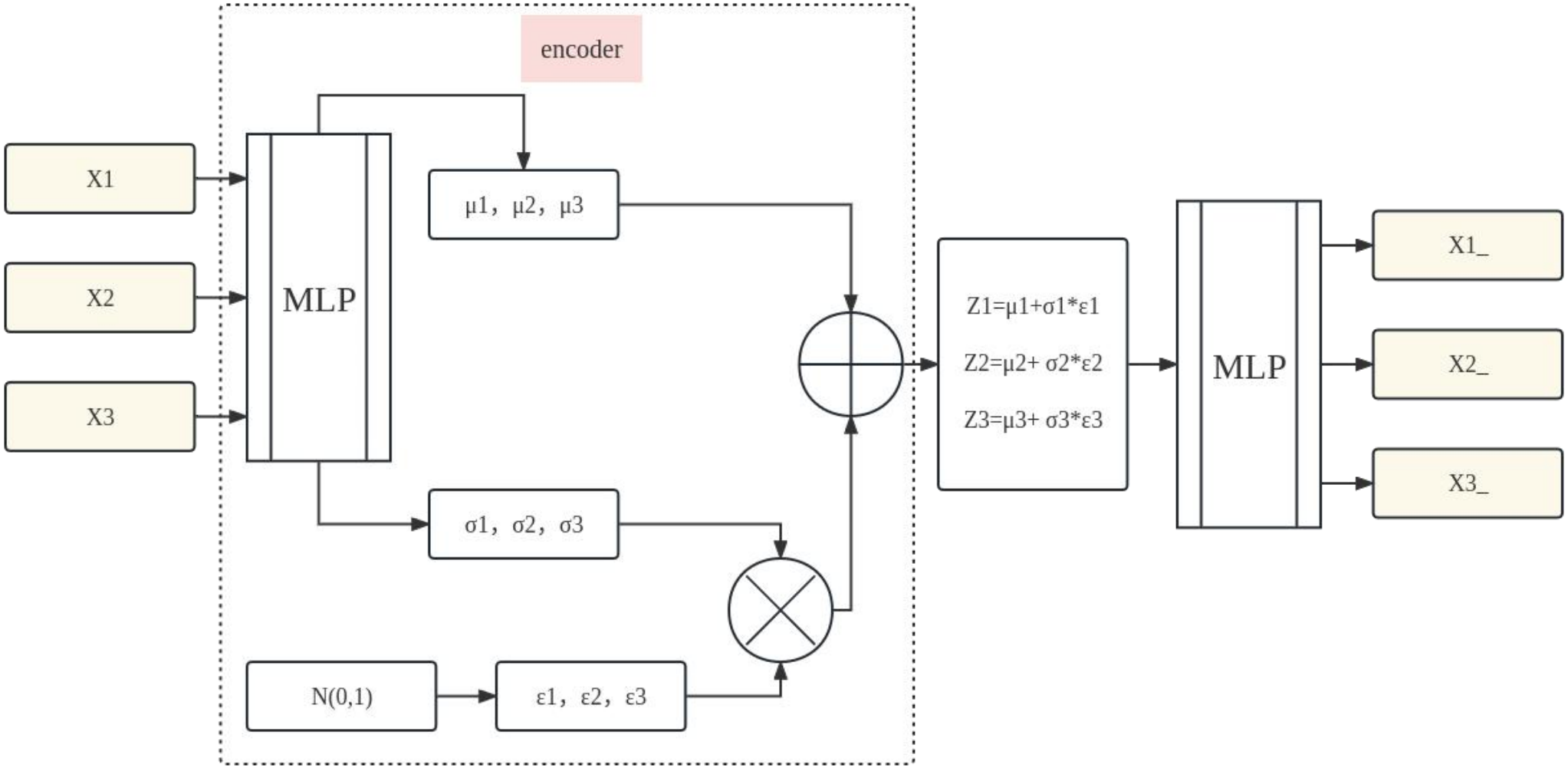
假设：样本服从某些常见的分布(eg:正态)





p(Z) ：先验分布，假设为正态分布

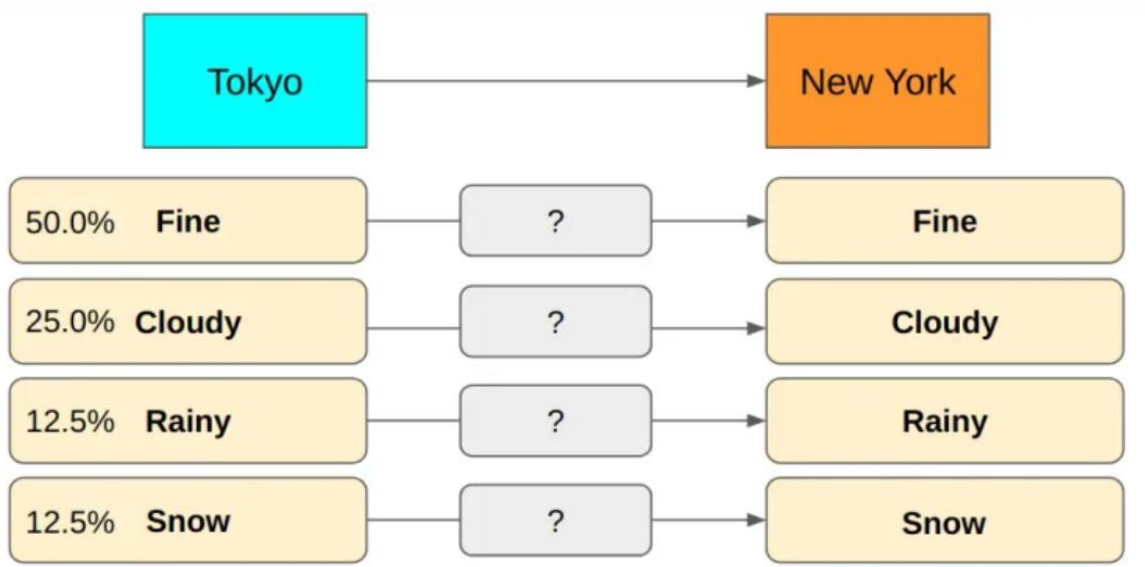
$p(X^R|Z)$:后验分布,由 Z 来生成 X的模型


encoder
X1
X2
X3
MLP
μ1, μ2, μ3
σ1, σ2, σ3
N(0,1)
ε1, ε2, ε3
Z1=μ1+σ1*ε1
Z2=μ2+ σ2*ε2
Z3=μ3+ σ3*ε3
MLP
X1_
X2_
X3_

2023/07/02

Huffman compression
哈夫曼编码

0	1	1	0	.....



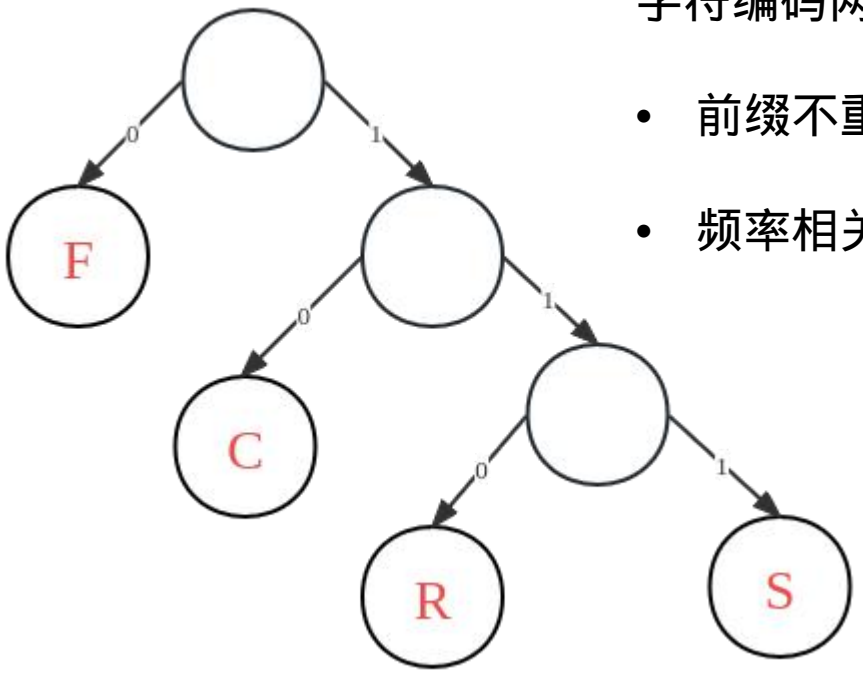
字符编码两原理：

- 前缀不重复（prefix-free）
- 频率相关

编码方式	F(50.0%)	C(25.0%)	R(12.5%)	S(12.5%)	编码长度期望
方式1	0	1	10	11	error
方式2	10	110	0	111	2*0.5+3*0.25+1*0.125+3*0.125=2.25
方式3	0	10	110	111	1*0.5+2*0.25+3*0.125+3*0.125=1.75

Huffman compression
哈夫曼编码

编码方式	F(50.0%)	C(25.0%)	R(12.5%)	S(12.5%)	编码长度期望
方式1	0	1	10	11	error
方式2	10	110	0	111	2*0.5+3*0.25+1*0.125+3*0.125=2.25
方式3	0	10	110	111	1*0.5+2*0.25+3*0.125+3*0.125=1.75



字符编码两原理：

- 前缀不重复（prefix-free）
- 频率相关

长度

F： $-\log_2 0.5=1$
C： $-\log_2 0.25=2$
R： $-\log_2 0.125=3$
S： $-\log_2 0.125=3$

信息熵就是平均最小编码长度

$$H(X) = -\sum_{i=1}^{b} P_i \log_2(P_i)$$

熵
信息熵
交叉熵

熵

- 服从某一特定概率分布事件的理论最小平均编码长度
- 熵在信息学中就是信息熵

$$H(X) = -\sum_{i=1}^{b} P_i \log_2(P_i)$$

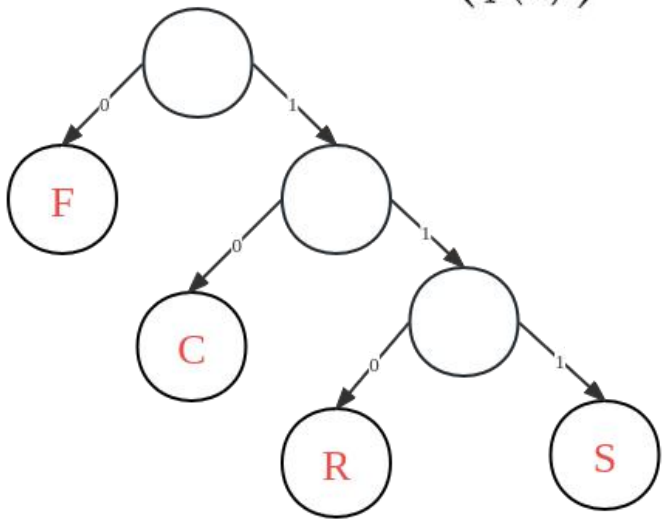
$$H(x) = -\int P(x) \log_2 P(x) dx$$

F(50.0%)	C(25.0%)	R(12.5%)	S(12.5%)

把来自一个分布q的消息使用另一个分布p的最佳代码传达的平均消息长度）称为交叉熵。

one-hot	F	C	R	S
实际值q	1	0	0	0
预测值p	1/2	1/4	1/8	1/8

$$H_p(q) = \sum_x q(x) \log_2 \left(\frac{1}{p(x)}\right) = -\sum_x q(x) \log_2 p(x)$$



用p表示q所需最佳编码平均长度

$H_p(q)=-\{1*\log_2 0.5+0*\log_2 0.25+0*\log_2 0.125+0*\log_2 0.125\}=1$

$H_q(q)=-\{1*\log_2 1+0*\log_2 0.25+0*\log_2 0.125+0*\log_2 0.125\}=0$

$H_q(p)=\infty$

KL散度

one-hot	F	C	R	S
实际值p	1	0	0	0
预测值p	1/2	1/4	1/8	1/8

实际值p最小编码长度

H(p) = -1*$\log_2 1$+.....=0
用预测值q编码实际值p期望长度
$H_q(p) = 1$

KL散度：熵和交叉熵之间的差

$$D_q(p) = H_q(p) - H(p)$$

KL散度：预测值为了表示实际值
编码长度期望变长了多少

设 $I = \int_{-\infty}^{+\infty} e^{-x^2} dx$，那么

$$I^2 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x+y)^2} dxdy$$

作极**坐标变换** $r^2 = x^2 + y^2$，$dxdy = rdrd\theta$：

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} rdrd\theta = \frac{1}{2} \int_0^{2\pi} d\theta \int_0^{\infty} e^{-r^2} dr^2 = \pi$$

所以

$$I = \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

实际值p信息熵

$H(p) = -\int p\log_2 p dx$

预测值q信息熵

$H(q) = -\int q\log_2 q dx$

预测值q编码实际值p期望长度

$H(p||q) = -\int p\log_2 q dx$

KL散度(q编码p - p编码p)

$KL(p||q) = H(p||q)-H(p)$

$= -\int p\log_2 q/p dx$

p,q均为高斯分布

$p\sim N(\mu_1, \delta_1)$, $q\sim N(\mu_2, \delta_2)$

$\int \exp(-x^2)dx = \pi^{1/2}$

$\int x^2\exp(-x^2)dx = \pi^{1/2}/2$

$H(p(x))=\int p(x)\log_2 p(x)dx$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$KL(p,q) = -\int p(x)\log q(x)dx + \int p(x)\log p(x)dx$$

$$= \frac{1}{2}\log(2\pi\sigma_2^2) + \frac{\sigma_1^2 + (\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}(1+\log 2\pi\sigma_1^2)$$

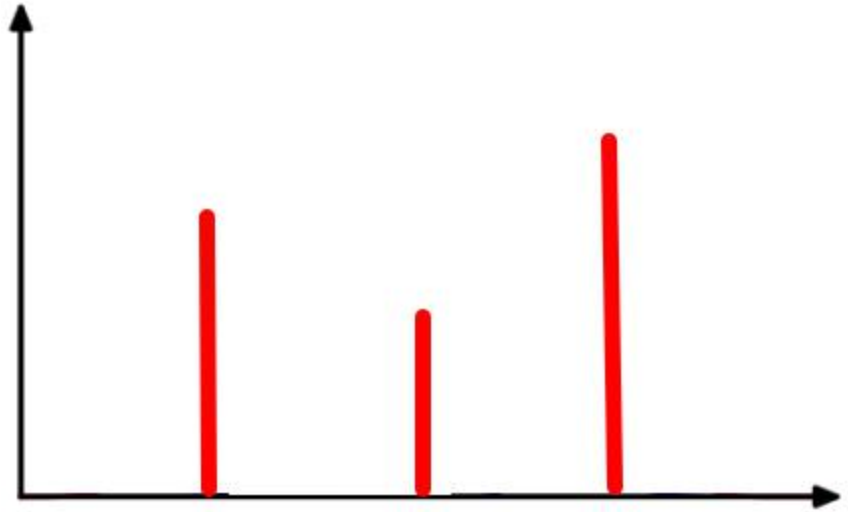
$$= \log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

$q\sim N(0,1)$ $p(\mu, \delta)$

$KL(p,q) = -\log\delta + \delta^2/2 + \mu^2/2 - 1/2$

```
# D_KL(Q(z|X) || P(z)); calculate in closed form as both dist. are Gaussian
# here we assume that \Sigma is a diagonal matrix, so as to simplify the computation
KLD = 0.5 * torch.sum(  torch.exp(log_var)   +   torch.pow(mu, 2)   - 1.    - log_var)
# 0.5*{ e^data  +  mu^2    -1    -      data}
# 0.5*data + 0.5*e^data  + 0.5*mu^2 -0.5
```

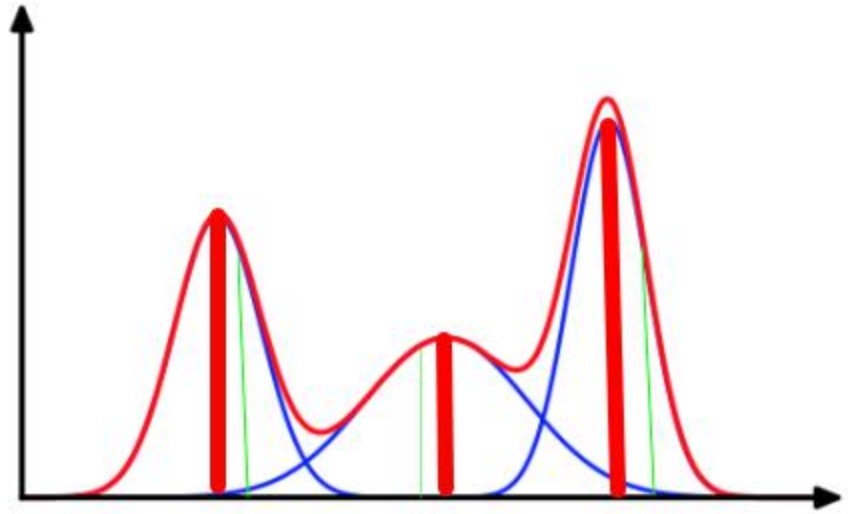
变分自编码器
VAE



目标1：构建从隐变量 Z生成目标数据 X 的模型

Loss1 =BCE(X , X^R)

X和X^R的二值交叉熵



变分自编码器
VAE

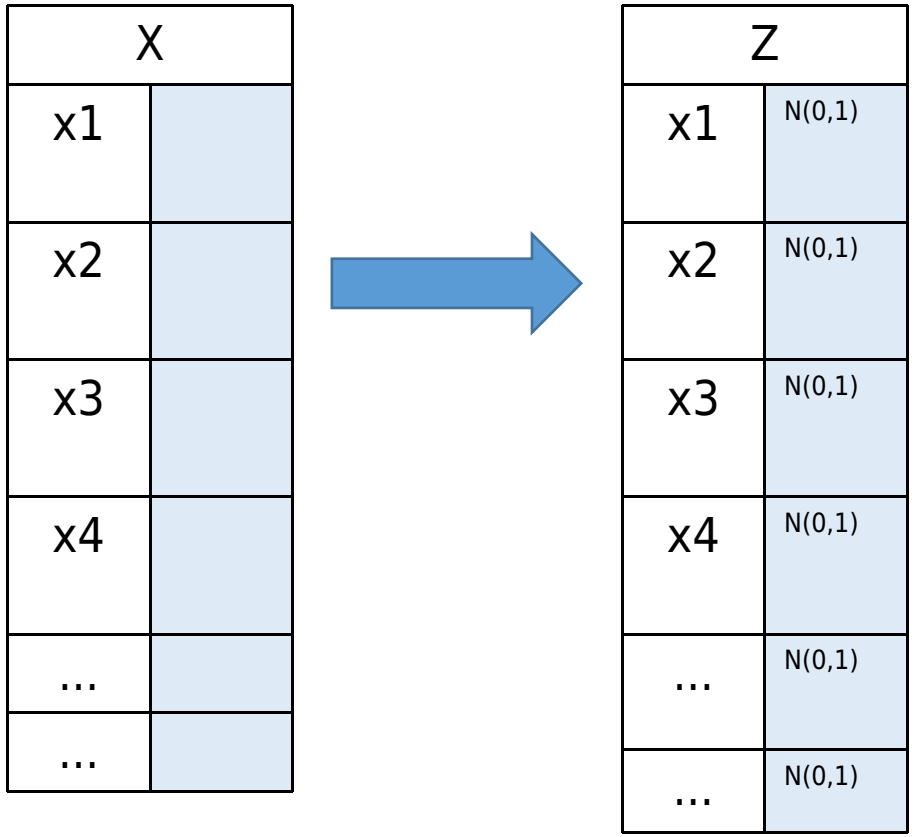
主要目标：构建从隐变量 Z生成目标数据 X 的模型

假设：样本服从某些常见的分布(eg:正态)

任意模型均可映射为高维标准正态分布

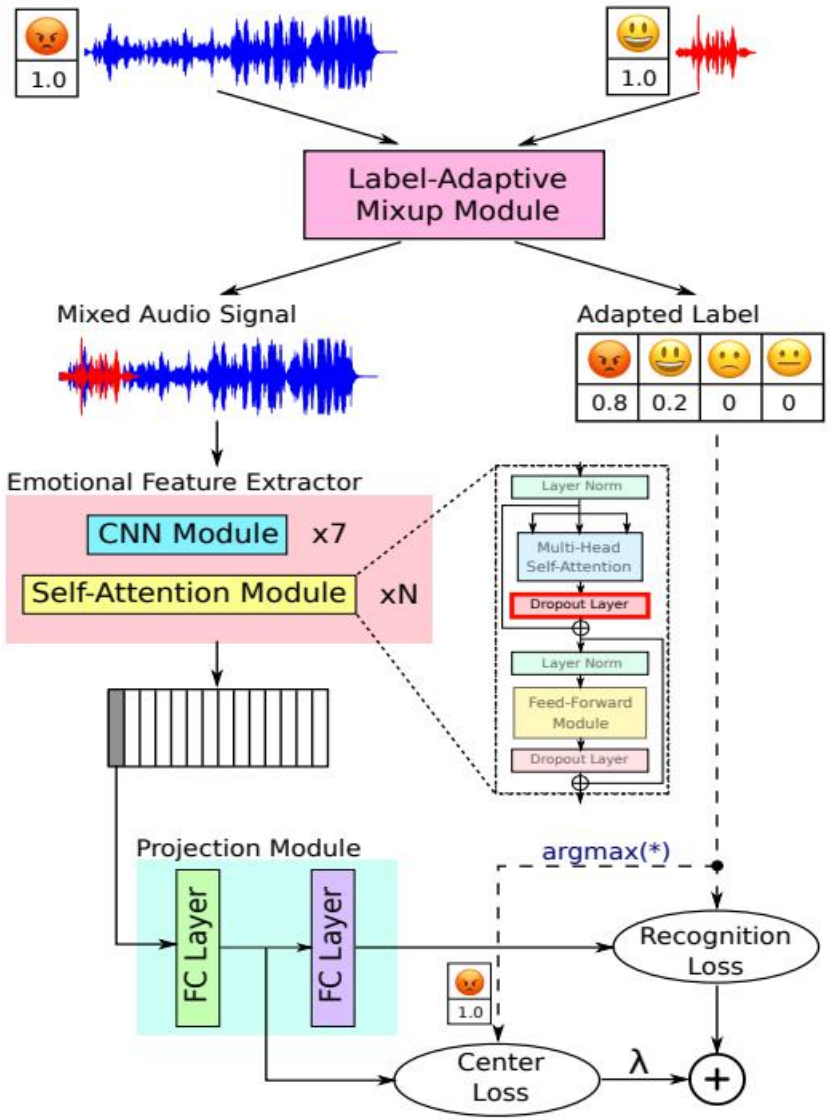
p(Z) ：先验分布，假设为标准正态分布

p(X|Z) :后验分布,由 Z 来生成 X的模型



2023/07/10

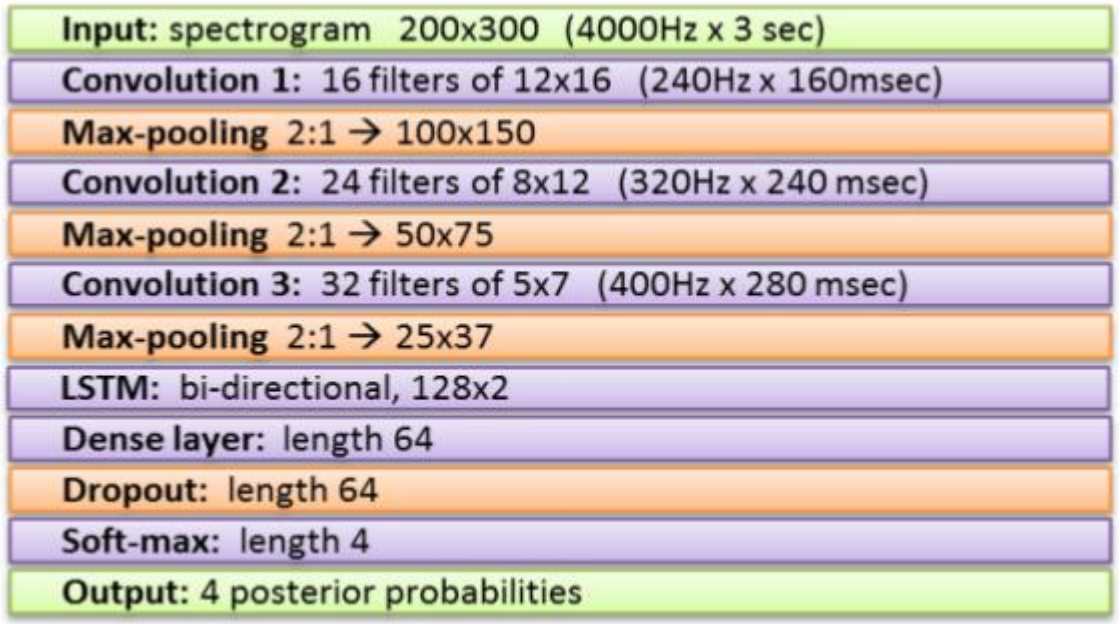
论文关键点
标签自适应混合方法
Label-Adaptive
Mixup

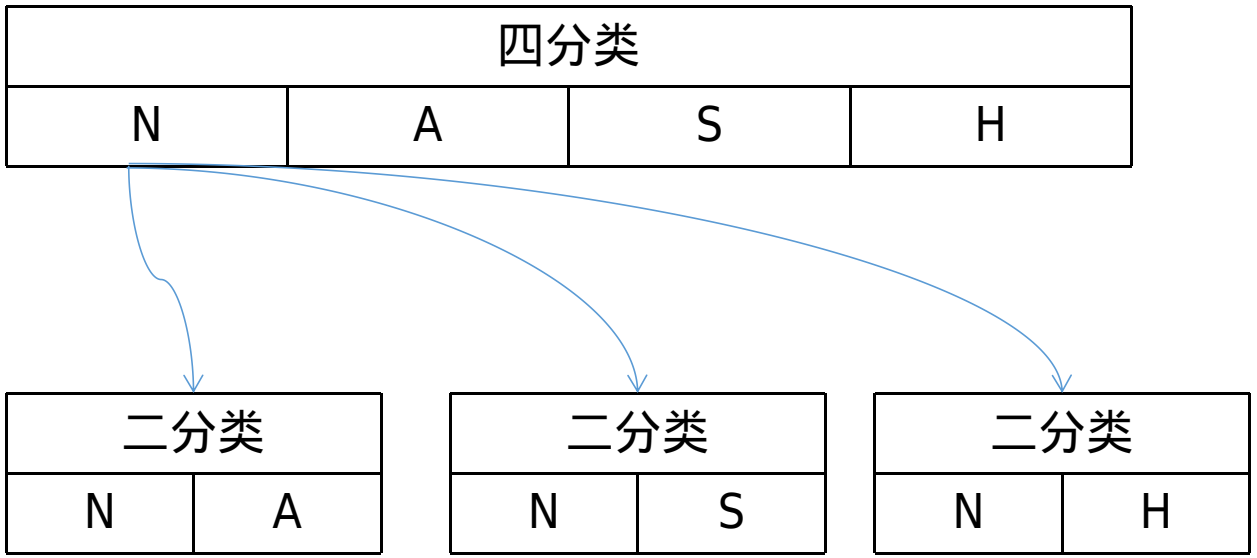


- 修改hu-bert
  提出标签自适应混合方法
  中心损失+识别损失

Method	Year	WA (%)	UA(%)
Human Performance [4]	2017	69.00	70.00
TDNN-LSTM-attn *et al.* [6]	2018	70.10	60.70
LSTM *et al.* [19]	2019	56.99	53.07
IS09-classification *et al.* [7]	2019	64.33	64.79
CNN-GRU-SeqCap *et al.* [20]	2019	72.73	59.71
HGFM *et al.* [21]	2020	66.60	70.50
ACNN *et al.* [5]	2020	67.28	67.94
ASR-SER *et al.* [22]	2020	68.60	69.70
Lightweight model *et al.* [1]	2020	70.39	71.72
SSL&CMKT fusion *et al.* [23]	2021	61.16	62.50
$\text{Audio}_{25,250}$+BERT *et al.* [2]	2021	69.44	70.90
Selective MTL *et al.* [24]	2022	56.87	59.47
MFCC+Spectrogram+W2E *et al.* [18]	2022	69.80	71.05
CNN-SeqCap *et al.* [3]	2022	70.54	56.94
Proposed	**2023**	**75.37**	**76.04**

加噪声(现实意义的)
多次分类





	Network	Overall accuracy	Class accuracy
1.	5 convolution layers 10Hz grid resolution	66.1%	56.6%
2.	5 convolution layers 20 Hz grid resolution	63.6%	53.4%
3.	3 convolution layers and LSTM 10 Hz grid resolution	**68.8%**	**59.4%**
4.	3 convolution layers and LSTM 20 Hz grid resolution	65.8%	56.6%
5.	Two-step predictor (refer to explanation below) 10 Hz grid resolution	**67.3%**	**62.0%**

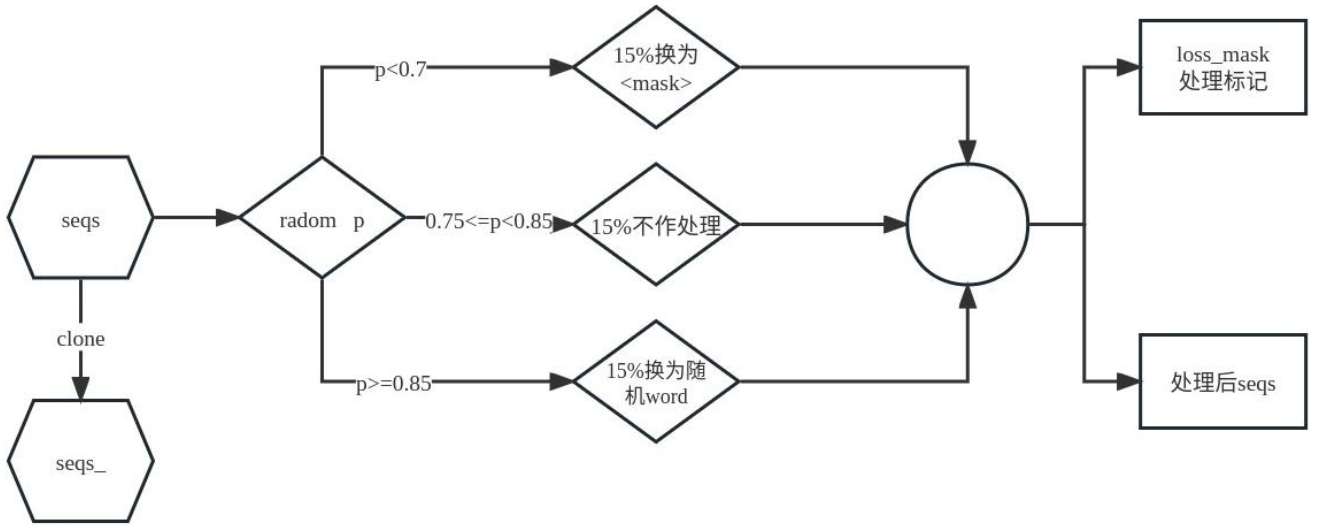
流形算法

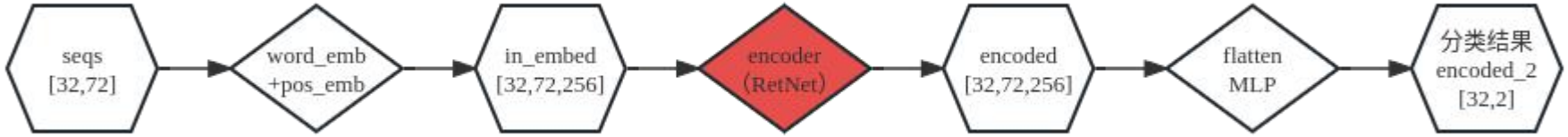
流形学习（manifold learning）假设数据在高维空间的分布位于某一更低维的流形

2023/10/20

Retnet

batch	list 1	tensor(32,72)	32样本72字符在词汇总表索引	seqs
	list 2	tensor(32,72)	前后句信息，0前句，1后句，2空符	segs
	list 3	tensor(32,2)	一行两句话各自长度	xlen
	list 4	tensor(32,1)	标签 array([0]) 0or1	nsp_labels



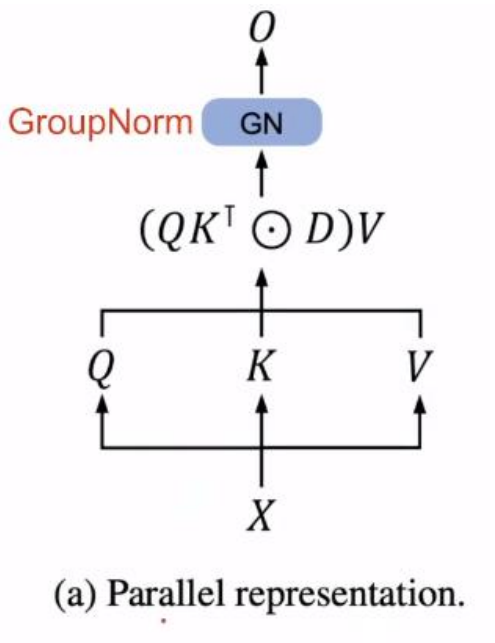


$$o_n = \sum_{m=1}^{n} \gamma^{n-m} (Q_n e^{in\theta})(K_m e^{im\theta})^{\dagger} v_m$$

$$Q = (XW_Q) \odot \Theta, \quad K = (XW_K) \odot \overline{\Theta}, \quad V = XW_V$$

$$\Theta_n = e^{in\theta}, \quad D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ 0, & n < m \end{cases}$$

$$\text{Retention}(X) = (QK^{\intercal} \odot D)V$$



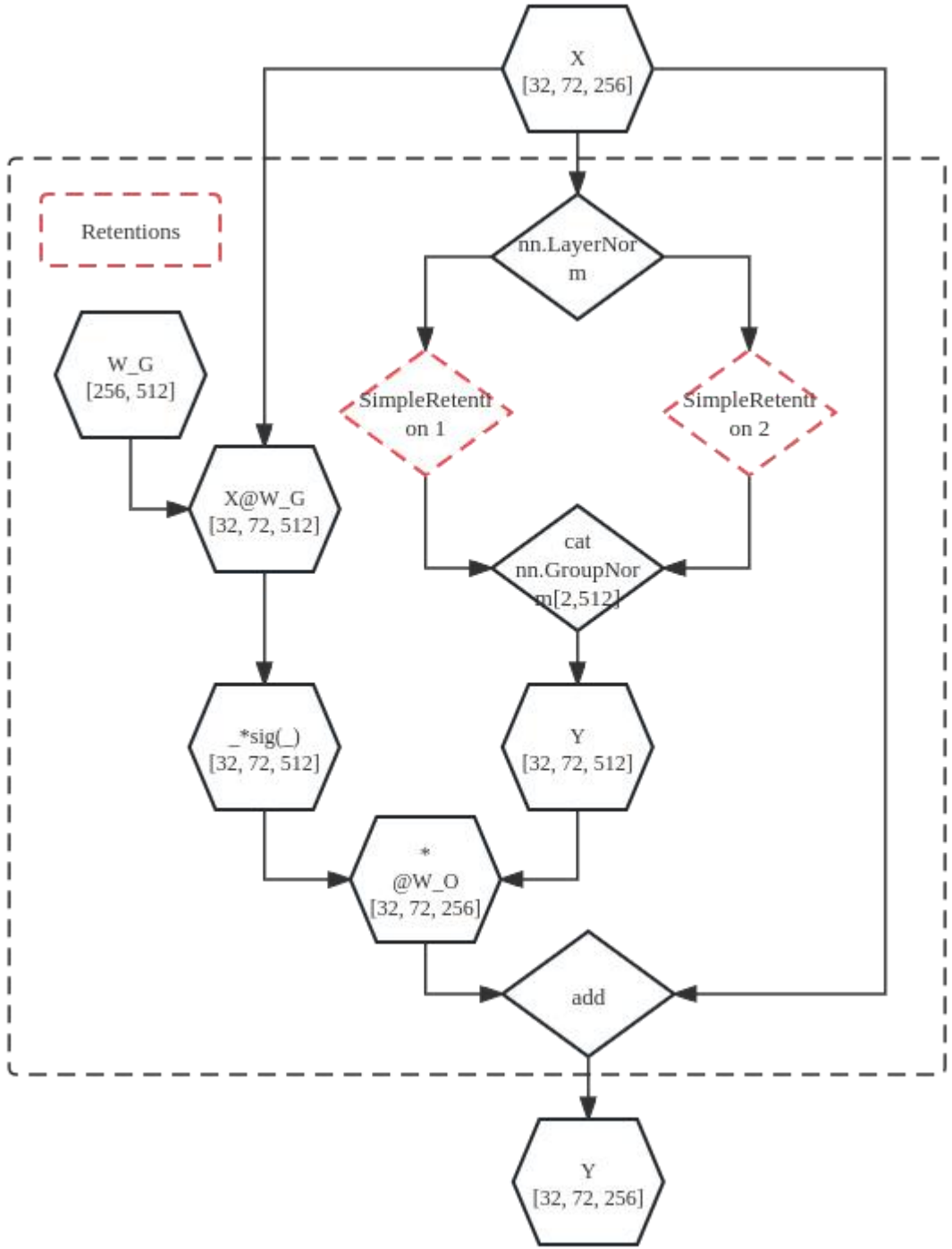
(a) Parallel representation.

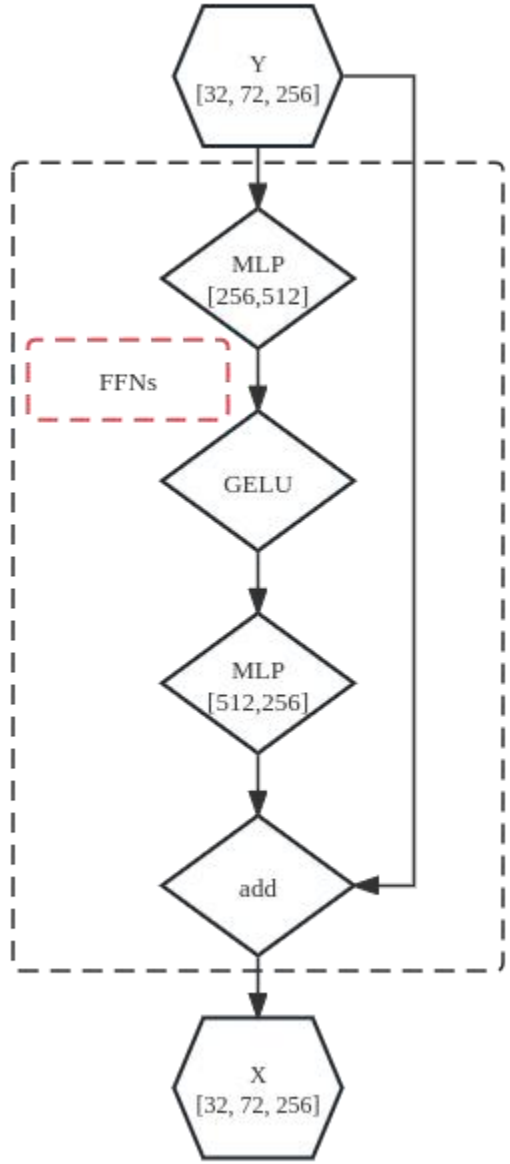
Retnet

Retentions

FFNs

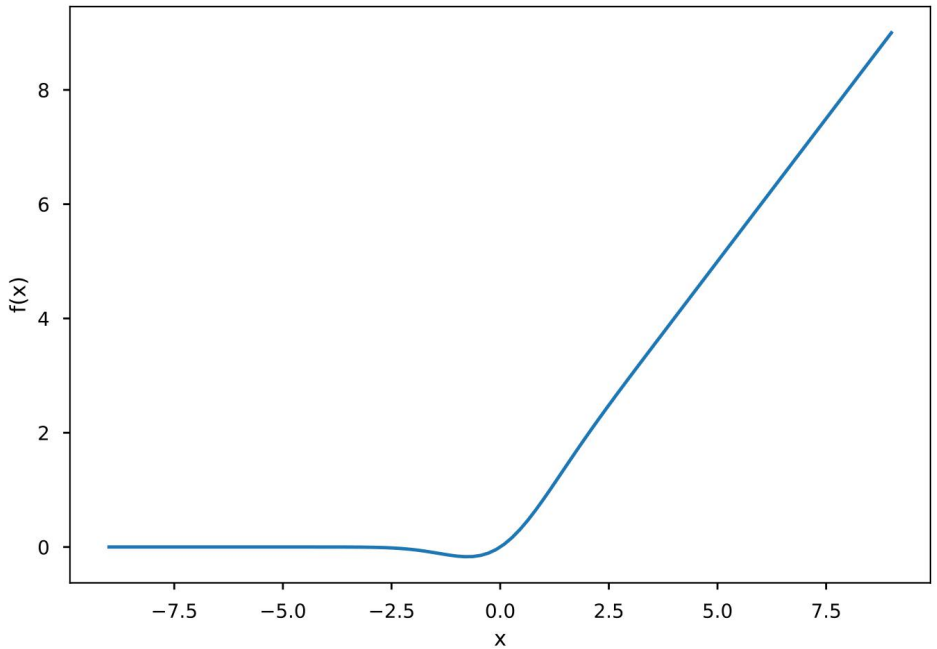
× layers (3)

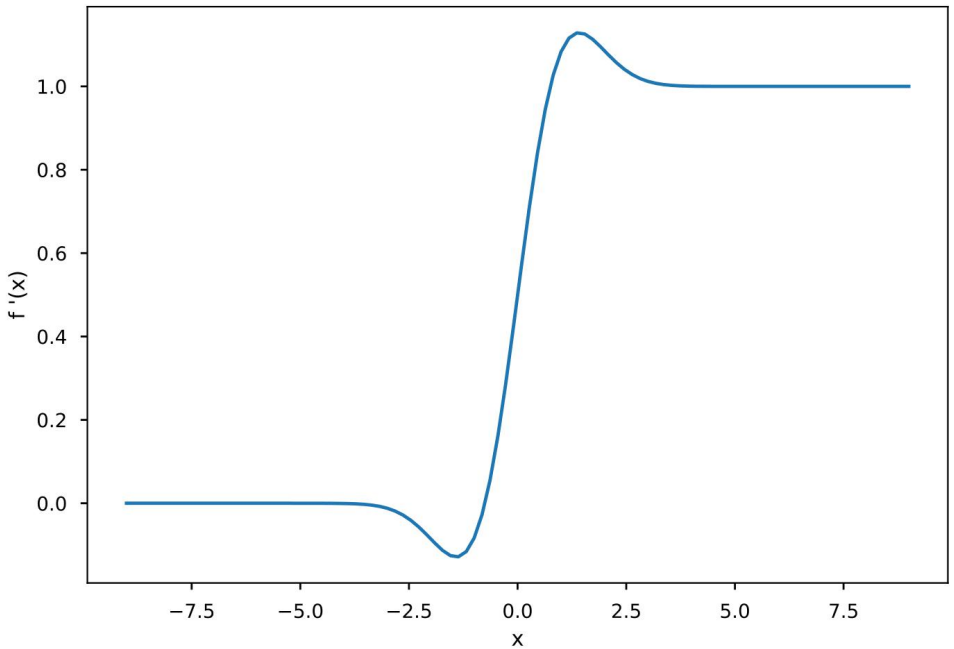




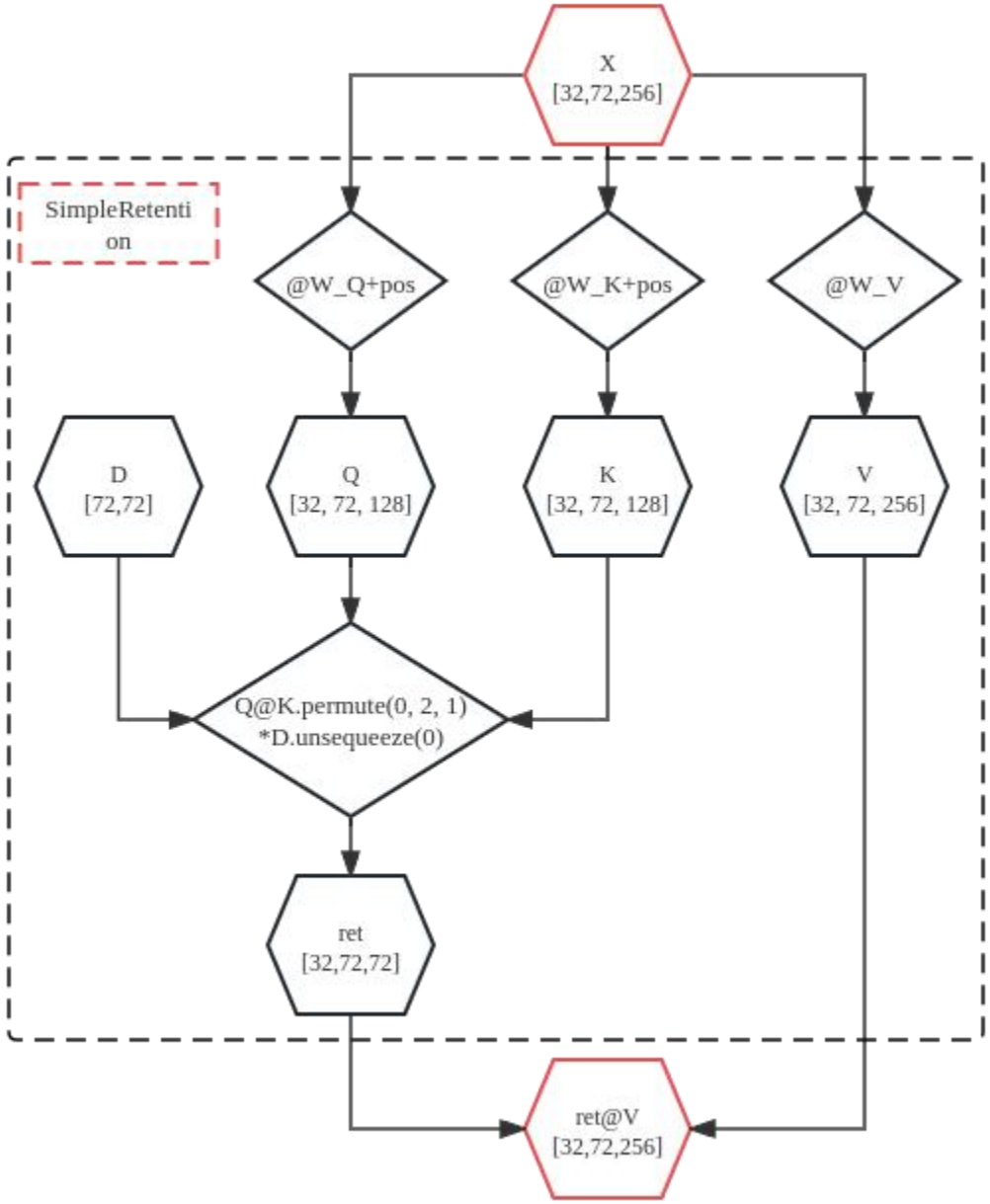
Retnet

GELU function and it's Derivative





Retnet

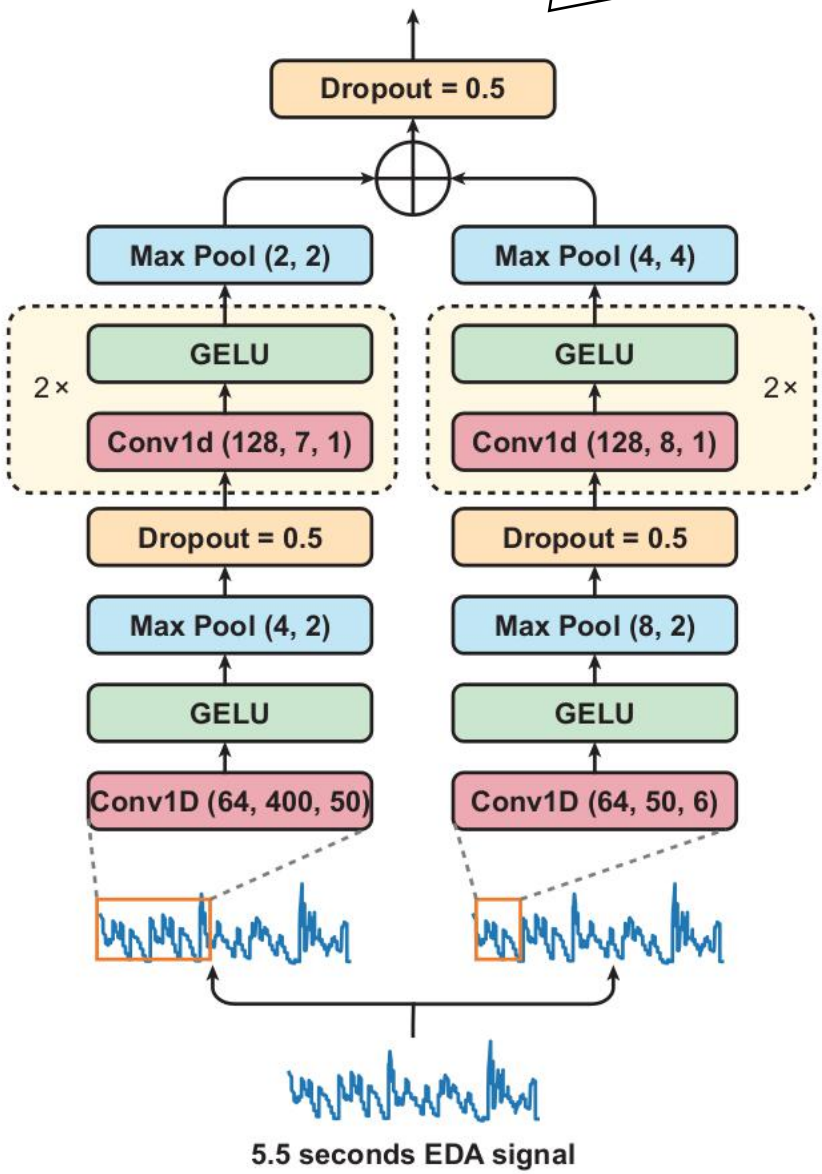

X
[32,72,256]
SimpleRetenti
on
@W_Q+pos
@W_K+pos
@W_V
D
[72,72]
Q
[32, 72, 128]
K
[32, 72, 128]
V
[32, 72, 256]
Q@K.permute(0, 2, 1)
*D.unsequeeze(0)
ret
[32,72,72]
ret@V
[32,72,256]

```
RetNet(
  (retentions): ModuleList(
    (0-2): 3 x MultiScaleRetention(
      (group_norm): GroupNorm(2, 512, eps=1e-05, affine=True)
      (retentions): ModuleList(
        (0-1): 2 x SimpleRetention(
          (xpos): XPOS()
        )
      )
    )
  )
  (ffns): ModuleList(
    (0-2): 3 x Sequential(
      (0): Linear(in_features=256, out_features=512, bias=True)
      (1): GELU(approximate='none')
      (2): Linear(in_features=512, out_features=256, bias=True)
    )
  )
  (layer_norms_1): ModuleList(
    (0-2): 3 x LayerNorm((256,), eps=1e-05, elementwise_affine=True)
  )
  (layer_norms_2): ModuleList(
    (0-2): 3 x LayerNorm((256,), eps=1e-05, elementwise_affine=True)
  )
)
```

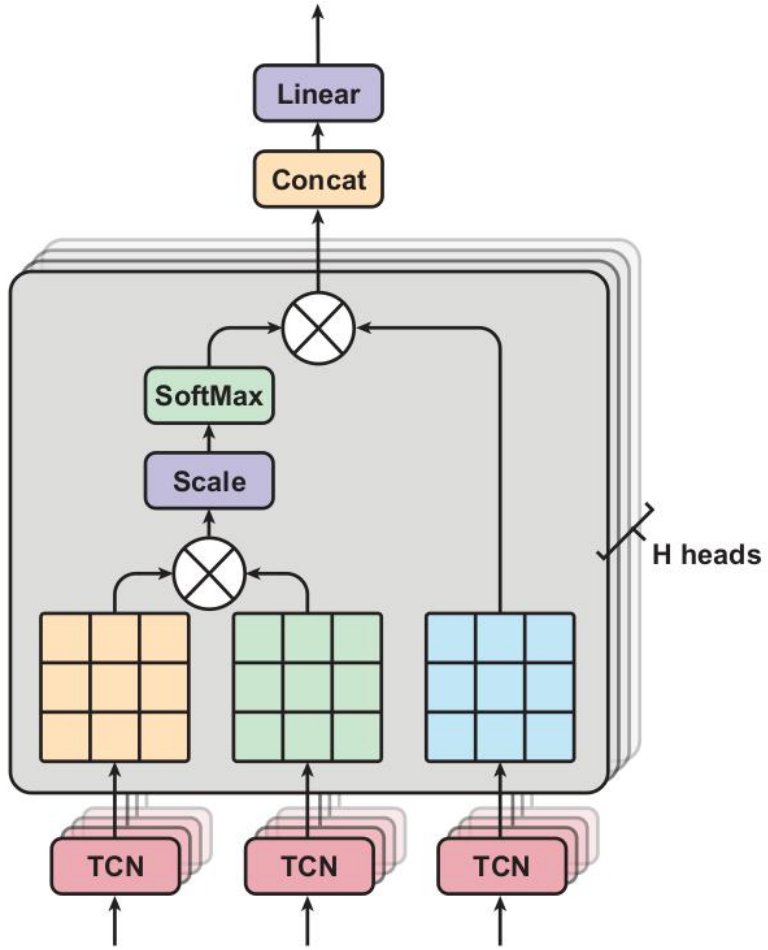
PainAtnNet

	样本			分类器
BioVid热疼痛数据库	皮肤电活动（EDA）	皮肤电导水平(SCL)		支持向量机（SVM）
		皮肤电导反应(SCR)		k近邻（KNN）
	心电图（ECG）			贝叶斯模型
	肌电图（EMG）			回归模型
	脑电图（EEG）			tree-based model
	video			

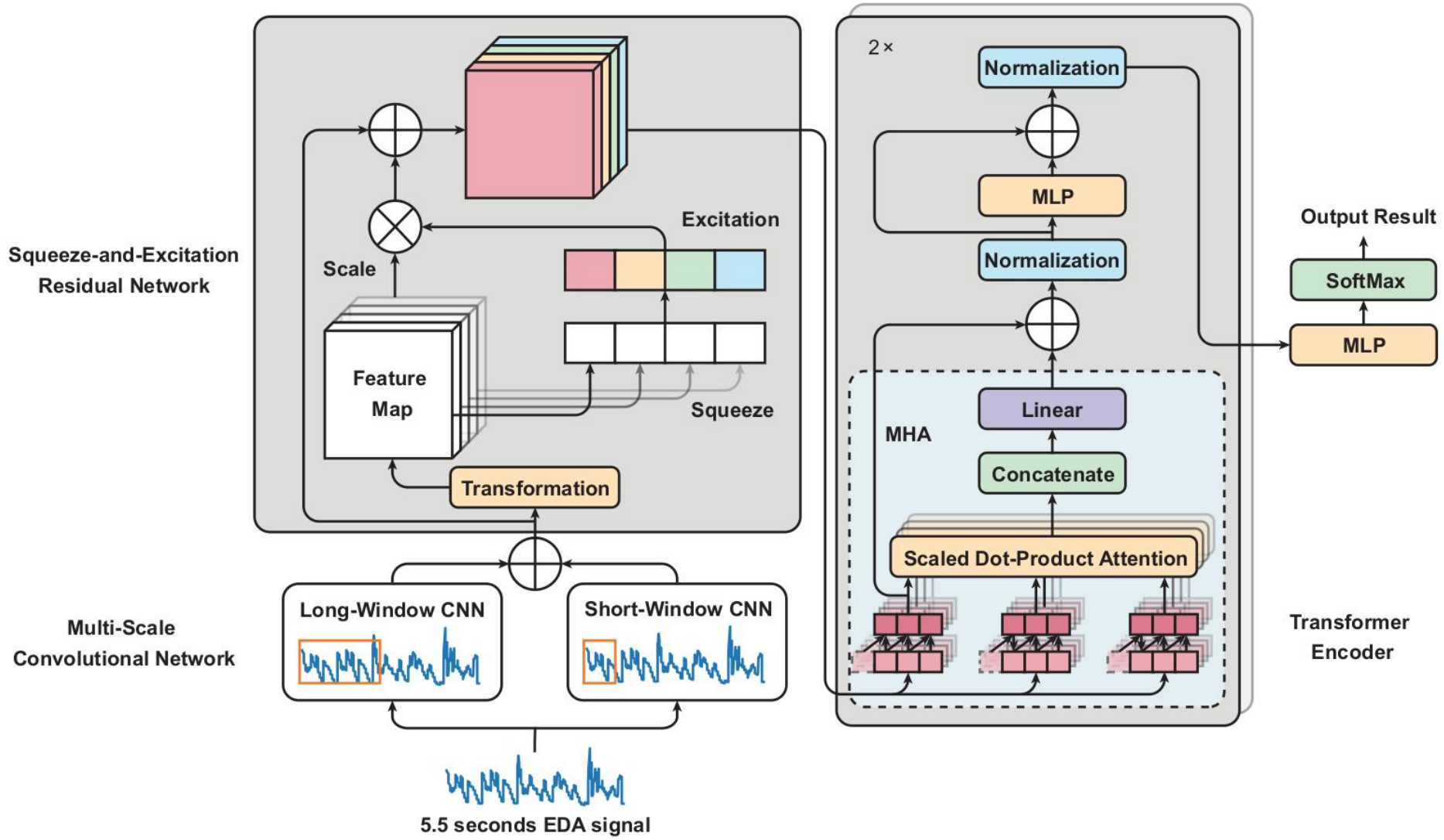
MSCN

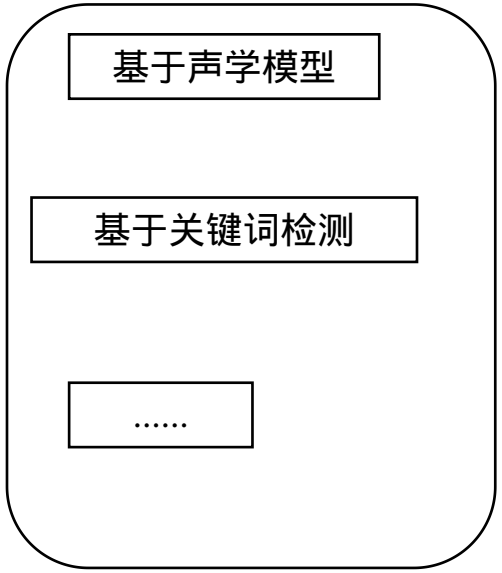


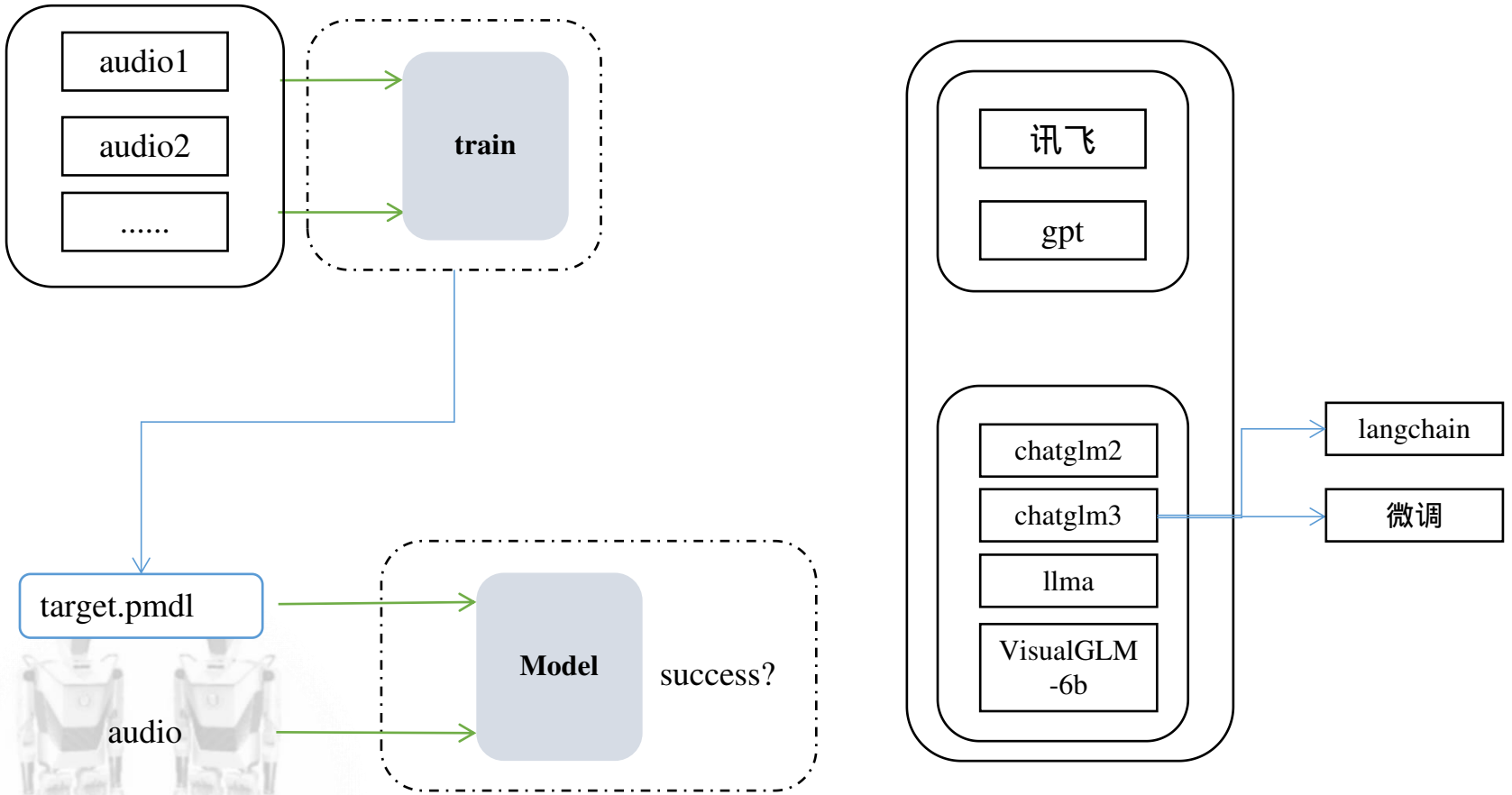
EDA信号本质上是非平稳的信号
通过利用窗口不同的卷积层进行卷积



PainAtnNet


Squeeze-and-Excitation
Residual Network
Scale
Excitation
Feature
Map
Squeeze
Transformation
Multi-Scale
Convolutional Network
Long-Window CNN
Short-Window CNN
5.5 seconds EDA signal
2×
Normalization
MLP
Normalization
MHA
Linear
Concatenate
Scaled Dot-Product Attention
Transformer
Encoder
Output Result
SoftMax
MLP


基于声学模型
基于关键词检测
.....


audio1
audio2
.....
train
target.pmdl
audio
Model
success?
讯飞
gpt
chatglm2
chatglm3
llma
VisualGLM
-6b
langchain
微调
vits
summer_speech

隐马尔可夫模型（HMM）中，我们不知道模型具体的状态序列，只知道状态转移的概率，即模型的状态转换过程是不可观察的。